

Diseases of White Blood Cells, Lymph Nodes, Spleen, and Thymus

CHAPTER CONTENTS

Normal Hematopoiesis, 531

Disorders of White Cells, 533

Leukopenia, 533

Neutropenia, Agranulocytosis, 533

Reactive Proliferations of White Cells and Lymph Nodes, 534

Leukocytosis, 534

Lymphadenitis, 535

Acute Nonspecific Lymphadenitis, 535

Chronic Nonspecific Lymphadenitis, 535

Hemophagocytic Lymphohistiocytosis, 536

Neoplastic Proliferations of White Cells: Overview, 536

Etiologic and Pathogenetic Factors in White Cell Neoplasia, 536

Chromosomal Translocations and Other Acquired Mutations, 537

Inherited Genetic Factors, 537

Viruses, 537

Chronic Inflammation, 537

Iatrogenic Factors, 538

Smoking, 538

Lymphoid Neoplasms, 538

Definitions and Classifications, 538

Precursor B- and T-Cell Neoplasms, 539

Acute Lymphoblastic Leukemia/Lymphoma, 539

Peripheral B-Cell Neoplasms, 543

Chronic Lymphocytic Leukemia/Small Lymphocytic

Lymphoma, 543

Follicular Lymphoma, 544

Diffuse Large B-Cell Lymphoma, 546

Burkitt Lymphoma, 547

Mantle Cell Lymphoma, 548

Marginal Zone Lymphomas, 548

Hairy Cell Leukemia, 549

Peripheral T- and NK-Cell Neoplasms, 550

Peripheral T-Cell Lymphoma, Not Otherwise Specified, 550

Nodal T-Follicular Helper Cell Lymphoma, 550

Anaplastic Large-Cell Lymphoma (ALK Positive), 551

Adult T-Cell Leukemia/Lymphoma, 551

Mycosis Fungoides and Sézary Syndrome, 551

Large Granular Lymphocytic Leukemia, 552

Extranodal NK-/T-Cell Lymphoma, 552

Plasma Cell Neoplasms and Related Disorders, 552

Multiple Myeloma, 553

Smoldering Myeloma, 555

Solitary Plasmacytoma, 555

Monoclonal Gammopathy of Undetermined Significance, 555

Lymphoplasmacytic Lymphoma, 555

Hodgkin Lymphoma, 556

Classification, 556

Pathogenic Lessons From Lymphoid Neoplasms, 560

Myeloid Neoplasms, 560

Clonal Hematopoiesis, 561

Acute Myeloid Leukemia, 562

Classification, 562

Immunophenotype, 563

Cytogenetics and Genetics, 563

Myelodysplastic Neoplasms, 564

Myeloproliferative Neoplasms, 566

Chronic Myeloid Leukemia, 566

Polycythemia Vera, 568

Essential Thrombocythemia, 569

Primary Myelofibrosis, 570

Langerhans Cell Histiocytosis, 571

Spleen, 572

Splenomegaly, 573

Nonspecific Acute Splenitis, 573

Congestive Splenomegaly, 573

Splenic Infarcts, 574

Neoplasms, 574

Congenital Anomalies, 574

Rupture, 574

Thymus, 574

Developmental Disorders, 575

Thymic Hyperplasia, 575

Thymoma, 575

The components of the hematopoietic system have been traditionally divided into the *myeloid tissues*, which include the bone marrow and the cells derived from it (e.g., red cells, platelets, granulocytes, and monocytes), and the *lymphoid tissues*, consisting of the thymus, lymph nodes, and spleen. It is important to recognize, however, that this subdivision is artificial with respect to both the normal physiology of hematopoietic cells and the diseases affecting them. For example, although bone marrow contains relatively few lymphocytes, it is the source of all lymphoid progenitors and the home of long-lived plasma cells and memory

lymphocytes. Similarly, neoplastic disorders of myeloid progenitor cells (myeloid leukemias) originate in the bone marrow but secondarily involve the spleen and (to a lesser degree) the lymph nodes. Some red cell disorders (e.g., immunohemolytic anemia, discussed in Chapter 14) result from the formation of autoantibodies, indicating a primary disorder of lymphocytes. Thus, it is not possible to draw neat lines between diseases involving the myeloid and lymphoid tissues.

Recognizing this difficulty, we somewhat arbitrarily divide diseases of the hematopoietic tissues into two chapters. In this chapter we

discuss white cell diseases and disorders affecting the spleen and thymus. In Chapter 14, we consider diseases of red cells and those affecting hemostasis. Before delving into specific diseases, we will briefly discuss the origins of hematopoietic cells, since many disorders of white cells and red cells involve disturbances of their normal development and maturation.

NORMAL HEMATOPOIESIS

Blood cell progenitors first appear during the third week of embryonic development in the yolk sac. Cells derived from the yolk sac are the source of long-lived tissue macrophages such as microglial cells in the brain and Kupffer cells in the liver (Chapter 3), but the contribution of the yolk sac to blood formation, mainly in the form of embryonic red cells, is only transient. Definitive *hematopoietic stem cells* (HSCs) arise several weeks later in the mesoderm of the intraembryonic aorta/gonad/mesonephros region. During the third month of embryogenesis, HSCs migrate to the liver, which becomes the chief site of blood

cell formation until shortly before birth. HSCs also take up residence in the fetal placenta; this pool of HSCs is of uncertain physiologic relevance but has clinical importance, as HSCs harvested at birth from umbilical cord blood are used in therapeutic HSC transplantation. By the fourth month of development, HSCs shift in location yet again to the bone marrow. By birth, marrow throughout the skeleton is hematopoietically active, and hepatic hematopoiesis dwindles to a trickle, persisting only in scattered foci that become inactive soon after birth. After puberty, hematopoiesis ceases in distal bones and becomes restricted to the axial skeleton. As a result, in normal adults only about half of the marrow space is hematopoietically active.

All formed elements of blood—red cells, granulocytes, monocytes, platelets, and lymphocytes—have a common origin from HSCs, pluripotent cells that sit at the apex of a hierarchy of bone marrow progenitors (Fig. 13.1). It is estimated that normal adult marrow contains between 50,000 and 200,000 HSCs, which are responsible for maintaining the production of hematopoietic cells throughout life. Although most evidence supporting this view comes from studies in

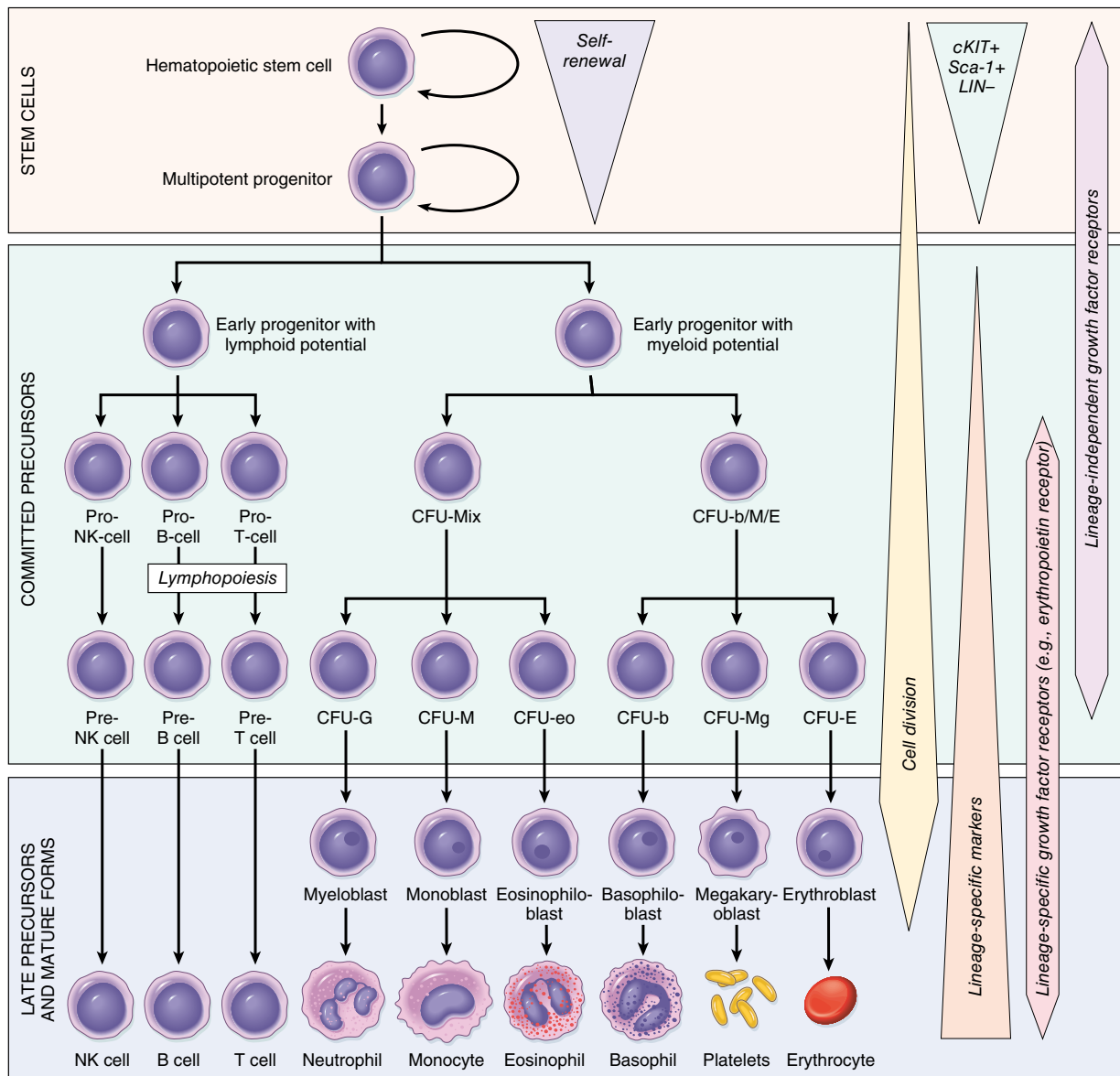


Fig. 13.1 Differentiation of blood cells. CFU, Colony forming unit; LIN-, negative for lineage-specific markers; NK, natural killer.

mice, human hematopoiesis is believed to proceed in a similar way. The development of mature blood cells from HSCs involves progressive commitment to increasingly specialized cell populations. HSCs give rise to several kinds of early progenitor cells with more restricted differentiation potential, such that they ultimately produce mainly myeloid cells or lymphoid cells. The origins of lymphoid cells are revisited when tumors derived from these cells are discussed. These early progenitors in turn give “birth” to progenitors that are further constrained in their differentiation potential. Some of these cells are referred to as *colony-forming units* (CFU; see Fig. 13.1) because they produce colonies composed of specific kinds of mature cells when grown in culture. From the various committed progenitors are derived the morphologically recognizable precursors, such as myeloblasts, proerythroblasts, and megakaryoblasts, which are the immediate progenitors of mature granulocytes, red cells, and platelets.

HSCs have two essential properties that are required for the maintenance of hematopoiesis: pluripotency and self-renewal. Pluripotency refers to the ability of a single HSC to generate all mature blood cells. When an HSC divides, at least one daughter cell must self-renew to avoid stem cell depletion. Self-renewing divisions occur within a specialized marrow niche, in which stromal cells and secreted factors nurture and protect the HSCs. As one might surmise from their ability to migrate during embryonic development, HSCs are not sessile. Particularly under conditions of stress, such as severe anemia or acute inflammation, HSCs are mobilized from the bone marrow and appear in the peripheral blood. In fact, HSCs used in transplantation are now mainly collected from the peripheral blood of donors treated with *granulocyte colony-stimulating factor* (G-CSF), one of the factors that can mobilize marrow HSCs from their stem cell niches.

The marrow response to short-term physiologic needs is regulated by hematopoietic growth factors through their effects on the committed progenitors. These growth factors are called *colony-stimulating factors* (CSFs) because they were discovered by their ability to support the growth of colonies of blood cells in vitro. Because mature blood elements are terminally differentiated cells with finite lifespans, their numbers must be constantly replenished. In current models of hematopoiesis, some divisions of HSCs give rise to cells referred to as *multipotent progenitors*, which are more proliferative than HSCs but have a lesser capacity for self-renewal (see Fig. 13.1). Division of multipotent progenitors gives rise to at least one daughter cell that leaves the stem cell pool and begins to differentiate. Once past this threshold, these newly committed cells lose the capacity for self-renewal and commence an inexorable journey down a road that leads to terminal differentiation and death. However, as these progenitors differentiate, they also proliferate rapidly in response to growth factors, expanding their numbers. Some growth factors, such as KIT ligand and FLT3 ligand, act through receptors that are expressed on very early committed progenitors. Others, such as erythropoietin, granulocyte-macrophage colony-stimulating factor (GM-CSF), G-CSF, and thrombopoietin, act through receptors that are expressed mainly on committed progenitors with more restricted differentiation potentials. Feedback loops involving these lineage-specific growth factors tune the marrow output, allowing the numbers of formed blood elements (red cells, white cells, and platelets) to be maintained within appropriate ranges (Table 13.1).

Many diseases alter the production of blood cells. The marrow is the ultimate source of most cells of the innate and adaptive immune system and responds to infectious or inflammatory challenges by increasing its output of granulocytes under the direction of specific growth factors and cytokines. By contrast, many other disorders are associated with defects in hematopoiesis that lead to deficiencies of one or more types of blood cells. Primary tumors of hematopoietic cells are among the most important diseases that interfere with marrow function, but

TABLE 13.1 Adult Reference Ranges for Blood Cells^a

Cell Type	Range
White cells ($\times 10^3/\mu\text{L}$)	4.8–10.8
Granulocytes (%)	40–70
Neutrophils ($\times 10^3/\mu\text{L}$)	1.4–6.5
Lymphocytes ($\times 10^3/\mu\text{L}$)	1.2–3.4
Monocytes ($\times 10^3/\mu\text{L}$)	0.1–0.6
Eosinophils ($\times 10^3/\mu\text{L}$)	0–0.5
Basophils ($\times 10^3/\mu\text{L}$)	0–0.2
Red cells ($\times 10^3/\mu\text{L}$)	
Adult males	4.3–5
Adult females	3.5–5
Platelets ($\times 10^3/\mu\text{L}$)	150–450

^aReference ranges vary among laboratories. The reference ranges for the laboratory providing the result should always be used.

certain genetic diseases, infections, toxins, and nutritional deficiencies, as well as chronic inflammation from any cause, may also decrease the production of blood cells by the marrow.

Tumors of hematopoietic origin are often associated with mutations that block progenitor cell maturation or abrogate their growth factor dependence. The net effect of such derangements is an unregulated clonal expansion of hematopoietic elements, which replace normal marrow progenitors and often spread to other hematopoietic tissues. In some instances, these tumors originate from transformed HSCs that retain the ability to differentiate along multiple lineages, whereas in other instances the origin is a more differentiated progenitor that has acquired an abnormal capacity for self-renewal (Chapter 7).

MORPHOLOGY

The bone marrow is a unique microenvironment that supports the orderly proliferation, differentiation, and release of blood cells. It is filled with a network of thin-walled sinusoids lined by a single layer of endothelial cells, which are underlain by a discontinuous basement membrane and adventitial cells. Within the interstitium lie clusters of hematopoietic cells and fat cells. Differentiated blood cells enter the circulation by transcellular migration through the endothelial cells.

The normal marrow is organized in subtle, but important, ways. For example, normal megakaryocytes lie next to sinusoids and extend cytoplasmic processes that bud off into the bloodstream to produce platelets, while red cell precursors often surround macrophages that dispose of nuclear remnants produced when red cells disgorge their nucleus prior to release into the bloodstream. Processes that distort the marrow architecture, such as deposits of metastatic cancer or granulomatous disorders, can cause the abnormal release of immature precursors into the peripheral blood, a finding referred to as **leukoerythroblastosis**.

Marrow aspirate smears provide the best assessment of the morphology of hematopoietic cells. The most mature marrow precursors can be identified based on their morphology alone. Immature precursors (“blast” forms) of different types are morphologically similar and can only be identified definitively using lineage-specific antibodies and histochemical markers (described later under white cell neoplasms). Bone marrow biopsies are a good way of estimating marrow activity. In normal adults the ratio of fat cells to hematopoietic elements is about 1:1. In hypoplastic states (e.g., aplastic anemia) the proportion of fat cells is greatly increased; conversely, fat cells often disappear when the marrow is involved by hematopoietic tumors and in diseases characterized by compensatory hyperplasias (e.g., hemolytic anemias) and neoplastic proliferations such as leukemias. Other disorders (e.g., metastatic cancers and granulomatous diseases) induce local marrow fibrosis. In such cases the marrow usually cannot be aspirated and the lesions are best seen in biopsies.

DISORDERS OF WHITE CELLS

Disorders of white blood cells can be classified into two broad categories: **proliferative disorders**, in which there is an expansion of leukocytes; and **leukopenias**, defined by a deficiency of leukocytes. Proliferations of white cells can be *reactive* or *neoplastic*. Reactive proliferations are typically seen in the setting of infections or inflammatory processes, when large numbers of leukocytes are needed for an effective host response. Neoplastic disorders, though less frequent, are more important clinically. In the following discussion, we first describe the leukopenic states and the common reactive disorders and then consider in some detail the malignant proliferations of white cells.

LEUKOPENIA

Leukopenia, a decrease in the number of circulating white cells, may be seen in a variety of disorders. A low white cell count most commonly results from reduced numbers of neutrophils (*neutropenia*, *granulocytopenia*). *Lymphopenia* is less common; in addition to congenital immunodeficiency diseases (Chapter 6), it is most often observed in advanced human immunodeficiency virus (HIV) infection, following therapy with glucocorticoids or cytotoxic drugs, autoimmune disorders, malnutrition, and certain acute viral infections. In the latter setting, lymphopenia stems from lymphocyte redistribution rather than a decrease in the number of lymphocytes in the body. Acute viral infections induce production of type I interferons, which activate T lymphocytes and change the expression of surface proteins that regulate T-cell migration. These changes result in the sequestration of activated T cells in lymph nodes and increased adherence to vascular endothelium, both of which contribute to lymphopenia. Granulocytopenia is more common and is often associated with diminished granulocyte function and thus merits further discussion.

Neutropenia, Agranulocytosis

Neutropenia, a reduction in the number of neutrophils in the blood, occurs in a wide variety of circumstances. *Agranulocytosis*, a marked reduction in neutrophils (usually $<100/\mu\text{L}$), has the serious consequence of making individuals susceptible to bacterial and fungal infections.

Pathogenesis

Neutropenia can be caused by (1) inadequate or ineffective granulopoiesis or (2) increased destruction or sequestration of neutrophils in the periphery. Inadequate or ineffective granulopoiesis is observed in the setting of:

- *Suppression of HSCs*, as occurs in aplastic anemia (Chapter 14) and infiltrative marrow disorders (e.g., tumors, granulomatous disease); in these conditions, granulocytopenia is accompanied by anemia and thrombocytopenia.
- *Suppression of committed granulocytic precursors* by exposure to certain drugs (discussed later).
- Disease states associated with *ineffective hematopoiesis*, such as megaloblastic anemia (Chapter 14) and myelodysplastic neoplasms, in which defective precursors die in the marrow.
- Rare *congenital conditions* (e.g., severe congenital neutropenia [Kostmann syndrome]), in which inherited defects in specific genes impair granulocytic differentiation.

Accelerated destruction or sequestration of neutrophils occurs with:

- *Immune mediated injury* to neutrophils, which can be idiopathic, associated with a well-defined immunologic disorder (e.g., systemic lupus erythematosus), or caused by exposure to drugs.
- *Splenomegaly*, in which splenic enlargement leads to increased numbers and activity of macrophages with resultant sequestration

and destruction of neutrophils in the spleen and modest neutropenia, sometimes associated with anemia and often with thrombocytopenia.

- *Increased peripheral utilization*, which can occur in overwhelming bacterial, fungal, or rickettsial infections.

The most common cause of agranulocytosis is drug toxicity.

Certain drugs, such as alkylating agents and antimetabolites used in cancer treatment, produce agranulocytosis in a predictable, dose-related fashion. Because such drugs cause a generalized suppression of hematopoiesis, production of red cells and platelets is also affected. Agranulocytosis may also occur as an idiosyncratic reaction to a large variety of agents including certain antibiotics, anticonvulsants, antiinflammatory drugs, antipsychotic drugs, and diuretics. The neutropenia induced by antipsychotic agents such as chlorpromazine and related phenothiazines results from a toxic effect on granulocytic precursors in the bone marrow. In contrast, agranulocytosis following administration of other drugs, such as sulfonamides, probably stems from antibody-mediated destruction of neutrophils through mechanisms similar to those involved in drug-induced immunohemolytic anemias (Chapter 14).

In some patients with acquired idiopathic neutropenia, autoantibodies directed against neutrophil-specific antigens destroy neutrophils. Severe neutropenia may also occur in association with monoclonal proliferations of large granular lymphocytes (so-called *LGL leukemia*). The mechanism of this neutropenia is not clear; suppression of granulocytic progenitors by products of the neoplastic cell (usually a CD8+ cytotoxic T cell) is considered most likely.

MORPHOLOGY

The alterations in the **bone marrow** vary with cause. With excessive destruction of neutrophils in the periphery, the marrow is usually hypercellular due to a compensatory increase in granulocytic precursors. Hypercellularity is also the rule with neutropenia caused by ineffective granulopoiesis, as occurs in megaloblastic anemia and myelodysplastic neoplasms. Agranulocytosis caused by agents that suppress or destroy granulocyte precursors is understandably associated with marrow hypocellularity.

Infections are a common consequence of agranulocytosis. Necrotizing ulcerative lesions of the gingiva, floor of the mouth, buccal mucosa, pharynx, or elsewhere in the oral cavity are quite characteristic. These are typically deep, undermined, and covered by gray to green-black necrotic membranes from which numerous bacteria or fungi can be isolated. Less frequently, similar ulcerative lesions occur in the skin, vagina, anus, or gastrointestinal tract. Severe life-threatening invasive bacterial or fungal infections may occur in the lungs, urinary tract, and kidneys. The neutropenic patient is at particularly high risk for deep fungal infections caused by *Candida* and *Aspergillus*. Sites of infection often show a massive growth of organisms with little leukocytic response.

Clinical Features

The signs and symptoms of neutropenia are related to infection and include malaise, chills, and fever, often followed by marked weakness and fatigability. With agranulocytosis, infections are often overwhelming and may cause death within hours to days.

Serious infections are most likely when the neutrophil count falls below $500/\mu\text{L}$. Because infections are often fulminant, broad-spectrum antibiotics must be given expeditiously whenever signs or symptoms appear. In some instances, such as following myelosuppressive chemotherapy, neutropenia is treated with G-CSF, a growth factor that stimulates the production of granulocytes from marrow precursors.

REACTIVE PROLIFERATIONS OF WHITE CELLS AND LYMPH NODES

Leukocytosis

Leukocytosis refers to an increase in the number of white cells in the blood. It is a common reaction to a variety of inflammatory states.

Pathogenesis

The peripheral blood leukocyte count is influenced by several factors, including:

- The size of the myeloid and lymphoid precursor and storage cell pools in the bone marrow, thymus, circulation, and peripheral tissues.
- The rate of release of white cells from the storage pools into the circulation.
- The proportion of white cells that are adherent to blood vessel walls at any time (the marginal pool).
- The rate of extravasation of white cells from the blood into tissues.

As discussed in Chapter 3, leukocyte homeostasis is maintained by cytokines, growth factors, and adhesion molecules through their effects on the proliferation, differentiation, and extravasation of leukocytes and their progenitors. [Table 13.2](#) summarizes the major mechanisms of neutrophilic leukocytosis and its causes, the most important of which is infection. In acute infection there is a rapid increase in the egress of mature granulocytes from the bone marrow pool, an alteration that may be mediated through the effects of tumor necrosis factor (TNF) and interleukin-1 (IL-1). If the infection or an inflammatory process is prolonged, IL-1, TNF, and other inflammatory mediators

stimulate macrophages, bone marrow stromal cells, and T cells to produce increased amounts of hematopoietic growth factors. These factors enhance the proliferation and differentiation of committed granulocytic progenitors and, over several days, cause a sustained increase in neutrophil production.

Some growth factors preferentially stimulate the production of a single type of leukocyte. For example, IL-5 mainly stimulates eosinophil production, while G-CSF induces neutrophilia. Such factors are differentially produced in response to various pathogenic stimuli, and, as a result, the five principal types of leukocytosis (neutrophilia, eosinophilia, basophilia, monocytosis, and lymphocytosis) tend to be observed in different clinical settings ([Table 13.3](#)).

In sepsis or severe inflammatory disorders (e.g., Kawasaki disease), leukocytosis is often accompanied by morphologic changes in neutrophils, such as toxic granulations, Döhle bodies, and cytoplasmic vacuoles ([Fig. 13.2](#)). *Toxic granules*, which are coarser and darker than normal neutrophilic granules, represent abnormal azurophilic (primary) granules. *Döhle bodies* are patches of dilated endoplasmic reticulum that appear as sky-blue cytoplasmic “puddles.”

In most instances it is not difficult to distinguish reactive and neoplastic leukocytoses, but uncertainties may arise in two settings. Acute viral infections, particularly in children, can cause the appearance of large numbers of activated lymphocytes that resemble neoplastic lymphoid cells. At other times, particularly in severe infections, many

TABLE 13.2 Mechanisms and Causes of Leukocytosis

Increased Marrow Production
Chronic infection or inflammation (growth factor–dependent)
Paraneoplastic (e.g., Hodgkin lymphoma; growth factor–dependent)
Myeloproliferative neoplasms (e.g., chronic myeloid leukemia; growth factor–independent)
Increased Release From Marrow Stores
Acute inflammation (e.g., with infection)
Chronic inflammation (multiple causes)
Decreased Margination
Exercise
Catecholamines
Decreased Extravasation Into Tissues
Glucocorticoids

TABLE 13.3 Causes of Leukocytosis

Type of Leukocytosis	Causes
Neutrophilic leukocytosis	Acute bacterial infections, especially those caused by pyogenic organisms; sterile inflammation caused by, for example, tissue necrosis (myocardial infarction, burns)
Eosinophilic leukocytosis (eosinophilia)	Allergic disorders such as asthma, hay fever; parasitic infestations; drug reactions; certain malignancies (e.g., Hodgkin and some non-Hodgkin lymphomas); autoimmune disorders (e.g., pemphigus, dermatitis herpetiformis) and some vasculitides; atheroembolic disease (transient)
Basophilic leukocytosis (basophilia)	Rare, often indicative of a myeloproliferative neoplasm (e.g., chronic myeloid leukemia)
Monocytosis	Chronic infections (e.g., tuberculosis), bacterial endocarditis, rickettsiosis, and malaria; autoimmune disorders (e.g., systemic lupus erythematosus); inflammatory bowel diseases (e.g., ulcerative colitis)
Lymphocytosis	Accompanies monocytosis in many disorders associated with chronic immunologic stimulation (e.g., tuberculosis, brucellosis); viral infections (e.g., hepatitis A, cytomegalovirus, Epstein-Barr virus); <i>Bordetella pertussis</i> infection

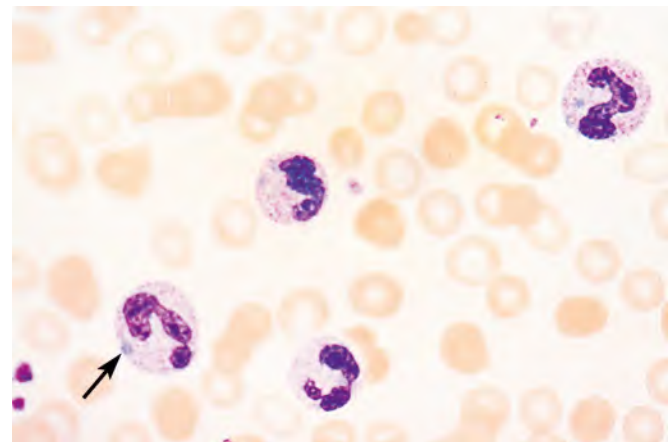


Fig. 13.2 Reactive changes in neutrophils. Neutrophils containing coarse purple cytoplasmic granules (toxic granulations) and blue cytoplasmic patches of dilated endoplasmic reticulum (Döhle bodies) (*arrow*) are observed in this peripheral blood smear prepared from a patient with bacterial sepsis.

immature granulocytes appear in the blood, mimicking myeloid leukemia (*leukemoid reaction*). Special laboratory studies (discussed later) are helpful in distinguishing reactive and neoplastic leukocytoses.

Lymphadenitis

Following their initial development from precursors in the central (also called primary) lymphoid organs—the bone marrow for B cells and the thymus for T cells—lymphocytes circulate through the blood and, under the influence of specific cytokines and chemokines, home to lymph nodes, spleen, tonsils, adenoids, and Peyer patches, which constitute the peripheral (secondary) lymphoid tissues. Lymph nodes, the most widely distributed and easily accessible lymphoid tissue, are frequently examined for diagnostic purposes. They are discrete encapsulated structures that contain separate B-cell and T-cell zones, each richly invested with phagocytes and antigen-presenting cells (see Fig. 6.8, Chapter 6).

The activation of resident immune cells leads to morphologic changes in lymph nodes. Within several days of antigenic stimulation, the primary follicles enlarge and develop pale-staining *germinal centers*, highly dynamic structures in which B cells acquire the capacity to make high-affinity antibodies against specific antigens. Paracortical T-cell zones may also undergo hyperplasia. The degree and pattern of morphologic change are dependent on the inciting stimulus and the intensity of the response. Trivial injuries and infections induce subtle changes, while more significant infections inevitably produce nodal enlargement and sometimes leave residual scarring. For this reason, lymph nodes in adults are almost never “normal” or “resting,” and it is often necessary to distinguish morphologic changes secondary to past exposures from those related to present disease. Infections and inflammatory stimuli often elicit regional or systemic immune reactions within lymph nodes. Some that produce distinctive morphologic patterns are described in other chapters. Most, however, cause stereotypical patterns of lymph node reaction designated acute and chronic nonspecific lymphadenitis.

Acute Nonspecific Lymphadenitis

Acute lymphadenitis in the cervical region is most often due to drainage of microbes or microbial products from infections of the teeth or tonsils, while in the axillary or inguinal regions it is most often caused by infections in the extremities. Acute lymphadenitis also occurs in mesenteric lymph nodes in the setting of acute appendicitis and other inflammatory conditions involving the gut (including self-limiting viral infections), a differential diagnosis that plagues the surgeon. Systemic viral infections (particularly in children) and bacteremia often produce acute generalized lymphadenopathy.

MORPHOLOGY

The nodes are swollen, gray-red, and engorged. Microscopically, there is prominence of large reactive germinal centers containing numerous mitotic figures. Macrophages often contain particulate debris derived from dead bacteria or necrotic cells. When pyogenic organisms are the cause, neutrophils are prominent, and the centers of the follicles may undergo necrosis; sometimes the entire node is converted to pus. With less severe reactions, scattered neutrophils infiltrate about the follicles and accumulate within the lymphoid sinuses. The endothelial cells lining the sinuses become activated and enlarge in size.

Nodes involved by acute lymphadenitis are swollen and painful. When abscess formation is extensive the nodes become fluctuant. The overlying skin is red. Sometimes, suppurative infections penetrate through the capsule of the node and track to the skin to produce draining sinuses. Healing of such lesions is associated with scarring.

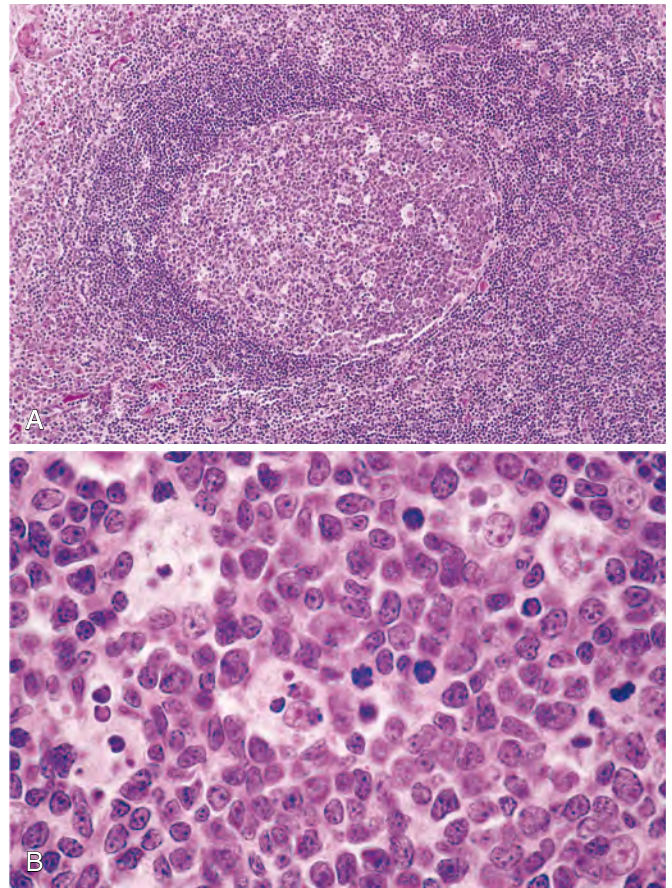


Fig. 13.3 Follicular hyperplasia. (A) Low-power view showing a reactive follicle and surrounding mantle zone. The dark-staining mantle zone is more prominent adjacent to the germinal center light zone in the left half of the follicle. The right half of the follicle consists of the dark zone. (B) High-power view of the dark zone shows several mitotic figures and numerous macrophages containing phagocytosed apoptotic cells (tingible bodies).

Chronic Nonspecific Lymphadenitis

A wide variety of chronic immunologic stimuli may produce nonspecific lymphadenitis. Several different patterns of morphologic change are seen, often within the same lymph node.

MORPHOLOGY

Follicular hyperplasia is caused by stimuli that activate humoral immune responses. It is defined by the presence of large oblong germinal centers (secondary follicles), which are surrounded by a collar of small resting naive B cells (the mantle zone) (Fig. 13.3). Germinal centers are polarized, consisting of two distinct regions: (1) a dark zone with proliferating blast-like B cells (centroblasts) and (2) a light zone composed of B cells with irregular or cleaved nuclear contours (centrocytes). Interspersed among the germinal center B cells is an inconspicuous network of antigen-presenting follicular dendritic cells and macrophages (often referred to as **tingible body macrophages**) containing the nuclear debris of B cells, which undergo apoptosis if they fail to produce an antibody with a high affinity for antigen.

Causes of follicular hyperplasia include rheumatoid arthritis, toxoplasmosis, and early HIV infection. This form of hyperplasia is morphologically similar to follicular lymphoma (discussed later). Features favoring a reactive (nonneoplastic) hyperplasia include (1) preservation of the lymph node architecture,

including the interfollicular T-cell zones and the sinusoids, (2) marked variation in the shape and size of the follicles, and (3) the presence of frequent mitotic figures, phagocytic macrophages, and recognizable light and dark zones, all of which tend to be absent from neoplastic follicles.

Paracortical hyperplasia is caused by stimuli that trigger T-cell-mediated immune responses, such as acute viral infections (e.g., infectious mononucleosis). The T-cell regions typically contain immunoblasts, activated T cells three to four times the size of resting lymphocytes that have round nuclei, open chromatin, prominent nucleoli, and moderate amounts of pale cytoplasm. The expanded T-cell zones encroach on and, in particularly exuberant reactions, may efface the B-cell follicles. In such cases, immunoblasts are so numerous that special studies may be needed to exclude a lymphoid neoplasm. In addition, there is often hypertrophy of sinusoidal and vascular endothelial cells, sometimes accompanied by infiltrating macrophages and eosinophils.

Sinus histiocytosis (also called *reticular hyperplasia*) is marked by an increase in the number and size of the endothelial cells that line lymphatic sinusoids and increased numbers of intrasinusoidal macrophages, which expand and distort the sinusoids. This form of hyperplasia may be particularly prominent in lymph nodes draining cancers such as carcinoma of the breast.

Characteristically, lymph nodes in chronic reactions are nontender, as enlargement occurs slowly over time and acute inflammation with associated tissue damage is absent. Chronic lymphadenitis is particularly common in inguinal and axillary nodes, which drain relatively large areas of the body and are frequently stimulated by immune reactions to trivial injuries and infections of the extremities.

Chronic immune reactions also can promote the appearance of organized collections of immune cells in nonlymphoid tissues. These collections are sometimes called *tertiary lymphoid structures*. A classic example is seen in chronic gastritis caused by *Helicobacter pylori*, in which aggregates of mucosal lymphocytes are present that simulate the appearance of Peyer patches. A similar phenomenon occurs in rheumatoid arthritis, in which B-cell follicles often appear in the inflamed synovium. Lymphotoxin, a cytokine required for the formation of normal secondary lymphoid organs, is probably involved in the establishment of these “extranodal” inflammation-induced collections of lymphoid cells.

Hemophagocytic Lymphohistiocytosis

Hemophagocytic lymphohistiocytosis (HLH) is a reactive condition marked by cytopenias and signs and symptoms of systemic inflammation related to macrophage activation. Some forms are familial and may appear early in life, even in infants, while other forms are sporadic and may affect people of any age.

Pathogenesis

The common feature of all forms of HLH is systemic activation of macrophages and CD8⁺ cytotoxic T cells. The activated macrophages phagocytose blood cell progenitors in the marrow and formed elements in the peripheral tissues, while the “stew” of mediators released from macrophages and lymphocytes suppress hematopoiesis and produce symptoms of systemic inflammation. These effects lead to cytopenias and a shock-like picture, sometimes referred to as “cytokine storm” or the systemic inflammatory response syndrome (Chapter 4).

Familial forms of HLH are associated with several different mutations, many of which impact the ability of cytotoxic T lymphocytes (CTLs) and natural killer (NK) cells to properly form or deploy cytotoxic granules. It is hypothesized that these defects prevent CTLs and NK cells from eliminating infected cells, particularly cells infected with viruses. Since the infected cells cannot be eliminated, they continue to activate CD8⁺ T cells and NK cells, which respond by turning on their second

effector mechanism, the secretion of cytokines, notably IFN γ , which is a major activator of macrophages. The result is uncontrolled macrophage activation, leading to production of many cytokines that have systemic effects. One common trigger for HLH is Epstein-Barr virus (EBV) infection, suggesting that in some instances HLH stems from a defect in the ability of CD8⁺ CTLs to kill EBV-infected cells. HLH is also a common complication of peripheral T-cell lymphoma (discussed later), a tumor of mature T cells that is marked by immune dysregulation. Regardless of the trigger, HLH is uniformly associated with extremely high levels of inflammatory mediators such as interferon- γ , TNF- α , IL-6, and IL-12.

Clinical Features

Most patients present with an acute febrile illness associated with splenomegaly and hepatomegaly. Hemophagocytosis is usually seen on bone marrow examination but is neither sufficient nor required to make the diagnosis. Laboratory studies typically reveal anemia, thrombocytopenia, and very high levels of plasma ferritin and soluble IL-2 receptor (sIL-2R), both indicative of severe inflammation, as well as elevated liver function tests and triglyceride levels, both related to hepatitis. Coagulation studies may show evidence of disseminated intravascular coagulation. If untreated, this picture can progress rapidly to multiorgan failure, shock, and death.

The mainstays of treatment are immunosuppressive drugs such as glucocorticoids, and etoposide, a chemotherapy drug. Patients with germline mutations that cause HLH or who have persistent/resistant disease are candidates for HSC transplantation. Without treatment, the prognosis is grim, particularly in those with familial forms of the disease, who typically survive for less than 2 months. With prompt treatment, with or without subsequent HSC transplantation, roughly half of patients survive, though many do so with significant sequelae, such as renal damage in adults and growth retardation and intellectual disability in children.

NEOPLASTIC PROLIFERATIONS OF WHITE CELLS: OVERVIEW

Malignancies are clinically the most important disorders of white cells. These diseases fall into three broad categories:

- *Lymphoid neoplasms* include a diverse group of tumors of B-cell, T-cell, and NK-cell origin. In many instances the phenotype of the neoplastic cell closely resembles that of a particular lymphocyte class or stage of maturation, a feature that is used in the diagnosis and classification of these disorders.
- *Myeloid neoplasms* arise from early hematopoietic progenitors. Three categories of myeloid neoplasia are recognized: *acute myeloid leukemias* (AMLs), in which immature progenitor cells accumulate in the bone marrow; *myelodysplastic neoplasms*, which are associated with ineffective hematopoiesis and resultant peripheral blood cytopenias; and *myeloproliferative neoplasms*, in which increased production of one or more terminally differentiated myeloid elements (e.g., granulocytes) usually leads to elevated peripheral blood counts.
- The *histiocytoses* are uncommon proliferative lesions of macrophages and dendritic cells. “Histiocyte” (literally, “tissue cell”) is an archaic morphologic term that is still used. A special type of immature dendritic cell, the Langerhans cell, gives rise to a spectrum of neoplastic disorders referred to as the *Langerhans cell histiocytoses*.

Etiologic and Pathogenetic Factors in White Cell Neoplasia

As with other cancers, the etiology of white blood cell neoplasms uniformly involves inherited or genetic alterations in cancer genes that produce the hallmarks of cancer. Possibly because hematopoietic cells

“naturally” possess some cancer hallmarks, such as the ability to migrate widely through the bloodstream and egress into tissues, in general hematologic neoplasms are genetically less complex than cancers derived from “solid tissues,” a factor that may contribute to their increased curability. In addition to driver mutations in cancer genes, several environmental factors also contribute to the development of specific hematologic malignancies. Before delving into the specific entities, we will first discuss these genetic and environmental factors.

Chromosomal Translocations and Other Acquired Mutations

Nonrandom chromosomal abnormalities, most commonly translocations, are present in most white cell neoplasms. Many of these alterations are associated with specific neoplasms and have a critical role in their genesis (Chapter 7).

- *Recurrently affected genes are often those that play crucial roles in the development, growth, or survival of the normal counterpart of the malignant cell.* Mutations in certain genes are so strongly associated with specific tumor types that they are required for the diagnosis. Certain mutations produce a “dominant-negative” protein that interferes with a normal function (a loss of function); in others, the result is an inappropriate increase in some normal activity (a gain of function).
- *Oncoproteins created by genomic aberrations often block normal maturation, turn on progrowth signaling pathways, or protect cells from apoptotic cell death.* Fig. 13.4 highlights several well-characterized driver mutations and their pathogenic consequences, in particular white cell neoplasms.
 - Many oncoproteins cause an arrest in differentiation, often at a stage when cells are proliferating rapidly. This mechanism is most evident in the acute leukemias, in which dominant-negative oncogenic mutations involving transcription factors interfere with early stages of lymphoid or myeloid cell differentiation.
 - Other mutations in transcriptional regulators directly enhance the self-renewal of tumor cells, giving them stem cell–like properties. These types of mutations often collaborate with mutations that constitutively activate tyrosine kinases, which in turn activate RAS and its two downstream signaling arms, the PI3K/AKT and MAPK pathways (Chapter 7), thereby driving cell growth.
 - Finally, mutations that inhibit apoptosis are prevalent in certain hematologic malignancies.
- *Proto-oncogenes are often activated in lymphoid cells by errors that occur during attempted antigen receptor gene diversification.* Among

lymphoid cells, potentially oncogenic mutations occur most frequently in germinal center B cells. After antigen stimulation, B cells enter germinal centers and upregulate the expression of *activation-induced cytidine deaminase* (AID), a specialized DNA-modifying enzyme that is essential for two types of immunoglobulin gene modifications: *class switching*, an intragenic recombination event in which the IgM heavy-chain constant gene segment is replaced with a different constant segment (e.g., IgG₃), leading to a switch in the class (isotype) of antibody produced, and *somatic hypermutation*, which creates point mutations within immunoglobulin genes that may increase antibody affinity for antigen (Chapter 6). Certain proto-oncogenes, such as *MYC*, are activated in germinal center B-cell lymphomas by translocations to the transcriptionally active immunoglobulin locus. Remarkably, AID expression induces *MYC::IGH* translocations in a small fraction of normal germinal center B cells, apparently because AID creates lesions in DNA that lead to chromosomal breaks. “Mistargeting” of AID is also implicated in point mutations that upregulate the expression and activity of BCL6, an oncogenic transcription factor with an important role in several B-cell malignancies. Another type of regulated genomic instability that is unique to precursor B and T cells is attributable to V(D)J recombinase, which cuts DNA at specific sites within the immunoglobulin and T-cell receptor loci, respectively. This process is essential for generating diversity in assembled antigen receptor genes but sometimes goes awry, leading to the joining of proto-oncogenes to antigen receptor gene regulatory elements. This mechanism underlies the generation of oncogenic chromosomal rearrangements in a variety of T and B cell neoplasms.

Inherited Genetic Factors

As discussed in Chapter 7, individuals with genetic diseases that promote genomic instability, such as Bloom syndrome, Fanconi anemia, and ataxia telangiectasia, are at increased risk of acute leukemia. In addition, both Down syndrome (trisomy 21) and type I neurofibromatosis are associated with an increased incidence of childhood leukemia.

Viruses

Three lymphotropic viruses—human T-cell leukemia virus-1 (HTLV-1), EBV, and human herpesvirus 8 (HHV8; also known as Kaposi sarcoma herpesvirus [KSHV])—have been implicated as causative agents in particular lymphomas. The possible mechanisms of transformation by viruses are discussed in Chapter 7. HTLV-1 is associated with adult T-cell leukemia/lymphoma. EBV is found in a subset of Burkitt lymphoma, 30% to 40% of Hodgkin lymphoma (HL), many B-cell lymphomas arising in the setting of T-cell immunodeficiency, and rare NK-cell lymphomas. In addition to Kaposi sarcoma (Chapter 11), HHV8 is associated with an unusual B-cell lymphoma that presents as a malignant effusion, often in the pleural cavity.

Chronic Inflammation

Several agents that cause localized chronic inflammation predispose to lymphoid neoplasia, which almost always arises within the inflamed tissue. Examples include the associations between *H. pylori* infection and gastric B-cell lymphomas (Chapter 17); gluten-sensitive enteropathy and intestinal T-cell lymphomas; and even breast implants, which are associated with an unusual subtype of T-cell lymphoma. The causal link between chronic inflammation and cancer development was discussed in Chapter 7. This can be contrasted with HIV infection, which is associated with an increased risk of B-cell lymphomas that may arise within virtually any organ. Early in the course, T-cell dysregulation by HIV infection causes a systemic hyperplasia of germinal center B cells

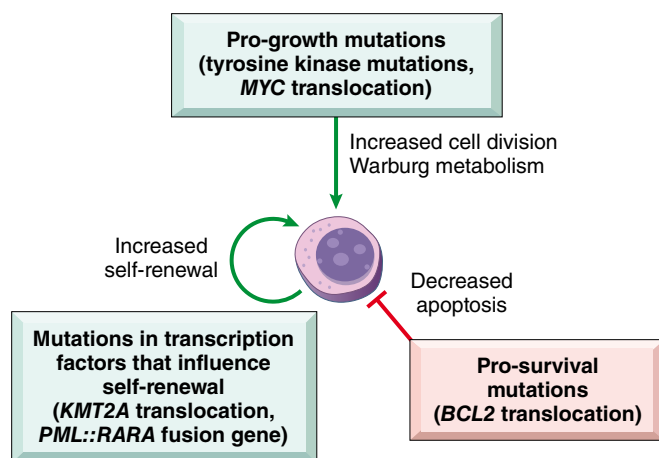


Fig. 13.4 Pathogenesis of white cell malignancies. Various tumors harbor mutations that principally affect maturation or enhance self-renewal, drive growth, or prevent apoptosis. Examples of each type of mutation are listed; details are provided later under specific tumor types.

that is associated with an increased incidence of germinal center B-cell lymphomas. In advanced infection (acquired immunodeficiency syndrome [AIDS]), severe T-cell immunodeficiency further elevates the risk for B-cell lymphomas, particularly those associated with EBV and KSHV/HHV8. These relationships are discussed in more detail in Chapter 6.

Iatrogenic Factors

Ironically, radiation therapy and certain forms of chemotherapy used to treat cancer increase the risk of subsequent myeloid and lymphoid neoplasms. This association stems from the mutagenic effects of ionizing radiation and chemotherapeutic drugs on hematolymphoid progenitor cells.

Smoking

The incidence of AML is increased 1.3- to 2-fold in smokers, presumably because of exposure to carcinogens, such as benzene, in tobacco smoke.

LYMPHOID NEOPLASMS

Taken together, the diverse lymphoid neoplasms constitute a complex, clinically important group of cancers, with about 100,000 new cases being diagnosed each year in the United States.

Definitions and Classifications

Neoplasms that present with widespread involvement of the bone marrow and (usually, but not always) the peripheral blood are called *leukemias*. Proliferations of white cells, typically lymphocytes, that usually present as discrete tissue masses are called *lymphomas*. Originally these terms were attached to what were considered distinct entities, but with time and increased understanding these divisions have blurred. Many entities called “lymphoma” occasionally have leukemic presentations, and evolution to “leukemia” is not unusual during the progression of incurable “lymphomas.” Conversely, tumors identical to “leukemias” sometimes arise as soft tissue masses without detectable bone marrow disease. Hence the terms leukemia and lymphoma merely reflect the usual tissue distribution of each disease at presentation.

Within the large group of lymphomas, *Hodgkin lymphoma* is segregated from other forms, which constitute the *non-Hodgkin lymphomas* (NHLs). Hodgkin lymphoma has distinctive pathologic features and is treated in a unique fashion. Another special group of tumors includes *plasma cell neoplasms*. These most often arise in the bone marrow and only infrequently involve lymph nodes or the peripheral blood.

The clinical presentation of lymphoid neoplasms is most often determined by the anatomic distribution of disease. Two-thirds of NHLs and virtually all Hodgkin lymphomas present as enlarged nontender lymph nodes (often >2 cm). The remaining NHLs present with symptoms related to the involvement of extranodal sites (e.g., skin, stomach, or brain). Lymphocytic leukemias most often come to attention because of signs and symptoms related to the suppression of normal hematopoiesis by tumor cells in the bone marrow, whereas the most common plasma cell neoplasm, multiple myeloma, causes bony destruction of the skeleton and often presents with pain due to pathologic fractures. Other symptoms are frequently caused by proteins secreted from tumor cells or from immune cells responding to the tumor. Specific examples include the plasma cell tumors, in which much of the pathophysiology is related to the secretion of whole antibodies or immunoglobulin fragments; Hodgkin lymphoma, which is often associated with fever related to the release of cytokines from nonneoplastic inflammatory cells; and peripheral T-cell lymphomas, tumors of

functional T cells that often release proinflammatory cytokines and chemokines.

Historically, few areas of pathology have evoked as much controversy as the classification of lymphoid neoplasms. Beginning in 1994, a single consensus classification system reigned for almost 3 decades, but regrettably (particularly for students) a recent schism has set the field back by producing two competing classification systems generated by different teams of pathologists and clinicians, the World Health Organization (WHO) classification and the International Consensus Classification (ICC). We will follow the WHO classification of lymphoid neoplasms (Table 13.4), occasionally referencing points of distinction from the ICC that students may need to understand and contend with.

The current WHO classification scheme uses morphologic, immunophenotypic, genotypic, and clinical features to sort the lymphoid neoplasms into five broad categories, separated according to the cell of origin:

1. Precursor B-cell neoplasms (neoplasms of immature B cells)
2. Peripheral B-cell neoplasms (neoplasms of mature B cells)
3. Precursor T-cell neoplasms (neoplasms of immature T cells)
4. Peripheral T-cell and NK-cell neoplasms (neoplasms of mature T cells and NK cells)
5. Hodgkin lymphomas (neoplasms of Reed-Sternberg cells and variants)

TABLE 13.4 World Health Organization Classification of Lymphoid Neoplasms^a

I. Precursor B-Cell Neoplasms
B-cell acute lymphoblastic leukemia/lymphoma (B-ALL)
II. Peripheral B-Cell Neoplasms
Chronic lymphocytic leukemia/small lymphocytic lymphoma
Splenic B cell lymphomas and leukemias
Marginal zone lymphoma
Mantle cell lymphoma
Follicular lymphoma
Diffuse large B-cell lymphoma
Burkitt lymphoma
Lymphoplasmacytic lymphoma
Plasma cell neoplasms
III. Precursor T-Cell Neoplasms
T-cell acute lymphoblastic leukemia/lymphoma (T-ALL)
IV. Peripheral T-Cell and NK-Cell Neoplasms
T-cell prolymphocytic leukemia
T-large granular lymphocytic leukemia
Mycosis fungoides
Sézary syndrome
Peripheral T-cell lymphoma, not otherwise specified
Anaplastic large-cell lymphoma
Nodal T-follicular helper cell lymphoma
Enteropathy-associated T-cell lymphoma
Subcutaneous panniculitis-like T-cell lymphoma
Hepatosplenic T-cell lymphoma
Adult T-cell leukemia/lymphoma
Extranodal NK/T-cell lymphoma
NK-cell leukemia
V. Hodgkin Lymphoma
Classic subtypes
Nodular sclerosis
Mixed cellularity
Lymphocyte-rich
Lymphocyte-depleted
Nodular lymphocyte predominant

^aSome rare entities not listed.

NK, Natural killer.

Before discussing the specific entities, some important principles relevant to lymphoid neoplasms should be emphasized.

- *Lymphoid neoplasia can be suspected based on clinical features, but histologic examination of lymph nodes or other involved tissues is required for diagnosis.* Analysis of lineage-specific protein (marker) expression and genetic alterations is an important complement to the morphologic studies. Markers recognized by antibodies that are helpful in the characterization of lymphomas and leukemias are listed in [Table 13.5](#).
- *Antigen receptor gene rearrangement generally precedes transformation of lymphoid cells; hence all daughter cells derived from the malignant progenitor share the same antigen receptor gene configuration and sequence and synthesize identical antigen receptor*

TABLE 13.5 Immune Cell Antigens Detected by Monoclonal Antibodies

Antigen Designation	Normal Cellular Distribution
Primarily T-Cell Associated	
CD1	Thymocytes and Langerhans cells
CD3	Thymocytes, mature T cells
CD4	Helper T cells, subset of thymocytes
CD5	T cells and small subset of B cells
CD8	Cytotoxic T cells, subset of thymocytes, and some NK cells
Primarily B-Cell Associated	
CD10	Pre-B cells and germinal center B cells
CD19	Pre-B cells and mature B cells but not plasma cells
CD20	Pre-B cells after CD19 and mature B cells but not plasma cells
CD21	EBV receptor; mature B cells and follicular dendritic cells
CD23	Activated mature B cells
CD79a	Marrow pre-B cells and mature B cells
Primarily Monocyte or Macrophage Associated	
CD11c	Granulocytes, monocytes, and macrophages; also expressed by hairy cell leukemias
CD13	Immature and mature monocytes and granulocytes
CD14	Monocytes
CD15	Granulocytes; Reed-Sternberg cells and variants
CD33	Myeloid progenitors and monocytes
CD64	Mature myeloid cells
Primarily NK-Cell Associated	
CD16	NK cells and granulocytes
CD56	NK cells and a subset of T cells
Primarily Stem Cell and Progenitor Cell Associated	
CD34	Pluripotent hematopoietic stem cells and progenitor cells of many lineages
Activation Markers	
CD30	Activated B cells, T cells, and monocytes; Reed-Sternberg cells and variants
Present on All Leukocytes	
CD45	All leukocytes; also known as leukocyte common antigen (LCA)

CD, Cluster of differentiation; EBV, Epstein-Barr virus; NK, natural killer.

proteins (either immunoglobulins or T-cell receptors). In contrast, normal immune responses are comprised of polyclonal populations of lymphocytes that express many different antigen receptors. Thus, analyses of antigen receptor genes and their protein products can be used to distinguish reactive (polyclonal) and malignant (monoclonal) lymphoid proliferations. In addition, each antigen receptor gene rearrangement produces a unique DNA sequence that constitutes a highly specific clonal marker, which can be used to detect small numbers of residual malignant lymphoid cells after therapy.

- *Most lymphoid neoplasms resemble some recognizable stage of B- or T-cell differentiation (Fig. 13.5), a feature that is used in their classification.* The majority (85% to 90%) of lymphoid neoplasms are of B-cell origin, with most of the remainder being T-cell tumors; tumors of NK-cell origin are rare.
- *Lymphoid neoplasms are often associated with immune abnormalities.* Both a loss of protective immunity (susceptibility to infection) and a breakdown of tolerance (autoimmunity) may be seen, sometimes in the same patient. In a further ironic twist, individuals with inherited or acquired immunodeficiency are themselves at high risk of developing certain lymphoid neoplasms, particularly those caused by oncogenic viruses (e.g., EBV).
- *Neoplastic B and T cells tend to recapitulate the behavior of their normal counterparts.* Like normal lymphocytes, neoplastic B and T cells express adhesion molecules and chemokine receptors that govern their homing to certain tissue sites, leading to characteristic patterns of involvement. For example, follicular lymphomas home to germinal centers in lymph nodes, whereas cutaneous T-cell lymphomas home to the skin. Variable numbers of neoplastic B and T lymphoid cells also recirculate through the lymphatics and peripheral blood to distant sites; as a result, most lymphoid tumors are widely disseminated at the time of diagnosis. Notable exceptions to this rule include Hodgkin lymphomas, which are sometimes restricted to one group of lymph nodes, and marginal zone lymphomas, which are often restricted to sites of chronic inflammation.
- *Hodgkin lymphoma spreads in an orderly stepwise fashion, whereas most forms of NHL disseminate widely and somewhat unpredictably early in their course.* Hence, while lymphoma staging provides useful prognostic information, it is of most utility in guiding therapy in Hodgkin lymphoma.

We begin our discussion of specific entities with neoplasms of immature lymphoid cells and then move to mature B-cell tumors, plasma cell neoplasms, and tumors of T cells and NK cells. Some of the salient molecular and clinical features of these neoplasms are summarized in [Table 13.6](#). We will finish by discussing Hodgkin lymphoma.

Precursor B- and T-Cell Neoplasms

Acute Lymphoblastic Leukemia/Lymphoma

Acute lymphoblastic leukemia/lymphomas (ALLs) are neoplasms composed of immature B (pre-B) or T (pre-T) cells, which are called lymphoblasts. About 85% are B-ALLs, which typically manifest as childhood acute leukemias. The less common T-ALLs tend to present in adolescent males as thymic lymphomas. There is, however, considerable overlap in the clinical behavior of B-ALL and T-ALL; for example, B-ALL uncommonly presents as a mass in the skin or a bone, and many T-ALLs present with or evolve to a leukemic picture. Because of their morphologic and clinical similarities, the various forms of ALL are discussed here together.

ALL is the most common cancer of children. Approximately 2500 new cases are diagnosed each year in the United States, most occurring

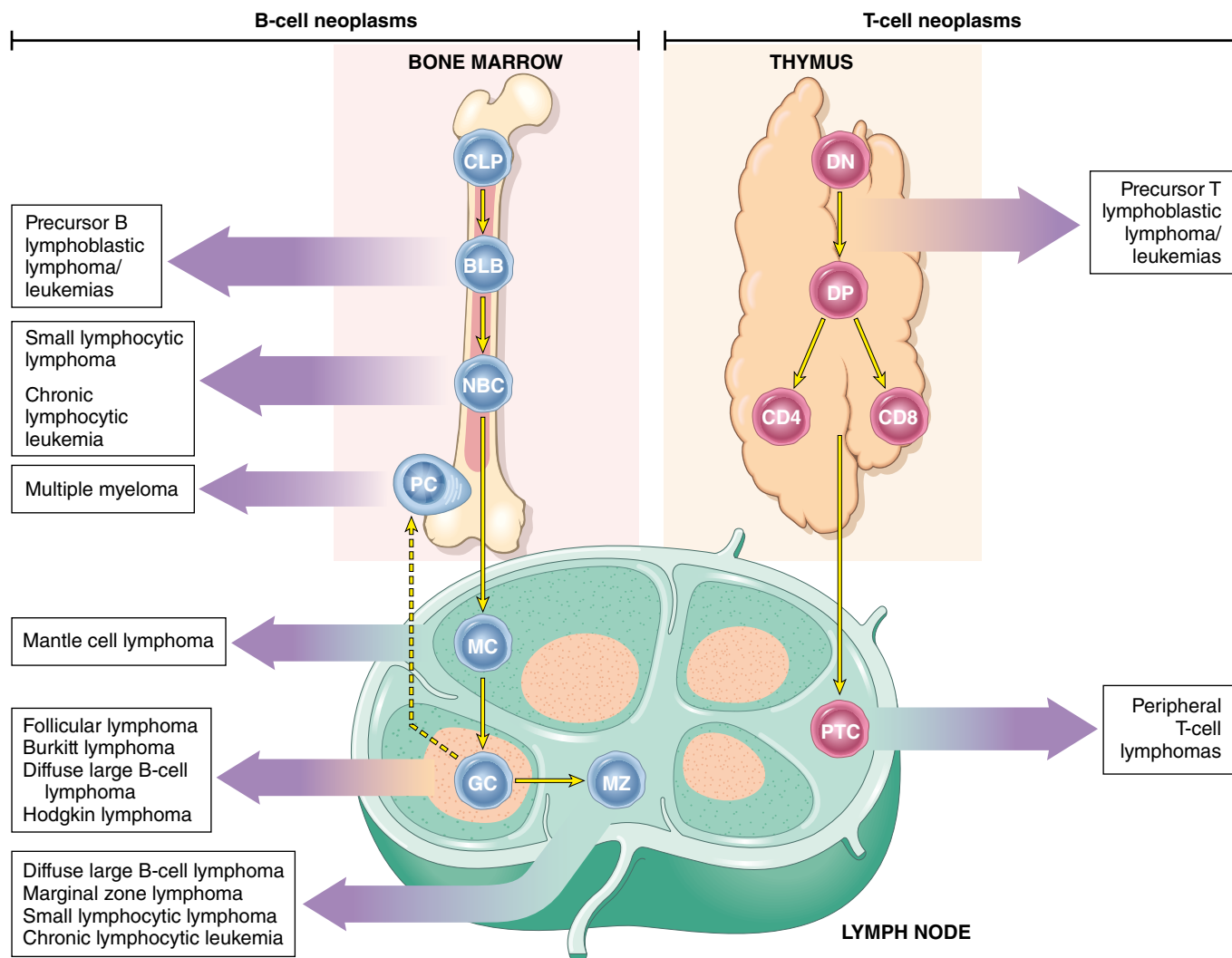


Fig. 13.5 Origin of lymphoid neoplasms. Stages of B- and T-cell differentiation from which specific lymphoid tumors emerge are shown. BLB, Pre-B lymphoblast; CLP, common lymphoid precursor; DN, CD4/CD8 double-negative pro-T cell; DP, CD4/CD8 double-positive pre-T cell; GC, germinal center B cell; MC, mantle B cell; MZ, marginal zone B cell; NBC, naive B cell; PC, plasma cell; PTC, peripheral T cell.

in individuals younger than 15 years. ALL is slightly more frequent in boys than in girls, and Hispanic/Latino children have the highest incidence of ALL of any ethnic group in the United States. B-ALL peaks in incidence at about the age of 3 years, perhaps because the number of normal bone marrow pre-B cells (the cell of origin) is greatest very early in life. Similarly, the peak incidence of T-ALL is in adolescence, the age when the thymus reaches maximum size. B-ALL and T-ALL also occur less frequently in adults of all ages.

Pathogenesis

Many of the chromosomal aberrations seen in ALL dysregulate the expression and function of transcription factors required for normal B- and T-cell development. Most T-ALLs have mutations in *NOTCH1*, a gene that is essential for T-cell development, while a high fraction of B-ALLs have mutations affecting genes such as *PAX5*, *TCF3*, *ETV6*, *RUNX1*, *BCR::ABL1*, *KMT2A*, and *PBX1*, all of which are required for the proper differentiation of early hematopoietic precursors. By disturbing the expression and function of “master” regulatory factors, these mutations promote maturation arrest and increased self-renewal,

a stem cell–like phenotype. Similar themes are relevant in the genesis of acute myeloid leukemia (discussed later).

In keeping with the multistep origin of cancer (Chapter 7), mutations in transcription factor genes are not sufficient to produce ALL. Other, complementary driver mutations often promote cell growth, such as mutations that increase tyrosine kinase activity (discussed later) and RAS signaling. Data from deep sequencing of ALL genomes suggest that fewer than 10 mutations are sufficient to produce full-blown ALL.

Approximately 90% of ALLs have numerical or structural chromosomal changes. Most common is hyperploidy (>50 chromosomes), but hypoploidy and a variety of balanced chromosomal translocations are also seen. Changes in chromosome numbers are of uncertain pathogenic significance but are important because they frequently correlate with immunophenotype and sometimes prognosis. For example, hyperdiploidy and hypodiploidy are seen only in B-ALL and are associated with better and worse prognoses, respectively. In addition, B-ALL and T-ALL are associated with aberrations affecting completely different sets of transcription factor genes; thus, while morphologically identical, they are genetically distinct.

TABLE 13.6 Summary of Major Types of Lymphoid Leukemias and Non-Hodgkin Lymphomas

Diagnosis	Cell of Origin	Genotype	Salient Clinical Features
Neoplasms of Precursor (Immature) B and T Cells			
B-cell acute lymphoblastic leukemia/lymphoma ^a	Bone marrow precursor B cell	Diverse chromosomal translocations; t(12;21) involving <i>RUNX1</i> and <i>ETV6</i> present in 25%	Predominantly children; symptoms relating to marrow replacement and pancytopenia; aggressive
T-cell acute lymphoblastic leukemia/lymphoma	Precursor T cell (often of thymic origin)	Diverse chromosomal translocations; <i>NOTCH1</i> mutations (50%–70%)	Predominantly adolescent males; thymic masses and variable bone marrow involvement; aggressive
Neoplasms of Mature B Cells			
Burkitt lymphoma ^a	Germinal center B cell	Translocations involving <i>MYC</i> and immunoglobulin loci, usually t(8;14); subset EBV-associated	Adolescents or young adults with extranodal masses; uncommonly presents as “leukemia”; aggressive
Diffuse large B-cell lymphoma ^b	Germinal center or post-germinal center B cell	Diverse chromosomal rearrangements, most often of <i>BCL6</i> (30%), <i>BCL2</i> (10%), or <i>MYC</i> (5%)	All ages, but most common in older adults; often appears as a rapidly growing mass; 30% extranodal; aggressive
Extranodal marginal zone lymphoma	Memory B cell	t(11;18), t(1;14), and t(14;18) creating <i>MALT1::IAP2</i> , <i>BCL10::IGH</i> , and <i>MALT1::IGH</i> fusion genes, respectively	Arises at extranodal sites in adults with chronic inflammatory diseases; may remain localized; indolent
Follicular lymphoma ^b	Germinal center B cell	t(14;18) creating <i>BCL2::IGH</i> fusion gene	Older adults with generalized lymphadenopathy and marrow involvement; indolent
Hairy cell leukemia	Memory B cell	Activating <i>BRAF</i> mutations	Older males with pancytopenia and splenomegaly; indolent
Mantle cell lymphoma	Naive B cell	t(11;14) creating cyclin D1:: <i>IGH</i> fusion gene	Older males with disseminated disease; moderately aggressive
Multiple myeloma/solitary plasmacytoma ^b	Post-germinal center bone marrow homing plasma cell	Diverse rearrangements involving <i>IGH</i> ; 13q deletions	Myeloma: older adults with lytic bone lesions, pathologic fractures, hypercalcemia, and renal failure; moderately aggressive Plasmacytoma: isolated plasma cell masses in bone or soft tissue; indolent
Chronic lymphocytic leukemia/small lymphocytic lymphoma	Naive B cell or memory B cell	Trisomy 12, deletions of 11q, 13q, and 17p; <i>NOTCH1</i> mutations; splicing factor mutations	Older adults with bone marrow, lymph node, spleen, and liver disease; autoimmune hemolysis and thrombocytopenia in a minority; indolent
Neoplasms of Mature T Cells or NK Cells			
Adult T-cell leukemia/lymphoma	Helper T cell	HTLV-1 provirus present in tumor cells	Adults with cutaneous lesions, marrow involvement, and hypercalcemia; occurs mainly in Japan, West Africa, and the Caribbean; aggressive
Peripheral T-cell lymphoma, not otherwise specified	Helper or cytotoxic T cell	No specific chromosomal abnormality	Mainly older adults; usually presents with lymphadenopathy; aggressive
Nodal T-follicular helper cell lymphoma	T-follicular helper cell	Frequent mutations in genes that are commonly mutated in myeloid neoplasms (<i>TET2</i> , <i>DNMT3A</i> , and <i>IDH2</i>)	Mainly older adults; usually presents with lymphadenopathy, often associated with systemic symptoms and immune abnormalities (e.g., auto-antibodies); aggressive
Anaplastic large-cell lymphoma	Cytotoxic T cell	Rearrangements of <i>ALK</i> (anaplastic large-cell lymphoma kinase) in a subset	Children and young adults, usually with lymph node and soft tissue disease; aggressive
Extranodal NK/T-cell lymphoma	NK-cell (common) or cytotoxic T cell (rare)	EBV-associated; no specific chromosomal abnormality	Adults with destructive extranodal masses, most commonly sinonasal; aggressive
Mycosis fungoides	Helper T cell	No specific chromosomal abnormality	Adult patients with cutaneous patches, plaques, nodules, or generalized erythema; indolent
Large granular lymphocytic leukemia	Two types: cytotoxic T cell and NK cell	Point mutations in <i>STAT3</i>	Adult patients with splenomegaly, neutropenia, and anemia, sometimes accompanied by autoimmune disease

^aMost common tumors in children.^bMost common tumors in adults.

EBV, Epstein-Barr virus; HTLV-1, human T-cell leukemia virus-1; Ig, immunoglobulin; NK, natural killer.

MORPHOLOGY

In leukemic presentations, **the marrow is hypercellular and packed with lymphoblasts**, which replace normal marrow elements. Mediastinal thymic masses occur in 50% to 70% of T-ALL, which is also more likely to be associated with lymphadenopathy and splenomegaly than B-ALL. In both B-ALL and T-ALL, the tumor cells have scant basophilic cytoplasm and nuclei somewhat larger than those of small lymphocytes (Fig. 13.6A). The nuclear chromatin is delicate and finely stippled, and nucleoli are usually small and sharply demarcated by a rim of condensed chromatin. The nuclear membrane is often deeply subdivided, imparting a convoluted appearance. In keeping with aggressive clinical behavior, the mitotic rate is high. As with other rapidly growing lymphoid tumors, interspersed macrophages ingesting apoptotic tumor cells may impart a “**starry sky**” appearance (shown in Fig. 13.15).

Because of their different responses to chemotherapy, ALL must be distinguished from acute myeloid leukemia, a neoplasm of immature myeloid cells that can cause identical signs and symptoms. **Compared with myeloblasts, lymphoblasts have more condensed chromatin, less conspicuous nucleoli, and smaller amounts of cytoplasm that usually lacks granules.** However, these morphologic distinctions are not absolute and definitive diagnosis relies on stains performed with antibodies specific for B- and T-cell antigens (Fig. 13.6B–C). Histochemical stains are also helpful, in that (in contrast to myeloblasts) lymphoblasts are myeloperoxidase-negative and often contain periodic acid–Schiff–positive cytoplasmic material.

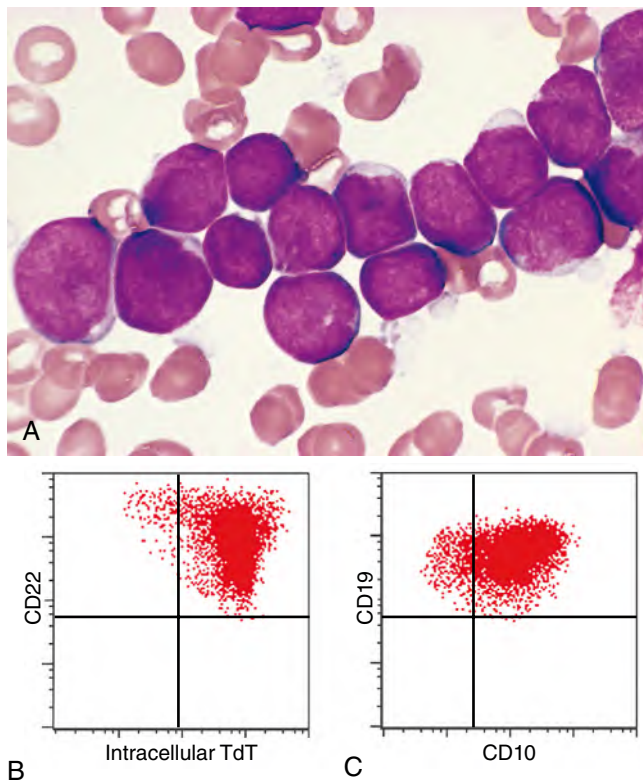


Fig. 13.6 (A) Acute lymphoblastic leukemia/lymphoma (ALL). Lymphoblasts with condensed nuclear chromatin, small nucleoli, and scant agranular cytoplasm. (B–C) Phenotype of ALL shown in (A) analyzed by flow cytometry. (B) The lymphoblasts represented by the red dots express terminal deoxynucleotidyl transferase (TdT) and the B-cell marker CD22. (C) The same cells are positive for two other markers, CD10 and CD19, commonly expressed on pre-B lymphoblasts. Thus, this is a B-ALL. (A, Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas. B–C, Courtesy Dr. Louis Picker, Oregon Health Science Center, Portland.)

Immunophenotype. Immunostaining for terminal deoxynucleotidyl transferase (TdT), a specialized DNA polymerase that is expressed only in pre-B and pre-T lymphoblasts, is positive in more than 95% of cases (Fig. 13.6B). B-ALLs are arrested at various stages of pre-B-cell development, which correlate with the expression of certain proteins. The lymphoblasts usually express the pan B-cell marker CD19 and the transcription factor PAX5 as well as CD10. In very immature B-ALLs, CD10 is negative. Alternatively, more mature “late pre-B” ALLs express CD10, CD19, CD20, and cytoplasmic IgM heavy chain (μ chain). Similarly, T-ALLs are arrested at various stages of pre-T-cell development. In most cases the cells are positive for CD1, CD2, CD5, and CD7. The more immature tumors are usually negative for surface CD3, CD4, and CD8, whereas “late” pre-T-cell tumors are positive for these markers.

Clinical Features

Although ALL and acute myeloid leukemia are genetically and immunophenotypically distinct, they are clinically very similar. In both, the accumulation of neoplastic “blasts” in the bone marrow suppresses normal hematopoiesis by physical crowding, competition for growth factors, and other poorly understood mechanisms. The common features and those characteristic of ALL are:

- **Abrupt stormy onset** within days to a few weeks of the first symptoms.
- **Symptoms related to depression of marrow function**, including fatigue due to anemia; fever, reflecting infections secondary to neutropenia; and bleeding due to thrombocytopenia.
- **Mass effects caused by neoplastic infiltration** (which are more common in ALL), including bone pain resulting from marrow expansion and infiltration of the subperiosteum; generalized lymphadenopathy, splenomegaly, and hepatomegaly; testicular enlargement; and, in T-ALL, complications related to compression of large vessels and airways in the mediastinum.
- **Central nervous system manifestations** such as headache, vomiting, and nerve palsies resulting from meningeal spread, all of which are also more common in ALL.

Treatment of pediatric ALL is one of the great success stories of oncology. With aggressive chemotherapy about 95% of children with ALL obtain a complete remission, and 75% to 85% are cured. Despite these achievements, however, ALL remains a leading cause of cancer deaths in children and less than half of adults are cured. Several factors are associated with a worse prognosis:

- Age younger than 2 years, largely because of the strong association of infantile ALL with translocations involving the *KMT2A* gene and CNS involvement
- Presentation in adolescence or adulthood
- Peripheral blood blast counts greater than 100,000, which probably reflects a high tumor burden

Favorable prognostic markers include:

- Age between 2 and 10 years
- A low white cell count
- Hyperdiploidy
- Trisomy of chromosomes 4, 7, and 10
- The presence of a t(12;21) translocation, which generates the *ETV6::RUNX1* fusion gene

Notably, the molecular detection of residual disease after therapy is predictive of a worse outcome in both B-ALL and T-ALL and is being used to guide treatment.

Although most chromosomal aberrations in ALL alter the function of transcription factors, the t(9;22) instead creates a fusion gene that encodes a constitutively active BCR-ABL tyrosine kinase (described in more detail under chronic myeloid leukemia). In B-ALL, the

BCR-ABL protein is usually 190 kDa in size and has stronger tyrosine kinase activity than the form of BCR-ABL that is found in chronic myeloid leukemia, in which a BCR-ABL protein of 210 kDa in size is usually seen. Treatment of t(9;22)-positive ALLs with BCR-ABL kinase inhibitors in combination with conventional chemotherapy is highly effective and has greatly improved the outcome for this molecular subtype of B-ALL in children and adults. Of interest, cryptic rearrangements involving genes encoding tyrosine kinases other than ABL also occur in B-ALL, particularly in adults, and these too may be therapeutic targets. The outlook for adults with ALL lacking “targetable” molecular lesions remains more guarded, in part because of differences in the molecular pathogenesis of adult and childhood ALL and because older adults cannot tolerate the intensive chemotherapy regimens that are curative in children. Dramatic responses in B-ALL have been achieved with chimeric antigen receptor T cells directed against the B-cell antigen CD19, but at high financial cost and with associated toxicities that have sometimes proved fatal.

KEY CONCEPTS

Acute Lymphoblastic Leukemia/Lymphoma

- Most common type of cancer in children; may be derived from either precursor B or T cells. B-ALL represents 75% to 80% of ALL cases.
- Highly aggressive tumors manifest with signs and symptoms of bone marrow failure or as rapidly growing masses.
- Tumor cells contain genetic lesions that block differentiation, leading to the accumulation of immature, nonfunctional blasts.
- A subset of tumors contains activating mutations in tyrosine kinases (e.g., BCR-ABL) that are important targets of therapy.

Peripheral B-Cell Neoplasms

Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

Chronic lymphocytic leukemia (CLL) and small lymphocytic lymphoma (SLL) differ only in the degree of peripheral blood lymphocytosis. Most affected patients have sufficient lymphocytosis to fulfill the diagnostic requirement for CLL (absolute lymphocyte count $>5000/\text{mm}^3$). **CLL is the most common leukemia of adults in the Western world.** There are about 15,000 new cases of CLL each year in the United States. The median age at diagnosis is 60 years, and there is a 2:1 male predominance. In contrast, SLL constitutes only 4% of NHLs. CLL/SLL is less common in Japan and other Asian countries than in the West.

Pathogenesis

Unlike most other lymphoid malignancies, chromosomal translocations are rare in CLL/SLL. The most common genetic anomalies are deletions of 13q14.3, 11q, and 17p and trisomy 12q. Molecular characterization of the region deleted on chromosome 13 has implicated two microRNAs, miR-15a and miR-16-1, as tumor suppressor genes. Loss of these miRs is believed to result in overexpression of the anti-apoptotic protein BCL2, which is uniformly observed in CLL/SLL. DNA sequencing has revealed that the immunoglobulin genes of some CLL/SLL are somatically hypermutated, whereas others are not, suggesting that the cell of origin may be either a post-germinal center memory B cell or a naive B cell. For unclear reasons, tumors with unmutated immunoglobulin segments (those putatively of naive B-cell origin) pursue a more aggressive course. Deep sequencing of CLL genomes has also revealed gain-of-function mutations involving the NOTCH1 receptor in 10% to 18% of tumors that confer a less favorable prognosis, as well as frequent mutations in genes that regulate RNA splicing.

The growth of CLL/SLL cells is largely confined to proliferation centers (described later), where tumor cells receive critical cues from

the microenvironment. Stromal cells in proliferation centers express a variety of factors that stimulate the activity of the transcription factors NF- κ B, which promotes cell survival, and MYC, which promotes cell growth. Other critical signals are generated by the B-cell receptor (membrane bound immunoglobulin), which triggers a signaling pathway that includes Bruton tyrosine kinase (BTK). The importance of BTK in B cells is emphasized by X-linked agammaglobulinemia (Chapter 6), a failure of B-cell development that is caused by loss-of-function mutations in the *BTK* gene. Notably, BTK inhibitors produce sustained therapeutic responses in a high fraction of CLL patients, indicating that CLL cells depend on B-cell receptor signaling and BTK activity for their growth and survival.

MORPHOLOGY

Lymph nodes are diffusely effaced by predominantly small lymphocytes 6 to 12 μm in diameter with round to slightly irregular nuclei, condensed chromatin, and scant cytoplasm (Fig. 13.7). Admixed are variable numbers of larger activated lymphocytes that often gather in loose aggregates referred to as **proliferation centers** that contain mitotically active cells. When present, **proliferation centers are pathognomonic for CLL/SLL**. The blood contains variable numbers of small round lymphocytes with scant cytoplasm (Fig. 13.8). Some of these cells are disrupted in the process of making smears, producing so-called **smudge cells**. In almost all cases, the bone marrow, spleen, and liver (Fig. 13.9) also are involved, albeit to widely varying degrees.

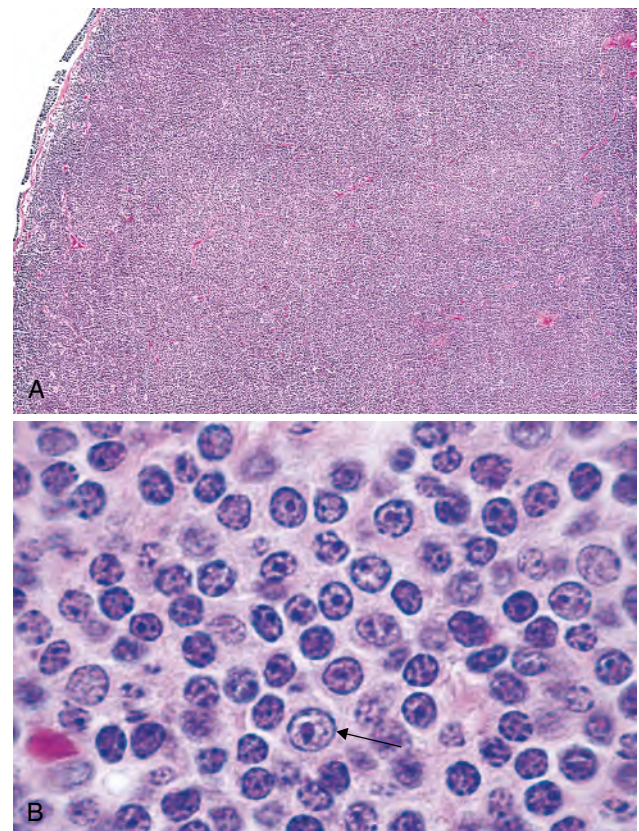


Fig. 13.7 Small lymphocytic lymphoma/chronic lymphocytic leukemia (lymph node). (A) Low-power view shows diffuse effacement of nodal architecture. (B) At high power most of the tumor cells are small round lymphocytes. A prolymphocyte, a larger cell with a centrally placed nucleolus, is also present in this field (arrow). (A, Courtesy Dr. José Hernandez, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

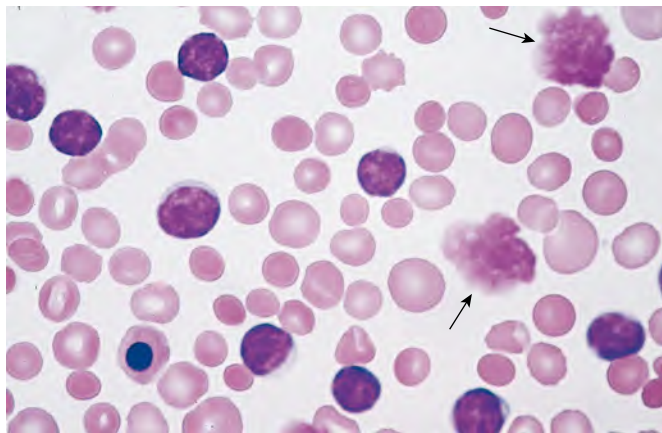


Fig. 13.8 Chronic lymphocytic leukemia. This peripheral blood smear is flooded with small lymphocytes with condensed chromatin and scant cytoplasm. A characteristic finding is the presence of disrupted tumor cells (smudge cells, arrows), two of which are present in this smear. A coexistent autoimmune hemolytic anemia (Chapter 14) explains the presence of spherocytes (hyperchromatic, round red cells). A nucleated erythroid cell is present in the lower left corner of the field. In this setting, circulating nucleated red cells could stem from premature release of progenitors in the face of severe anemia, marrow infiltration by tumor (leukoerythroblastosis), or both.

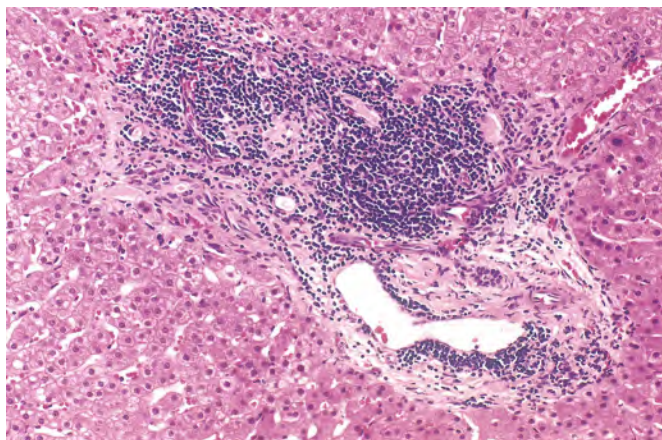


Fig. 13.9 Small lymphocytic lymphoma/chronic lymphocytic leukemia involving the liver. Low-power view of a typical periportal lymphocytic infiltrate. (Courtesy Dr. Mark Fleming, Department of Pathology, Children's Hospital, Boston.)

Immunophenotype. CLL/SLL has a distinctive immunophenotype. The tumor cells express the pan B-cell markers CD19 and CD20, as well as CD23 and CD5, the latter a marker that is found on T cells and a small subset of normal B cells. Low-level expression of surface immunoglobulin (usually IgM, or IgM and IgD) is also typical, as is high-level expression of BCL2.

Clinical Features

Patients are often asymptomatic at diagnosis. When symptoms appear, they are nonspecific and include easy fatigability, weight loss, and anorexia. Generalized lymphadenopathy and hepatosplenomegaly are present in 50% to 60% of symptomatic patients. The leukocyte count is highly variable; leukopenia can be seen in individuals with SLL and marrow involvement, while counts in excess of 200,000/mm³ are sometimes seen in CLL patients with heavy tumor burdens. A small

monoclonal immunoglobulin “spike” is present in the blood of some patients. At the other end of the spectrum are asymptomatic patients with monoclonal B cells in their peripheral blood that number too few to merit the diagnosis of CLL. This condition, referred to as monoclonal lymphocytosis of uncertain significance, is considered a precursor lesion and has about a 1% risk per year of progressing to CLL.

CLL/SLL disrupts normal immune function through uncertain mechanisms. Hypogammaglobulinemia is common and contributes to an increased susceptibility to infections, particularly those caused by bacteria. Conversely, 10% to 15% of patients develop hemolytic anemia or thrombocytopenia due to autoantibodies made by nonneoplastic B cells.

The course and prognosis are extremely variable and depend primarily on the clinical stage. Overall median survival is approximately 10 years, but some patients rapidly deteriorate and die in 1 to 2 years, while others live for several decades with clinically benign disease. In addition to high clinical stage, other variables that correlate with a worse outcome include (1) the presence of deletions of 11q and 17p (the latter involving *TP53*), (2) a lack of somatic hypermutation, (3) the expression of ZAP-70, a protein that augments signals produced by the immunoglobulin receptor, and (4) the presence of *NOTCH1* mutations. Treatment choices are evolving and include BCL2 inhibitors, anti-CD20 antibodies, and BTK inhibitors.

Another factor that impacts patient survival is the tendency of CLL/SLL to transform to a more aggressive tumor. Most commonly this takes the form of a transformation to diffuse large B-cell lymphoma (DLBCL), so-called *Richter syndrome* (approximately 5% to 10% of patients). Richter syndrome is frequently heralded by the development of a rapidly enlarging mass within a lymph node or the spleen and is often associated with acquisition of new abnormalities involving *TP53* or *MYC*. It is an ominous event, with most patients surviving less than 2 years.

Follicular Lymphoma

Follicular lymphoma is the most common form of indolent NHL in the United States, trailing only DLBCL (discussed later) in frequency among lymphomas. It affects 15,000 to 20,000 individuals each year in the United States. The incidence of follicular lymphoma is lower in Europe than the United States, and it is uncommon in Asian populations. It usually present in middle age and affects males and females equally.

Pathogenesis

Follicular lymphoma is strongly associated with chromosomal translocations involving BCL2. Its hallmark is a (14;18) translocation that juxtaposes the *IGH* locus on chromosome 14 and the *BCL2* locus on chromosome 18. The t(14;18) is seen in up to 90% of follicular lymphomas and leads to overexpression of BCL2 (see Fig. 13.12). BCL2 antagonizes apoptosis (Chapters 2 and 7) and promotes the survival of follicular lymphoma cells. Notably, while normal germinal centers contain numerous B cells undergoing apoptosis, follicular lymphoma is characteristically devoid of apoptotic cells. Deep sequencing of follicular lymphoma genomes has identified mutations in the *KMT2D* gene in about 90% of cases and in the *EZH2* gene in about 25% of cases. Notably, *KMT2D* and *EZH2* encode different histone-modifying enzymes, suggesting that epigenetic abnormalities such as changes in the patterns of histone marks have an important role in this neoplasm.

Particularly early in the disease, follicular lymphoma cells growing in lymph nodes are found within a network of reactive follicular dendritic cells admixed with macrophages and T cells. Expression profiling studies have shown that differences in the genes expressed by these stromal cells are predictive of outcome, implying that the response of follicular lymphoma cells to therapy is influenced by the surrounding microenvironment.

MORPHOLOGY

In most cases, a nodular or nodular and diffuse growth pattern is observed in involved lymph nodes (Fig. 13.10A). Two principal cell types are present in varying proportions: (1) small cells with irregular or cleaved nuclear contours and scant cytoplasm, referred to as **centrocytes** (small, cleaved cells), and (2) larger cells with open nuclear chromatin, several nucleoli, and modest amounts of cytoplasm, referred to as **centroblasts** (Fig. 13.10B). In most follicular lymphomas, centrocytes are in the majority. Peripheral blood involvement sufficient to produce lymphocytosis (usually $<20,000$ cells/ μL) is seen in about 10% of cases. Bone marrow involvement occurs in 85% of cases and characteristically takes the form of paratrabecular lymphoid aggregates. The splenic white pulp (Fig. 13.11) and hepatic portal triads are also frequently involved.

Immunophenotype. The neoplastic cells closely resemble normal germinal center B cells, expressing CD19, CD20, CD10, surface immunoglobulin, and BCL6. Unlike CLL/SLL and mantle cell lymphoma, CD5 is not expressed. BCL2 is expressed in more than 90% of cases, in distinction to normal follicular center B cells, which are BCL2-negative (Fig. 13.12).

Clinical Features

Follicular lymphoma tends to present with painless, generalized lymphadenopathy. Involvement of extranodal sites such as the gastrointestinal

tract, central nervous system, or testis is relatively uncommon. Although incurable, it usually follows an indolent waxing and waning course. Survival (median, 7 to 9 years) is not improved by aggressive therapy; hence the usual approach is to palliate patients with immunotherapy (e.g., anti-CD20 antibody) combined with “gentle” chemotherapy when they become symptomatic. Of interest, EZH2 inhibitors appear to be highly active in treating the subset of follicular lymphomas with *EZH2* gain-of-function mutations.

Histologic transformation occurs in 30% to 50% of follicular lymphomas, most commonly to DLBCL. Follicular lymphomas show evidence of ongoing somatic hypermutation, which may promote transformation by causing point mutations or chromosomal aberrations. Transformed tumors often acquire new aberrations that increase the expression of *MYC*, which you will recall drives Warburg metabolism and rapid cell growth. Overall outcomes for patients with transformed tumors is varied;

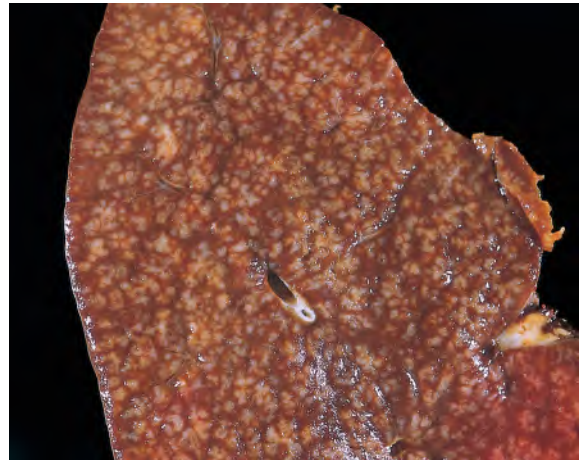


Fig. 13.11 Follicular lymphoma (spleen). Prominent nodules represent white pulp follicles expanded by follicular lymphoma cells. Other indolent B-cell lymphomas (small lymphocytic lymphoma, mantle cell lymphoma, marginal zone lymphoma) can produce an identical pattern of involvement. (Courtesy Dr. Jeffrey Jorgenson, Department of Hematopathology, MD Anderson Cancer Center, Houston.)

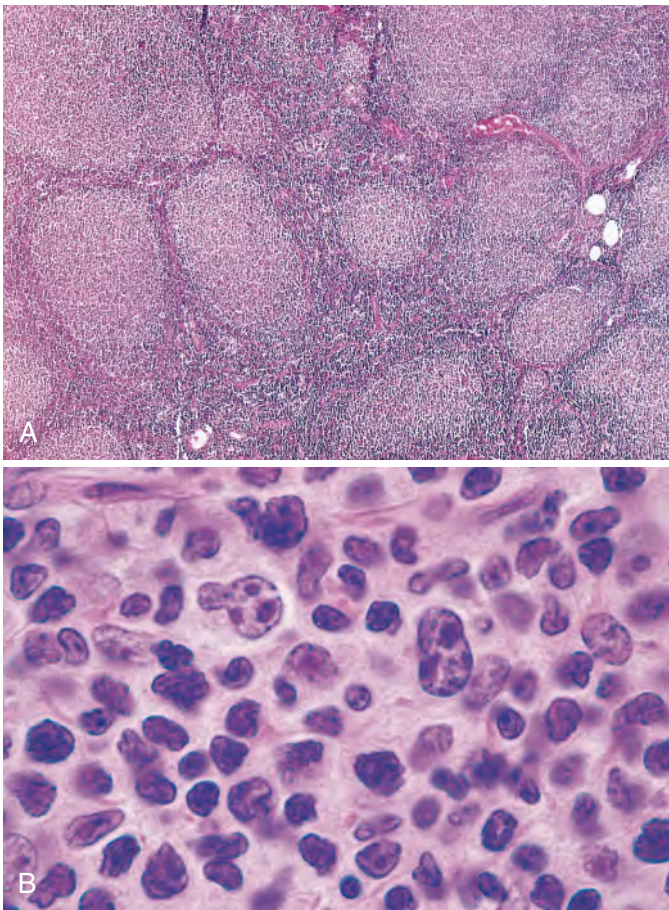


Fig. 13.10 Follicular lymphoma (lymph node). (A) Nodular aggregates of lymphoma cells are present throughout lymph node. (B) At high magnification, small lymphoid cells with condensed chromatin and irregular or cleaved nuclear outlines (centrocytes) are mixed with a population of larger cells with nucleoli (centroblasts). (A, Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

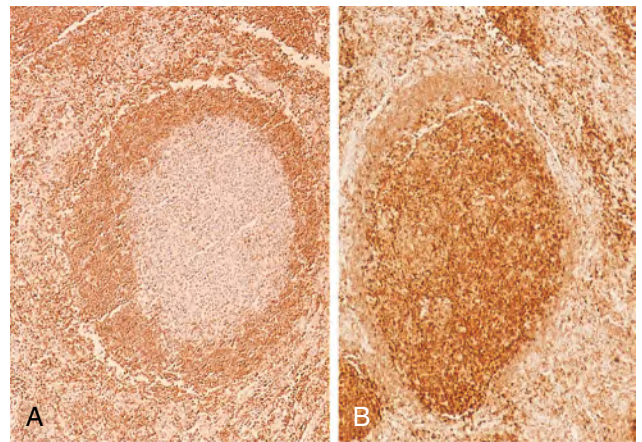


Fig. 13.12 BCL2 expression in reactive and neoplastic follicles. BCL2 protein was detected by using an immunohistochemical technique that produces a brown stain. In reactive follicles (A), BCL2 is present in mantle zone cells but not follicular center B cells, whereas follicular lymphoma cells (B) show strong BCL2 staining. (Courtesy Dr. Jeffrey Jorgenson, Department of Hematopathology, MD Anderson Cancer Center, Houston.)

the median survival is 3 to 5 years but is particularly poor for patients with tumors that acquire *MYC* translocations.

Diffuse Large B-Cell Lymphoma

DLBCL is the most common form of NHL. Each year in the United States there are about 25,000 new cases. There is a slight male predominance. The median patient age is about 60 years, but DLBCL also occurs in young adults and children.

Pathogenesis

Genetic, gene expression profiling, and immunohistochemical studies indicate that DLBCL is molecularly heterogeneous. One frequent pathogenic event is dysregulation of *BCL6*, a DNA-binding zinc-finger transcriptional repressor that is required for the formation of normal germinal centers. About 30% of DLBCLs contain various translocations that have in common a breakpoint in *BCL6* at chromosome 3q27. Acquired mutations in *BCL6* promoter sequences that abrogate *BCL6* autoregulation (an important negative-regulatory mechanism) are seen even more frequently. It is hypothesized that both types of lesions are inadvertent byproducts of somatic hypermutation that result in overexpression of *BCL6*, which has several important consequences. *BCL6* represses the expression of factors that normally serve to promote germinal center B-cell differentiation, growth arrest, and apoptosis, and each of these effects is believed to contribute to the development of DLBCL. Mutations similar to those found in *BCL6* are also seen in multiple other oncogenes, including *MYC*, suggesting that somatic hypermutation in DLBCL cells is “mistargeted” to a wide variety of loci.

Another 10% to 20% of tumors are associated with the t(14;18) (discussed earlier under Follicular Lymphoma), which leads to the overexpression of the anti-apoptotic protein *BCL2*. Tumors with *BCL2* rearrangements usually lack *BCL6* rearrangements, suggesting that these rearrangements define two distinct molecular classes of DLBCL. Some tumors with *BCL2* rearrangements may arise from unrecognized underlying follicular lymphomas, which frequently transform to DLBCL. Roughly 5% of DLBCLs are associated with translocations involving *MYC*; these tumors may have a distinctive biology (discussed later under Burkitt lymphoma). Finally, based on sequencing of DLBCL genomes, it has been proposed that DLBCL can be subdivided into four distinct genetic subtypes defined by different combinations of acquired aberrations in the genes encoding *BCL6*; the B-cell signaling CD79B and MYD88; the histone methylase EZH2; and the NOTCH1 and NOTCH2 receptors.

MORPHOLOGY

The common features are a relatively **large cell size** (usually four to five times the diameter of a small lymphocyte) and a **diffuse pattern of growth** (Fig. 13.13). Other morphologic features show substantial variation. Most commonly, the tumor cells have a round or oval nucleus that appears vesicular due to margination of chromatin to the nuclear membrane, but large multilobated or irregular nuclei are prominent in some cases. Nucleoli may be two to three in number and located adjacent to the nuclear membrane or single and centrally placed. The cytoplasm is usually moderately abundant and may be pale or basophilic. More anaplastic tumors may contain multinucleated cells with large inclusion-like nucleoli that resemble Reed-Sternberg cells (the malignant cell of Hodgkin lymphoma).

Immunophenotype. These mature B-cell tumors express CD19 and CD20 and show variable expression of germinal center B-cell markers such as CD10 and *BCL6*. Most have surface immunoglobulin. High-level expression of both *MYC* and *BCL2* proteins is seen in some cases.

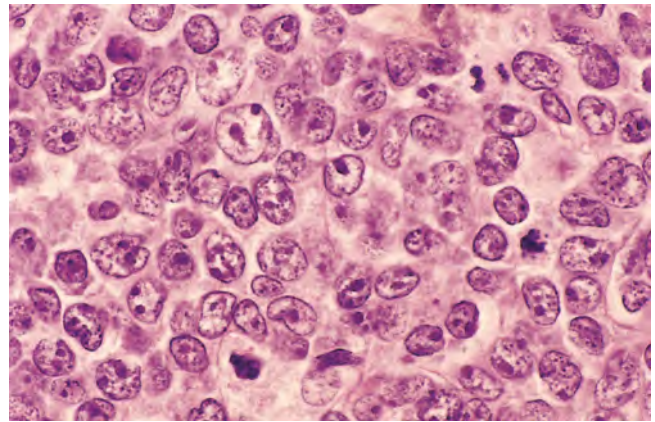


Fig. 13.13 Diffuse large B-cell lymphoma. Tumor cells have large nuclei, open chromatin, and prominent nucleoli. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Special Subtypes. Several subtypes of DLBCL are sufficiently distinctive to merit brief discussion.

- **Immunodeficiency-associated large B-cell lymphoma** occurs in the setting of T-cell immunodeficiency (e.g., in advanced HIV infection and in recipients of organ or HSC transplants). The neoplastic B cells are usually infected with EBV, which plays a critical pathogenic role. Restoration of T-cell immunity may lead to regression of these proliferations.
- **Primary effusion lymphoma** presents as a malignant pleural or ascitic effusion, mostly in patients with advanced HIV infection or in older adults. The tumor cells are often anaplastic in appearance and typically fail to express surface B- or T-cell markers but have clonal *IGH* gene rearrangements. In all cases the tumor cells are infected with HHV8, which appears to have a causal role.
- **Primary mediastinal large B-cell lymphoma** virtually always presents as a mediastinal mass, sometimes of sufficient size to cause compression of the superior vena cava and swelling of the face and neck (*superior vena cava syndrome*). The tumor cells often have amplification of the gene encoding the immune checkpoint molecule PD-L1, leading to PD-L1 overexpression, which is believed to have an important role in allowing these tumors to escape the host immune response.
- **Intravascular large B-cell lymphoma** is an unusual tumor that preferentially grows within small venules throughout the body rather than producing tumor masses, presumably because the tumor cells lack the ability to extravasate into tissues. Patients often present with nonlocalizing central nervous system signs and symptoms due to involvement of vessels in the brain. The diagnosis can be made through skin biopsy but is often not suspected in life, and many cases are only recognized at autopsy.
- **Diffuse large B-cell lymphoma with *MYC* and *BCL2* rearrangements** is a small subset of DLBCLs that tends to be exceptionally aggressive, exhibiting high rates of growth due to *MYC* overexpression and resistance to therapy due to the anti-apoptotic actions of *BCL2*. These tumors may arise *de novo* or through transformation of an underlying follicular lymphoma.

Clinical Features

DLBCL typically presents as a rapidly enlarging mass at a nodal or extranodal site. It can arise virtually anywhere in the body including the Waldeyer ring (i.e., the oropharyngeal lymphoid tissue comprised of tonsils and adenoids). Primary or secondary involvement of the liver



Fig. 13.14 Diffuse large B-cell lymphoma involving the spleen. The isolated large mass is typical. In contrast, indolent B-cell lymphomas usually produce multifocal expansion of white pulp (see Fig. 13.11). (Courtesy Dr. Mark Fleming, Department of Pathology, Children's Hospital, Boston.)

and spleen may take the form of large destructive masses (Fig. 13.14). Extranodal sites include the gastrointestinal tract, skin, bone, brain, and other tissues. Bone marrow involvement is relatively uncommon and usually occurs late in the course. Rarely, a leukemic picture emerges.

DLBCLs are aggressive tumors that are fatal in weeks to months without treatment. With intensive combination chemotherapy, 60% to 80% of patients achieve a complete remission, and 40% to 50% are cured. Adjuvant therapy with anti-CD20 antibody improves both the initial response and the overall outcome. Individuals with limited disease fare better than those with widespread disease or bulky tumor masses. DLBCLs with *MYC* translocations have a worse prognosis than those without and may be better treated with chemotherapy regimens that are now standard for Burkitt lymphoma. CAR-T cells directed against the B-cell antigen CD19 are now available for the treatment of patients with relapsed refractory DLBCL.

Burkitt Lymphoma

Within the category of Burkitt lymphoma fall (1) African (endemic) Burkitt lymphoma, (2) sporadic (nonendemic) Burkitt lymphoma, and (3) a subset of aggressive lymphomas occurring in individuals infected with HIV. Burkitt lymphomas occurring in these three settings are histologically identical but have distinct clinical, genotypic, and virologic characteristics.

Pathogenesis

All forms of Burkitt lymphoma are associated with translocations of the *MYC* gene on chromosome 8 that lead to increased *MYC* protein levels. *MYC* is a master transcriptional regulator that increases the expression of genes that are required for aerobic glycolysis, the so-called Warburg effect (Chapter 7). When nutrients such as glucose and glutamine are available, Warburg metabolism allows cells to biosynthesize all the building blocks—nucleotides, lipids, proteins—that are needed for growth and cell division. In line with the importance of *MYC* in regulating proliferation, Burkitt lymphoma is among the fastest-growing human tumors. The *MYC* translocation partner is usually the *IGH* locus [t(8;14)], but may also be the immunoglobulin κ [t(2;8)] or λ [t(8;22)] light chain locus. The breakpoints in the *IGH* locus in sporadic Burkitt lymphoma are usually found in the class switch regions, whereas the breakpoints in endemic Burkitt lymphoma tend to lie within more 5' V(D)J sequences. The basis for this subtle molecular distinction is not known, but both types of translocations can be induced in germinal center B cells by AID, a

specialized DNA-modifying enzyme required for both immunoglobulin class switching and somatic hypermutation (see earlier). The net effect of these translocations is similar; the *MYC* coding sequence is repositioned adjacent to strong immunoglobulin enhancer elements, which drive increased *MYC* expression. In addition, the translocated *MYC* allele often harbors point mutations that stabilize *MYC* protein and further increase its activity.

Sequencing of the genomes of Burkitt lymphoma cells has revealed that most tumors have mutations that increase the activity of the transcription factor TCF3 (also known as E2A), an important regulator of gene expression in germinal center B cells. It is believed that TCF3 drives the expression of a set of genes, including cyclin D, that collaborate with *MYC* to enable the very rapid growth that characterizes Burkitt lymphoma.

Essentially all endemic Burkitt lymphomas are latently infected with EBV, which also is present in about 25% of HIV-associated tumors and 15% to 20% of sporadic cases. The configuration of the EBV DNA is identical in all tumor cells within individual cases, indicating that infection precedes transformation. Although this places EBV at the “scene of the crime,” its precise role in the genesis of Burkitt lymphoma remains speculative (Chapter 7).

MORPHOLOGY

Involved tissues are effaced by a diffuse infiltrate of intermediate-sized lymphoid cells 10 to 25 μm in diameter with round or oval nuclei, coarse chromatin, several nucleoli, and a moderate amount of cytoplasm (Fig. 13.15). **The tumor exhibits a high mitotic index and contains numerous apoptotic cells, the nuclear remnants of which are phagocytosed by interspersed benign macrophages.** These phagocytes have abundant clear cytoplasm, creating a characteristic “starry sky” pattern. When the bone marrow is involved, aspirates reveal tumor cells with slightly clumped nuclear chromatin, two to five distinct nucleoli, and **dark blue cytoplasm containing clear cytoplasmic vacuoles.**

Immunophenotype. These are tumors of mature B cells that express surface IgM, CD19, CD20, CD10, and BCL6, a phenotype consistent with a germinal center B-cell origin. Unlike most other tumors of germinal center origin, Burkitt lymphoma almost always fails to express the anti-apoptotic protein BCL2, which may explain the high rate of apoptosis that is characteristic of this neoplasm.

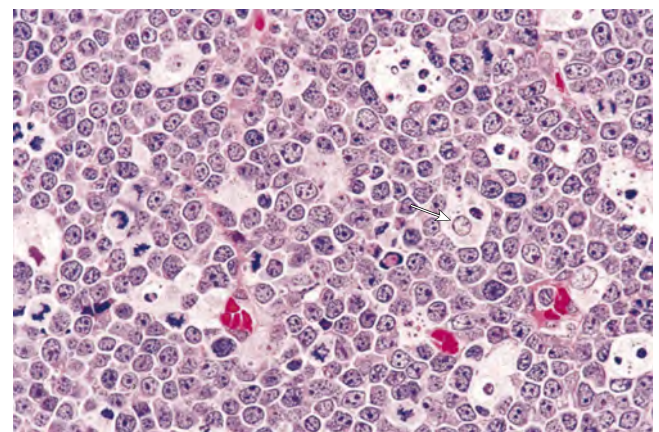


Fig. 13.15 Burkitt lymphoma. Numerous pale tingible body macrophages are evident (arrow), producing a “starry sky” appearance. The tumor cells have multiple small nucleoli and a high mitotic index. The lack of significant variation in nuclear shape and size lends a monotonous appearance. The red “material” is entrapped red cells.

Clinical Features

Both endemic and sporadic Burkitt lymphomas are found mainly in children or young adults; overall, Burkitt lymphoma accounts for about 30% of childhood NHLs in the United States. Most tumors manifest at extranodal sites. Endemic Burkitt lymphoma often presents as a mass involving the mandible and shows an unusual predilection for involvement of abdominal viscera, particularly the kidneys, ovaries, and adrenal glands. In contrast, sporadic Burkitt lymphoma most often appears as a mass involving the ileocecum and peritoneum. Involvement of the bone marrow and peripheral blood is uncommon, especially in endemic cases.

Burkitt lymphoma is very aggressive but responds well to intensive chemotherapy. Most children and young adults can be cured. The outcome is more guarded in older adults.

Mantle Cell Lymphoma

Mantle cell lymphoma is an uncommon lymphoid neoplasm that makes up about 2.5% of NHL in the United States and 7% to 9% of NHL in Europe. It usually presents in the fifth to sixth decades of life and shows a male predominance. As the name implies, the tumor cells closely resemble the normal mantle zone B cells that surround germinal centers.

Pathogenesis

Virtually all mantle cell lymphomas have an (11;14) translocation involving the *IGH* locus on chromosome 14 and the cyclin D1 locus on chromosome 11 that leads to overexpression of cyclin D1. The resulting up-regulation of cyclin D1 promotes G₁- to S-phase progression during the cell cycle, as was described in Chapter 7.

MORPHOLOGY

At diagnosis most patients have generalized lymphadenopathy, and 20% to 40% have peripheral blood involvement. Frequent sites of extranodal involvement include the bone marrow, spleen, liver, and gut. Occasionally, mucosal involvement of the small bowel or colon produces polyp-like lesions (**lymphomatoid polyposis**); of all forms of NHL, mantle cell lymphoma is most likely to spread in this fashion.

Nodal tumor cells may surround reactive germinal centers to produce a nodular appearance at low power or diffusely efface the node (Fig. 13.16A). Typically, the proliferation consists of a homogeneous population of **small lymphocytes with irregular to occasionally deeply clefted (cleaved) nuclear contours** (Fig. 13.16B). Large cells resembling centroblasts and proliferation centers are absent, distinguishing mantle cell lymphoma from follicular lymphoma and CLL/SLL, respectively. In most cases the nuclear chromatin is condensed, nucleoli are inconspicuous, and the cytoplasm is scant. Occasionally, tumors composed of intermediate-sized cells with more open chromatin and a brisk mitotic rate are observed; immunophenotyping is necessary to distinguish these “blastoid” variants from ALL.

Immunophenotype. Mantle cell lymphomas express high levels of cyclin D1, CD19, and CD20 and moderately high levels of surface immunoglobulin (usually IgM and IgD with κ or λ light chain). This tumor is usually CD5+ and CD23-, which helps to distinguish it from CLL/SLL. The *IGH* genes lack somatic hypermutation, supporting an origin from a naive B cell.

Clinical Features

The most common presentation, as is true of many lymphomas, is painless lymphadenopathy. Symptoms related to involvement of the spleen (present in approximately 50% of cases) and gut are also common. Mantle cell lymphoma is moderately aggressive and incurable;

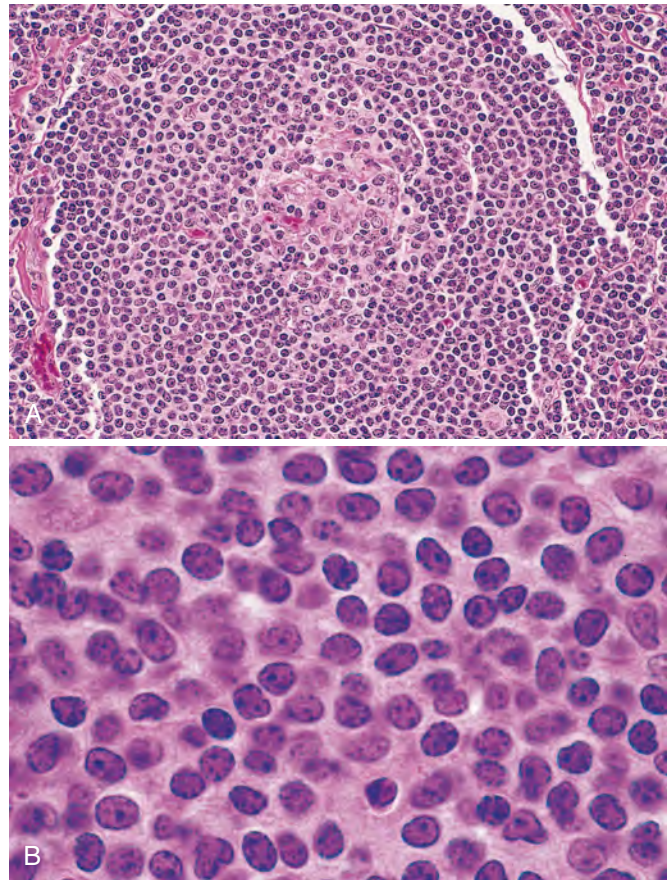


Fig. 13.16 Mantle cell lymphoma. (A) At low power, neoplastic lymphoid cells surround a small, atrophic germinal center, producing a mantle zone pattern of growth. (B) High-power view shows a homogeneous population of small lymphoid cells with somewhat irregular nuclear outlines, condensed chromatin, and scant cytoplasm. Large cells resembling prolymphocytes (seen in chronic lymphocytic leukemia) and centroblasts (seen in follicular lymphoma) are absent.

the median survival is 8 to 10 years. The blastoid variant, a high rate of proliferation, and *TP53* mutations are associated with shorter survival.

Marginal Zone Lymphomas

The category of marginal zone lymphomas encompasses a heterogeneous group of B-cell tumors that arise within lymph nodes, spleen, or extranodal tissues. The extranodal tumors were initially recognized at mucosal sites and are often referred to as mucosa-associated lymphoid tumors (or MALTomas). In most cases, the tumor cells show evidence of somatic hypermutation and are considered to be of memory B-cell origin.

Although all marginal zone lymphomas share certain features, those occurring at extranodal sites deserve special attention because of their unusual pathogenesis and three exceptional characteristics.

- They often arise within tissues involved by chronic inflammatory disorders of autoimmune or infectious etiology; examples include the salivary gland in Sjögren syndrome, the thyroid gland in Hashimoto thyroiditis, and the stomach in *Helicobacter* gastritis.
- They remain localized for prolonged periods, spreading systemically only late in their course.
- They may regress if the inciting agent (e.g., *H. pylori*) is eradicated.

These characteristics suggest that **extranodal marginal zone lymphomas arising in chronically inflamed tissues lie on a continuum**

between reactive lymphoid hyperplasia and full-blown lymphoma. The disease begins as a polyclonal immune reaction. With the acquisition of unknown initiating mutations, a B-cell clone emerges that still depends on antigen-stimulated T-helper cells for signals that drive growth and survival. At this stage, withdrawal of the responsible antigen causes tumor involution. A clinically relevant example is found in gastric MALToma, in which antibiotic therapy directed against *H. pylori* often leads to tumor regression (Chapter 17). With time, however, tumors may acquire additional mutations that render their growth and survival antigen-independent, such as (11;18), (14;18), or (1;14) chromosomal translocations, which are relatively specific for extranodal marginal zone lymphomas. All of these translocations upregulate the expression and function of BCL10 or MALT1, protein components of a signaling complex that activates NF- κ B and promotes the growth and survival of B cells. With further clonal evolution, spread to distant sites and transformation to DLBCL may occur. This theme of polyclonal-to-monoclonal transition during lymphomagenesis is also applicable to the pathogenesis of EBV-induced lymphoma and is discussed more fully in Chapter 7.

Hairy Cell Leukemia

This rare but distinctive B-cell neoplasm constitutes about 2% of all leukemias. It is predominantly a disease of middle-aged White males, with a median age of 55 years and a male-to-female ratio of 5:1.

Pathogenesis

Hairy cell leukemia is associated in greater than 90% of cases with activating point mutations in the serine/threonine kinase BRAF, which lies immediately downstream of RAS in the MAPK signaling cascade (Chapter 7). The specific mutation, a valine-to-glutamate substitution at residue 600, is also found at high frequencies in many other neoplasms, including melanoma and Langerhans cell histiocytosis (discussed later).

MORPHOLOGY

Hairy cell leukemia derives its picturesque name from the appearance of the leukemic cells, which have **fine hair-like projections** that are best recognized under the phase-contrast microscope (Fig. 13.17). On routine peripheral blood smears, hairy cells have round, oblong, or reniform nuclei and moderate amounts of pale blue cytoplasm with threadlike or bleb-like extensions. The number of circulating cells is highly variable. The marrow is involved by a diffuse interstitial infiltrate of cells with oblong or reniform nuclei, condensed chromatin, and pale cytoplasm. Because these cells are enmeshed in an extracellular matrix composed of reticulin fibrils, they usually are inspirable (a clinical difficulty referred to as a “**dry tap**”) and are seen only in marrow biopsies. The splenic red pulp is usually heavily infiltrated, leading to obliteration of white pulp and a beefy red gross appearance. Hepatic portal triads are also involved frequently.

Immunophenotype. Hairy cell leukemias typically express the pan-B-cell markers CD19 and CD20; surface immunoglobulin (usually IgG); and certain relatively distinctive markers, such as CD11c, CD25, CD103, and annexin A1.

Clinical Features

Clinical manifestations result largely from infiltration of the bone marrow, liver, and spleen. Splenomegaly, often massive, is the most common and sometimes the only abnormal physical finding. Hepatomegaly is less common and not as marked; lymphadenopathy is rare. Pancytopenia resulting from marrow involvement and sequestration of cells in the enlarged spleen is seen in more than one-half of cases. About one-third of those affected present with infections. There is an increased incidence

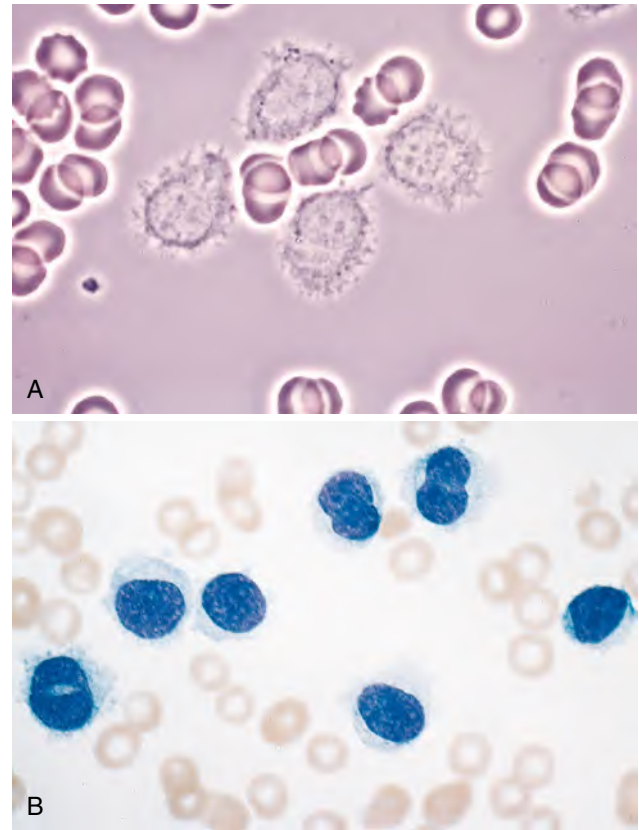


Fig. 13.17 Hairy cell leukemia (peripheral blood smear). (A) Phase-contrast microscopy shows tumor cells with fine hairlike cytoplasmic projections. (B) In stained smears, these cells have round or folded nuclei and modest amounts of pale blue, agranular cytoplasm.

of atypical mycobacterial infections, possibly related to monocytopenia of uncertain origin, which is commonly seen in this disease.

Hairy cell leukemia follows an indolent course. For unclear reasons, it is exceptionally sensitive to “gentle” chemotherapeutic regimens, which produce long-lasting remissions. Tumors often relapse after 5 or more years, yet generally respond well when retreated with the same agents, a feature that is highly unusual among human cancers. BRAF inhibitors appear to produce excellent responses in tumors that have not responded to conventional chemotherapy. The overall prognosis is excellent.

KEY CONCEPTS

Common Forms of Lymphoid Leukemia and Lymphoma Small Lymphocytic Lymphoma/Chronic Lymphocytic Leukemia

- Most common leukemia of adults.
- Tumor of mature B cells that usually manifests with bone marrow and lymph node involvement.
- Indolent course, commonly associated with immune abnormalities, including an increased susceptibility to infection and autoimmune disorders.

Follicular Lymphoma

- Most common indolent lymphoma of adults.
- Tumor cells recapitulate the growth pattern of normal germinal center B cells; most cases are associated with a (14;18) translocation that results in overexpression of BCL2.

Diffuse Large B-Cell Lymphoma

- Most common lymphoma of adults.
- Heterogeneous group of mature B-cell tumors that shares a large cell morphology and aggressive clinical behavior.
- Rearrangements or mutations of *BCL6* gene are recognized associations; one-third carry a (14;18) translocation involving *BCL2* and may arise from follicular lymphomas; about 5% have translocations involving *MYC*.

Burkitt Lymphoma

- Very aggressive tumor of mature B cells that usually arises at extranodal sites.
- Strongly associated with translocations involving the *MYC* proto-oncogene.
- Tumor cells often are latently infected by EBV.

Mantle Cell Lymphoma

- Tumor of naive B cells that pursues a moderately aggressive course.
- Highly associated with translocations involving the cyclin D1 gene.

Marginal Zone Lymphoma

- Indolent tumors of antigen-primed B cells that arise at sites of chronic immune stimulation.
- Often remain localized for long periods of time.

Hairy Cell Leukemia

- Morphologically distinct, very indolent tumor of mature B cells that involves spleen and marrow.
- Highly associated with mutations in the BRAF serine/threonine kinase.

Peripheral T- and NK-Cell Neoplasms

These categories include a heterogeneous group of neoplasms having phenotypes resembling mature T cells or NK cells. Peripheral T-cell tumors make up about 5% to 10% of NHLs in the United States and Europe, while NK-cell tumors are rare. By contrast, for unknown reasons both T- and NK-cell tumors occur more frequently in East Asia. Only the most common entities and those of particular pathogenetic interest will be discussed.

Peripheral T-Cell Lymphoma, Not Otherwise Specified

Although the WHO classification includes several distinct peripheral T-cell neoplasms, many T-cell lymphomas are not easily categorized and are lumped into a “wastebasket” diagnosis, *peripheral T-cell lymphoma, not otherwise specified* (PTCL-NOS). As might be expected, no morphologic feature is pathognomonic, but certain findings are characteristic. These tumors efface lymph nodes diffusely and are typically composed of a pleomorphic mixture of variably sized malignant T cells (Fig. 13.18). There is often a prominent infiltrate of reactive cells, such as eosinophils and macrophages, probably attracted by tumor-derived cytokines. Brisk neoangiogenesis may also be seen.

By definition, peripheral T-cell lymphomas are derived from mature T cells. They usually express CD2, CD3, CD5, and either $\alpha\beta$ or $\gamma\delta$ T-cell receptors. Some also express CD4 or CD8; such tumors are taken to be of helper or cytotoxic T-cell origin, respectively. However, many tumors have phenotypes that do not resemble any known normal T cell. In difficult cases where the differential diagnosis lies between lymphoma and a florid reactive process, DNA analysis is used to confirm the presence of clonal T-cell receptor gene rearrangements.

Most patients present with generalized lymphadenopathy, sometimes accompanied by eosinophilia, pruritus, fever, and weight loss. Although cures of peripheral T-cell lymphoma have been reported, these tumors as a group have a significantly worse prognosis than comparably aggressive mature B-cell neoplasms (e.g., DLBCL).

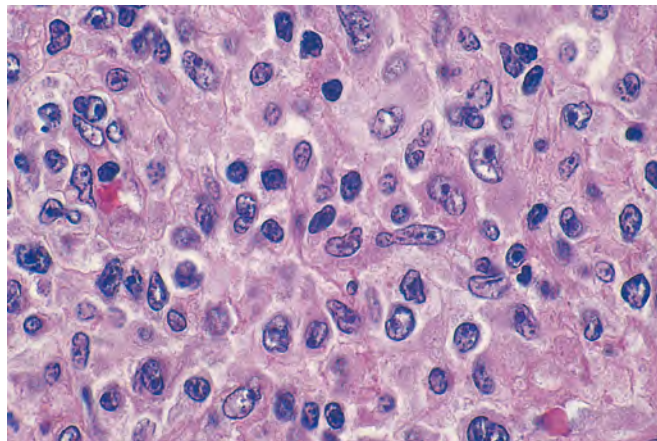


Fig. 13.18 Peripheral T-cell lymphoma, not otherwise specified (lymph node). A spectrum of small, intermediate, and large lymphoid cells, many with irregular nuclear contours, is visible.

Nodal T-Follicular Helper Cell Lymphoma

This relatively common subset of peripheral T-cell lymphoma composed of transformed T-follicular helper (Tfh) cells is associated with immune dysregulation, underlying clonal hematopoiesis, and EBV-driven B-cell proliferations. As you will recall, Tfh cells home to germinal centers and support the growth and maturation of germinal center B cells, the source of B cells and plasma cells making high-affinity antibodies. Tumors of Tfh cells somehow derange normal germinal center B-cell maturation, resulting in hypergammaglobulinemia and increased risk for development of autoantibodies and autoimmune cytopenias.

Several other associations with nodal T-follicular helper cell lymphoma are notable:

- Many patients with nodal T-follicular helper cell lymphoma also have clonal hematopoiesis (discussed later). Two of the most commonly mutated genes in nodal T-follicular helper cell lymphoma, *TET2* and *DNMT3A*, are the two most commonly mutated genes in clonal hematopoiesis, and in patients with nodal T-follicular helper cell lymphoma, it is common for the lymphoma to have the same mutations that are found in differentiated myeloid cells (e.g., monocytes). Thus, it appears that nodal T-follicular helper cell lymphoma often originates from a mutated hematopoietic stem cell.
- Lymph nodes involved by nodal T-follicular helper cell lymphoma often contain increased numbers of EBV-positive B cells, which sometimes give rise to full-blown EBV-positive B cell lymphomas. Remarkably, in some instances these tumors also share the same *DNMT3A* and *TET2* mutations, indicating that they, too, originate from a mutated hematopoietic stem cell.

Patients with T-follicular helper cell lymphoma present with lymphadenopathy, often accompanied by symptoms related to immune abnormalities, infection, or release of inflammatory cytokines. Lymph nodes show a population of CD4+ T cells expressing markers of Tfh cells, such as PD-1, CXCL13, CXCR5, CD10, and BCL6, often present in a vaguely nodular or follicular pattern due to an associated expansion of follicular dendritic cells. Between nodules there is often an increase in vascularity and an expansion of EBV-positive immunoblasts, the presence of which may be explained by acquired defects in adaptive immunity.

Most affected patients are older adults. The outcome is variable, but responses to chemotherapy are usually transient and most patients succumb to their disease within 5 years of diagnosis.

Anaplastic Large-Cell Lymphoma (ALK Positive)

This uncommon entity is defined by the presence of rearrangements in the *ALK* gene on chromosome 2p23. These rearrangements break the *ALK* locus and lead to the formation of chimeric genes encoding ALK fusion proteins, constitutively active tyrosine kinases that trigger the RAS and JAK/STAT signaling pathways.

As the name implies, this tumor is typically composed of large anaplastic cells, some containing horseshoe-shaped nuclei and voluminous cytoplasm (so-called *hallmark cells*) (Fig. 13.19A). The tumor cells often cluster about venules and infiltrate lymphoid sinuses, sometimes mimicking the appearance of metastatic carcinoma. ALK is not expressed in normal lymphocytes; thus, the detection of ALK protein in tumor cells (Fig. 13.19B) is a reliable indicator of an *ALK* gene rearrangement.

T-cell lymphomas with *ALK* rearrangements tend to occur in children or young adults, frequently involve soft tissues, and carry a very good prognosis (unlike other aggressive peripheral T-cell neoplasms). The cure rate with chemotherapy is 75% to 80%. Inhibitors of ALK have been developed and often produce excellent responses in tumors that do not respond to conventional chemotherapy.

T-cell lymphomas that are morphologically similar but lack ALK rearrangements (termed ALK[−] anaplastic large-cell lymphoma) tend

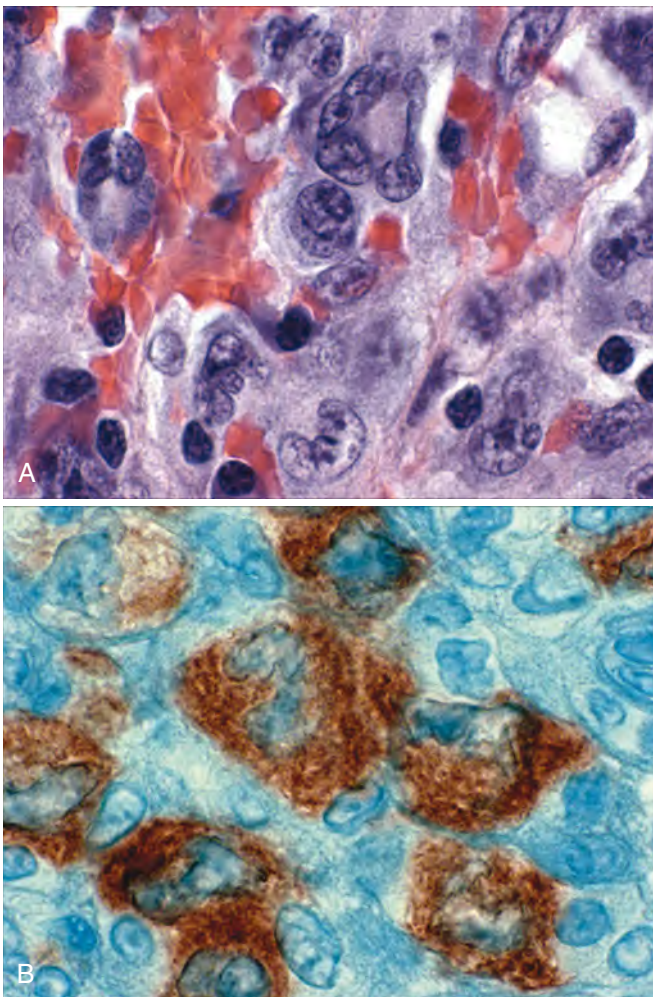


Fig. 13.19 Anaplastic large-cell lymphoma. (A) Several hallmark cells with horseshoe-like or embryoid nuclei and abundant cytoplasm lie near the center of the field. (B) Immunohistochemical stain demonstrating the presence of ALK fusion protein. (Courtesy Dr. Jeffery Kutok, Department of Pathology, Brigham and Women's Hospital, Boston.)

to occur in older adults and have a substantially worse prognosis. Both ALK⁺ and ALK[−] tumors usually express CD30, a member of the TNF receptor family. Notably, recombinant antibodies linked to toxins that recognize CD30 have significant antitumor activity against CD30⁺ T-cell lymphomas and Hodgkin lymphoma, another CD30⁺ tumor (described later).

Adult T-Cell Leukemia/Lymphoma

This neoplasm of CD4⁺ T cells is observed only in adults infected by HTLV-1 (Chapter 7). It occurs mainly in regions where HTLV-1 is endemic, namely southern Japan, West Africa, and the Caribbean basin. Common findings include skin lesions, generalized lymphadenopathy, hepatosplenomegaly, peripheral blood lymphocytosis, and hypercalcemia. In addition to adult T-cell leukemia/lymphoma, HTLV-1 infection may give rise to a progressive demyelinating disease of the central nervous system and spinal cord (Chapter 28).

The appearance of the tumor cells varies, but cells with multilobated nuclei ("cloverleaf" or "flower" cells) are frequently observed (Fig. 13.20). The tumor cells always contain clonal HTLV-1 proviruses, suggesting a direct pathogenic role for the virus. However, while several proviral proteins have been implicated in the development of adult T-cell leukemia/lymphoma (Chapter 7), the tumor develops in only a small fraction of those who are infected, usually after a latent period of decades. Details of the transformation process remain poorly understood but acquired mutations in tumor suppressor genes such as *TP53* and *CDKN2A* and acquired chromosomal copy variants are common in aggressive forms of adult T-cell leukemia/lymphoma.

Most patients present with rapidly progressive disease that is fatal within months to 1 year, despite aggressive chemotherapy. Less commonly, the tumor involves only the skin and follows a much more indolent course, like that of mycosis fungoides (described next).

Mycosis Fungoides and Sézary Syndrome

Mycosis fungoides and Sézary syndrome are tumors of CD4⁺ helper T cells that home to the skin. In both entities, the tumor cells express adhesion molecules such as cutaneous lymphocyte antigen and the chemokine receptor CCR4 that also contribute to the homing of normal CD4⁺ T cells to the skin. However, mycosis fungoides and Sézary syndrome have distinct clinical features.

- In *mycosis fungoides*, the cutaneous lesions typically progress through three somewhat distinct stages, an inflammatory *premycotic phase*, a *plaque phase*, and a *tumor phase* (Chapter 25). Histologically, the epidermis and upper dermis are infiltrated by neoplastic T cells, which often have a cerebriform appearance due to marked infolding of the nuclear membrane. While sensitive molecular analyses have shown that the tumor cells circulate through

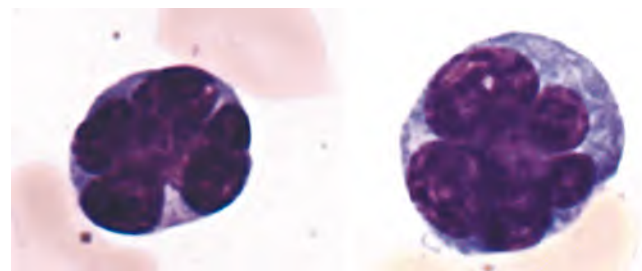


Fig. 13.20 Adult T-cell leukemia/lymphoma, peripheral blood. These tumor cells highlight the characteristic markedly irregular nuclear contours. (From Fletcher CD. *Diagnostic Histopathology of Tumors*. 5th ed. Elsevier; 2021.)

the blood, marrow, and lymph nodes even early in the course, overt extracutaneous disease is seen only late in the course.

- In *Sézary syndrome*, skin involvement is manifested as *generalized exfoliative erythroderma* and is accompanied by a leukemia of “Sézary” cells with characteristic cerebriform nuclei and generalized lymphadenopathy. In contrast to mycosis fungoides, the skin lesions rarely proceed to tumefaction.

Mycosis fungoides is an indolent tumor, with a median survival of roughly 10 years. Transformation to aggressive T-cell lymphoma occurs occasionally as a terminal event. The outlook is more guarded for patients with Sézary syndrome, only a minority of whom are alive 5 years after diagnosis.

Large Granular Lymphocytic Leukemia

T- and NK-cell variants of this rare neoplasm are recognized, both of which occur mainly in adults. Individuals with T-cell disease usually present with mild-to-moderate lymphocytosis and splenomegaly. Lymphadenopathy and hepatomegaly are usually absent. NK-cell disease often presents in an even more subtle fashion, with little or no lymphocytosis or splenomegaly.

Approximately 30% to 40% of large granular lymphocytic leukemias have acquired mutations in the transcription factor STAT3, which functions downstream of cytokine receptors. These mutations occur in both T- and NK-cell forms of the disease and result in cytokine-independent activation of STAT3, which appears to have a major role in the pathogenesis of these proliferations.

The tumor cells are large lymphocytes with abundant pale blue cytoplasm and a few coarse azurophilic granules, best seen in peripheral blood smears. The marrow, spleen, and liver usually contain a sparse interstitial lymphocytic infiltrate that is difficult to appreciate without immunohistochemical stains. T-cell variants express CD3 on their surfaces, whereas NK-cell variants lack surface CD3 and express NK markers such as CD56.

Despite the relative paucity of marrow infiltration, **neutropenia and anemia dominate the clinical picture**. Neutropenia may be accompanied by arrested maturation of granulocyte progenitors in the marrow. Rarely, *pure red cell aplasia* is seen. There is also an increased incidence of rheumatologic disorders. Some patients with *Felty syndrome*, a triad of rheumatoid arthritis, splenomegaly, and neutropenia, have this disorder as an underlying cause. The basis for these varied clinical abnormalities is unknown, but autoimmunity, provoked in some way by the tumor, seems likely.

The course is variable, being largely dependent on the severity of the cytopenias and their responsiveness to low-dose chemotherapy or steroids. In general, tumors of T-cell origin pursue an indolent course, whereas NK-cell tumors behave more aggressively.

Extranodal NK-/T-Cell Lymphoma

This neoplasm is rare in the United States and Europe but constitutes as many as 3% of NHLs in Asia. It most frequently presents as a destructive nasopharyngeal mass; less common sites of presentation include the testis and the skin. The tumor cell infiltrate typically surrounds and invades small vessels, leading to extensive *ischemic necrosis*. In touch preparations, *large azurophilic granules* are seen in the cytoplasm of the tumor cells that resemble those found in normal NK cells.

Extranodal NK-/T-cell lymphoma is highly associated with EBV infection. Within individual patients, all the tumor cells contain identical EBV episomes, indicating that the tumor originates from a single EBV-infected cell. How EBV gains entry is uncertain, since the tumor cells do not express CD21, the surface protein that serves as the B-cell EBV receptor. Most tumors are CD3[−], lack T-cell receptor gene rearrangements,

and express NK-cell markers, supporting an NK-cell origin. Multiple chromosomal aberrations are often present, but none are specific. Recurrent mutations have been reported in genes encoding components of the JAK/STAT pathway, epigenetic regulators, and *TP53*.

Most extranodal NK-/T-cell lymphomas are highly aggressive neoplasms that respond well to radiation therapy but tend to be relatively resistant to chemotherapy. Thus, the prognosis is generally poor in patients with advanced disease. In this setting, allogeneic hematopoietic stem cell transplantation offers some hope, as does treatment with immune checkpoint inhibitors; like other virus-driven cancers, extranodal NK-/T-cell lymphoma responds to immune checkpoint inhibitors, albeit with the usual attendant risks (e.g., cytokine storm, autoimmune disease) associated with these agents.

KEY CONCEPTS

Peripheral T- and NK-Cell Neoplasms

- Nodal T-follicular helper cell lymphoma: Aggressive T-cell tumor of T_{fh} cells associated with immune dysregulation, autoantibody formation, clonal hematopoiesis, and EBV-positive B-cell proliferations.
- Anaplastic large-cell lymphoma: Aggressive T-cell tumor associated in a subset with translocations that lead to constitutive activation of the ALK tyrosine kinase.
- Adult T-cell leukemia/lymphoma: Aggressive tumor of CD4⁺ T cells that is uniformly associated with HTLV-1 infection.
- Mycosis fungoides and Sézary syndrome: Tumors of CD4⁺ helper T cells that home to the skin.
 - Mycosis fungoides remains localized to skin for prolonged periods and follows an indolent course.
 - Sézary syndrome presents with diffuse erythroderma, lymphadenopathy, and peripheral blood involvement, and pursues a more aggressive course than mycosis fungoides.
- Large granular lymphocytic leukemia: Indolent tumor of cytotoxic T cells or NK cells that is associated with mutations in the transcription factor STAT3 and with autoimmune phenomena and cytopenias.
- Extranodal NK-/T-cell lymphoma: Aggressive tumor, usually derived from NK cells, that is strongly associated with EBV infection.

Plasma Cell Neoplasms and Related Disorders

These B-cell proliferations contain neoplastic plasma cells that virtually always secrete monoclonal immunoglobulin or immunoglobulin fragments, which serve as tumor markers and often have pathologic consequences. Collectively, plasma cell neoplasms account for about 15% of deaths caused by lymphoid neoplasms. The most common and deadly of these neoplasms is multiple myeloma, of which there are about 15,000 new cases per year in the United States.

A monoclonal immunoglobulin identified in the blood is referred to as an *M component*, in reference to myeloma. Because complete M components have molecular weights of 160,000 or higher, they are restricted to plasma and extracellular fluid and excluded from the urine in the absence of glomerular damage. However, **neoplastic plasma cells often synthesize excess light chains along with complete immunoglobulin**. Occasionally only light chains are produced, and rare tumors secrete only heavy chains. In most patients with plasma cell tumors, the level of free light chains in the blood is elevated and markedly skewed toward one light chain (e.g., κ) at the expense of the other (e.g., λ). Because free light chains are small, they are also excreted in the urine, where they are referred to as *Bence Jones proteins*.

Terms used to describe the abnormal immunoglobulins associated with plasma cell neoplasms include *monoclonal gammopathy* and

paraproteinemia. These abnormal proteins are associated with the following clinicopathologic entities:

- **Multiple myeloma (plasma cell myeloma)**, the most important plasma cell neoplasm, usually presents as tumor masses scattered throughout the skeletal system. *Solitary plasmacytoma* is an infrequent variant that presents as a single mass in bone or soft tissue. *Smoldering myeloma* refers to another uncommon variant defined by a lack of symptoms and a high plasma M component.
- *Waldenström macroglobulinemia* is a syndrome in which high levels of IgM lead to symptoms related to hyperviscosity of the blood. It occurs in older adults, most commonly in association with lymphoplasmacytic lymphoma (described later).
- *Heavy-chain disease* is a rare monoclonal gammopathy seen in association with a diverse group of disorders, including lymphoplasmacytic lymphoma and an unusual small bowel marginal zone lymphoma that occurs in malnourished populations (so-called *Mediterranean lymphoma*). The common feature is the synthesis and secretion of free heavy-chain fragments.
- *Immunoglobulin-related (AL) amyloidosis* results from a monoclonal proliferation of plasma cells secreting light chains (usually of λ isotype) that are deposited as amyloid. Some patients have overt multiple myeloma, but others have only a minor clonal population of plasma cells in the marrow and have signs and symptoms are related to amyloid deposition in tissues.
- *Monoclonal gammopathy of undetermined significance (MGUS)* is applied to patients without signs or symptoms who have small to moderately large M components in their blood. MGUS is very common in older adults and may transform to multiple myeloma or other symptomatic plasma cell neoplasms.

With this background, we now turn to some of these specific entities; primary amyloidosis is discussed in Chapter 6.

Multiple Myeloma

Multiple myeloma is a plasma cell neoplasm commonly associated with lytic bone lesions, hypercalcemia, renal failure, and acquired immune abnormalities. Although bony disease dominates, it can spread late in its course to lymph nodes and extranodal sites. Multiple myeloma causes 1% of cancer deaths in Western countries. Its incidence is higher in males and people of African descent. It is chiefly a disease of older adults, with a peak age of incidence of 65 to 70 years.

Pathogenesis

Multiple myeloma is genetically heterogeneous and is associated with frequent rearrangements involving the *IGH* locus on chromosome 14q32 and various proto-oncogenes. Included among the rearranged genes are the cell cycle–regulatory genes cyclin D1 and cyclin D3. Deletions of chromosome 17p that involve the *TP53* tumor suppressor locus also occur and are associated with a poor outcome. Highly aggressive forms of the disease such as plasma cell leukemia are associated with rearrangements involving *MYC*. Deep sequencing of myeloma genomes has identified frequent mutations involving genes encoding components of the NF- κ B pathway, which supports B-cell survival.

The proliferation and survival of myeloma cells are dependent on several cytokines, most notably IL-6, which is an important growth factor for plasma cells. It is produced by the tumor cells themselves and by resident marrow stromal cells. High serum levels of IL-6 are seen in patients with active disease and are associated with a poor prognosis. Myeloma cell growth and survival are also augmented by direct physical interactions with bone marrow stromal cells, a phenomenon that is an area of intense research interest.

Factors produced by neoplastic plasma cells mediate bone destruction, the major pathologic feature of multiple myeloma.

One important factor appears to be myeloma-derived MIP1 α (also known as CCL3), a chemokine that augments osteoclast formation through several different mechanisms. Other factors released from tumor cells, such as modulators of the Wnt pathway, are potent inhibitors of osteoblast function. The net effect is a marked increase in bone resorption, which leads to hypercalcemia and pathologic fractures.

MORPHOLOGY

Multiple myeloma usually presents as destructive plasma cell tumors (plasmacytomas) involving the axial skeleton. The bones most often affected (in descending order of frequency) are the vertebral column, ribs, skull, pelvis, femur, clavicle, and scapula. Lesions begin in the medullary cavity, erode cancellous bone, and progressively destroy the bony cortex, often leading to pathologic fractures; these are most common in the vertebral column but may occur in any affected bone. **The bone lesions appear radiographically as punched-out defects, usually 1 to 4 cm in diameter (Fig. 13.21),** and consist of soft, gelatinous, red tumor masses. Less commonly, widespread myelomatous bone disease produces diffuse demineralization (osteopenia) rather than focal defects.

Even away from overt tumor masses, the marrow contains an increased number of plasma cells, which usually constitute more than 30% of the cellularity. The plasma cells may infiltrate in a subtle fashion or completely replace normal marrow elements. Like their benign counterparts, malignant plasma cells have a perinuclear clearing due to a prominent Golgi apparatus and an eccentrically placed nucleus (Fig. 13.22). Relatively normal-appearing plasma cells, **plasma-blasts** with vesicular nuclear chromatin and a prominent single nucleolus, or **bizarre, multinucleated cells** may predominate. Other cytologic variants stem from the dysregulated synthesis and secretion of immunoglobulin, which often leads to intracellular accumulation of intact or partially degraded protein. Such variants include **flame cells** with fiery red cytoplasm, **Mott cells** with multiple grapelike cytoplasmic droplets, and cells containing a variety of other inclusions, including fibrils, crystalline rods, and globules. The globular inclusions are referred to as **Russell bodies** (if cytoplasmic) or **Dutcher bodies** (if nuclear). In advanced disease, plasma cell infiltrates may be present in the spleen, liver, kidneys, lungs, lymph nodes, and other soft tissues.

Commonly, high levels of M protein in the blood cause red cells to stick to one another in linear arrays in smears, a finding referred to as **rouleaux formation**. Rouleaux formation is characteristic but not specific, as it may be seen in other conditions in which immunoglobulin levels are elevated, such as lupus erythematosus and early HIV infection. Rarely, tumor cells flood the peripheral blood, giving rise to **plasma cell leukemia**.

Bence Jones proteins are excreted in the kidney and contribute to a form of renal disease called **myeloma kidney**. This important complication is discussed in detail in Chapter 20.

Immunophenotype. The neoplastic plasma cells are positive for CD138, an adhesion molecule also known as syndecan-1, and often express CD56, a feature that is helpful in identifying small populations of tumor cells. They also contain either cytoplasmic κ or λ immunoglobulin light chain.

Clinical Features

The clinical features of multiple myeloma stem from (1) the effects of plasma cell growth in tissues, particularly the bones; (2) the production of excessive immunoglobulin of a single type, which often has abnormal physicochemical properties; and (3) the suppression of normal humoral immunity.



Fig. 13.21 Multiple myeloma of the skull (radiograph, lateral view). The sharply punched-out bone lesions are most obvious in the calvaria.

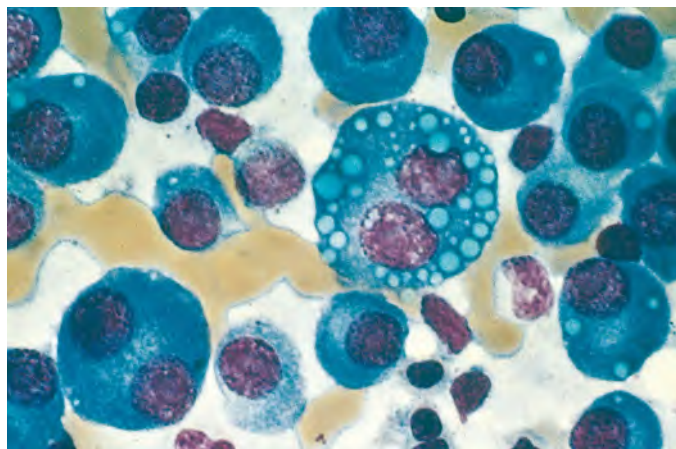
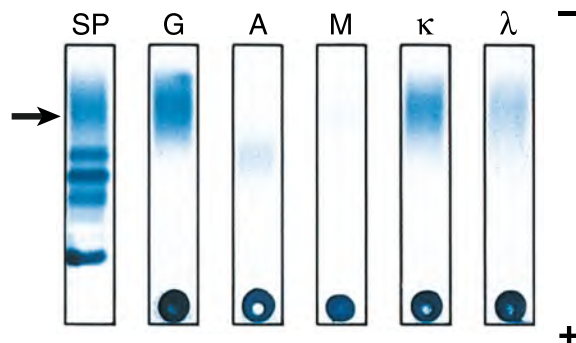


Fig. 13.22 Multiple myeloma (bone marrow aspirate). Normal marrow cells are largely replaced by plasma cells including forms with multiple nuclei, prominent nucleoli, and cytoplasmic droplets containing immunoglobulin.

Bone resorption often leads to pathologic fractures and chronic pain. The attendant hypercalcemia can give rise to neurologic manifestations, such as confusion, weakness, lethargy, constipation, and polyuria, and contributes to renal dysfunction. Decreased production of normal immunoglobulins sets the stage for recurrent bacterial infections. Cellular immunity is relatively unaffected. Of great significance is renal insufficiency, which trails only infections as a cause of death. The pathogenesis of renal failure (Chapter 20), which occurs in up to 50% of patients, is multifactorial. However, the single most important factor seems to be *Bence Jones proteinuria*, as the excreted light chains are toxic to renal tubular epithelial cells. Certain light chains (particularly those of the $\lambda 6$ and $\lambda 3$ families) are prone to cause amyloidosis of the AL type

Normal serum



Patient serum

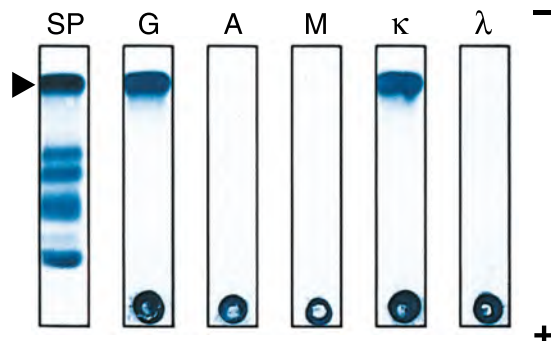


Fig. 13.23 M protein detection in multiple myeloma. Serum protein electrophoresis (SP) is used to screen for a monoclonal immunoglobulin (Ig) (M protein). Polyclonal IgG in normal serum (*arrow*) appears as a broad band; in contrast, serum from a patient with multiple myeloma contains a single sharp protein band (*arrowhead*) in this region of the electrophoretogram. The suspected M protein is confirmed and characterized by immunofixation, in which proteins are trapped in the gel with antibodies specific for IgG (G), IgA (A), IgM (M), or kappa (κ), or lambda (λ) light chain and then visualized with a protein stain. Note the sharp band in the patient serum is cross-linked by antisera specific for IgG heavy chain and κ light chain, indicating the presence of an IgG κ M protein. Levels of polyclonal IgG, IgA (A), and λ light chain are also decreased in the patient serum relative to normal, a finding typical of multiple myeloma. (Courtesy Dr. David Sacks, Department of Pathology, Brigham and Women's Hospital, Boston.)

(Chapter 6), which can exacerbate renal dysfunction and deposit in other tissues as well.

In 99% of patients, laboratory analyses reveal increased levels of immunoglobulin in the blood and/or light chains (Bence Jones proteins) in the urine. Monoclonal immunoglobulins are usually first detected as abnormal protein "spikes" in serum or urine electrophoresis and then further characterized by immunofixation (Fig. 13.23). Most myelomas are associated with more than 3 g/dL of serum immunoglobulin and/or more than 6 mg/dL of urine Bence Jones protein. The most common monoclonal immunoglobulin (M protein) is IgG (approximately 55% of patients), followed by IgA (approximately 25% of cases). Myelomas expressing IgM, IgD, or IgE occur but are rare. Excessive production and aggregation of M proteins, usually of the IgA or IgG₃ subtype, leads to symptoms related to hyperviscosity (described under lymphoplasmacytic lymphoma) in about 7% of patients. Both free light chains and serum M protein are observed together in 60% to 70% of patients. However, in about 20% of patients only free light chains are present, and around 1% of myelomas are nonsecretory; hence the absence of M proteins does not completely exclude the diagnosis.

The clinicopathologic diagnosis of multiple myeloma relies on identification of clonal plasma cells in the marrow and the presence of *CRAB criteria* (hypercalcemia, renal dysfunction, anemia, and bone lesions). It can be strongly suspected when the distinctive radiographic changes are present, but definitive diagnosis requires a bone marrow examination and tests assessing calcium levels, renal function, blood counts, and serum and urine immunoglobulins.

The prognosis is variable. About 60% of patients are alive 5 years after diagnosis. Patients with multiple bony lesions, if untreated, rarely survive for more than 6 to 12 months, whereas patients with “smoldering myeloma” may be asymptomatic for many years. Translocations involving cyclin D1 are associated with a good outcome, whereas deletions of 13q, deletions of 17p, and the t(4;14) all portend a more aggressive course.

Outcomes for patients with myeloma have improved steadily over the past 2 decades with advances in therapy. Myeloma cells are sensitive to inhibitors of the proteasome, a cellular organelle that degrades unwanted and misfolded proteins (Chapter 2). Because of the high levels of immunoglobulin synthesis, misfolding of immunoglobulin heavy and light chains may occur even in normal plasma cells, and this tendency is exacerbated in myeloma cells in which heavy and light chain synthesis is unbalanced. If these misfolded proteins are not degraded in proteasomes, they trigger apoptosis. Proteasome inhibitors induce cell death by exploiting this vulnerability, and also seem to slow bone resorption through effects on stromal cells. Thalidomide and related compounds such as lenalidomide also have activity against myeloma. Interestingly, this also appears to involve changes in protein degradation, as lenalidomide redirects and activates certain ubiquitin ligases, thereby targeting proteins for proteolysis that are required for myeloma growth. Bisphosphonates, drugs that inhibit bone resorption, reduce pathologic fractures and limit hypercalcemia. HSC transplantation prolongs life but has not been proven to be curative. CAR-T cell therapy against a plasma cell-specific surface antigen, B-cell maturation antigen (BCMA), produce deep remissions in a high fraction of patients but at the risk of significant toxicities, including cytokine release syndrome (cytokine “storm”), neurotoxicity, and infection related to hypogammaglobulinemia.

Smoldering Myeloma

This entity defines a middle ground between multiple myeloma and MGUS. Plasma cells make up 10% to 30% of the marrow cellularity, and the serum M protein level is greater than 3 g/dL, but patients are asymptomatic. About 75% of patients progress to multiple myeloma over a 15-year period.

Solitary Plasmacytoma

About 3% to 5% of plasma cell neoplasms present as a solitary lesion of bone or soft tissue. The bone lesions tend to occur in the same locations as in multiple myeloma. Extraosseous lesions are often located in the lungs, oronasopharynx, or nasal sinuses. Modest elevations of M proteins in the blood or urine may be found in some patients. Solitary osseous plasmacytoma almost inevitably progresses to multiple myeloma, but this can take 10 to 20 years or longer. In contrast, extraosseous plasmacytomas, particularly those involving the upper respiratory tract, are frequently cured by local resection.

Monoclonal Gammopathy of Undetermined Significance

MGUS is the most common plasma cell disorder, occurring in about 3% of persons older than 50 years of age and in about 5% of individuals older than 70 years of age. By definition, patients are asymptomatic and the serum M protein level is less than 3 g/dL. **Approximately 1% of patients with MGUS develop a symptomatic plasma cell neoplasm, usually multiple myeloma, per year.** The clonal plasma cells in MGUS contain many of the same chromosomal translocations and

deletions that are found in multiple myeloma, indicating that MGUS is an early stage of myeloma development. As with smoldering myeloma, progression to multiple myeloma is unpredictable; hence, periodic assessment of serum M component levels and Bence Jones proteinuria is warranted.

Lymphoplasmacytic Lymphoma

Lymphoplasmacytic lymphoma is a B-cell neoplasm of older adults that usually presents in the sixth or seventh decade of life. Although superficially resembling CLL/SLL, it differs in that a substantial fraction of the tumor cells undergo terminal differentiation to plasma cells. Most commonly, the plasma cell component secretes monoclonal IgM, often in amounts sufficient to cause a hyperviscosity syndrome known as *Waldenström macroglobulinemia*. Unlike multiple myeloma, complications stemming from the secretion of free light chains (e.g., renal failure and amyloidosis) are relatively rare, and bone destruction does not occur.

Pathogenesis

Virtually all cases of lymphoplasmacytic lymphoma are associated with acquired mutations in *MYD88*. *MYD88* encodes an adaptor protein that participates in signaling events that activate NF- κ B, which may promote the growth and survival of the tumor cells.

MORPHOLOGY

Typically, the marrow contains an infiltrate of lymphocytes, plasma cells, and plasmacytoid lymphocytes in varying proportions, often accompanied by mast cell hyperplasia (Fig. 13.24). Some tumors also contain a population of larger lymphoid cells with more vesicular nuclear chromatin and prominent nucleoli. Periodic acid–Schiff–positive inclusions containing immunoglobulin are frequently seen in the cytoplasm (**Russell bodies**) or the nucleus (**Dutcher bodies**) of some of the plasma cells. At diagnosis the tumor has usually disseminated to the lymph nodes, spleen, and liver. Infiltration of the nerve roots, meninges, and more rarely the brain can also occur with disease progression.

Immunophenotype. The lymphoid component expresses B-cell markers such as CD20 and surface immunoglobulin, whereas the plasma cell component secretes the same immunoglobulin that is expressed on the surface of the lymphoid cells, usually IgM.

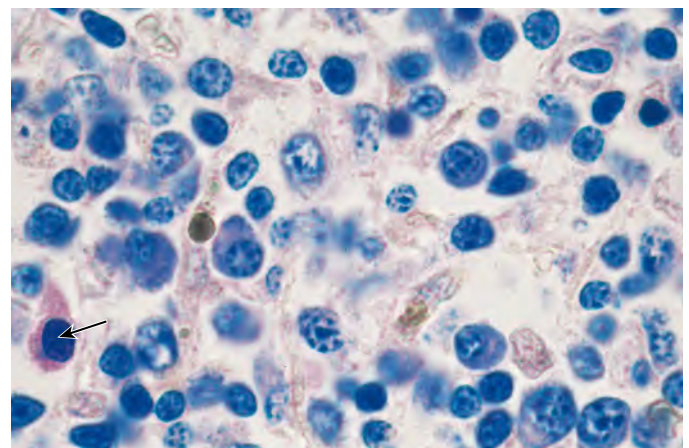


Fig. 13.24 Lymphoplasmacytic lymphoma. Bone marrow biopsy shows a characteristic mixture of small lymphoid cells exhibiting various degrees of plasma cell differentiation. In addition, a mast cell with purplish red cytoplasmic granules is present at the left side of the field (arrow).

Clinical Features

The dominant presenting complaints are nonspecific and include weakness, fatigue, and weight loss. Approximately half the patients have lymphadenopathy, hepatomegaly, and splenomegaly. Anemia caused by marrow infiltration is common. About 10% of patients have autoimmune hemolysis caused by cold agglutinins, IgM antibodies that bind to red cells at temperatures of less than 37°C (Chapter 14).

Patients with IgM-secreting tumors have additional signs and symptoms stemming from the physicochemical properties of IgM. Because of its large size, at high concentrations IgM greatly increases the viscosity of the blood, giving rise to a *hyperviscosity syndrome* characterized by:

- **Visual impairment** associated with venous congestion, reflected by striking tortuosity and distention of retinal veins; retinal hemorrhages and exudates may also contribute to the visual problems.
- **Neurologic problems** such as headaches, dizziness, deafness, and stupor, all stemming from sluggish blood flow.
- **Bleeding** related to the formation of complexes between IgM and clotting factors as well as interference with platelet function.
- **Cryoglobulinemia** resulting from the precipitation of IgM at low temperatures, producing symptoms such as Raynaud phenomenon and cold urticaria.

Lymphoplasmacytic lymphoma is an indolent disorder. Because most IgM is intravascular, symptoms caused by the high IgM levels (e.g., hyperviscosity and hemolysis) can be alleviated by plasmapheresis. Tumor growth can be controlled with low doses of chemotherapeutic drugs and immunotherapy with anti-CD20 antibody. Like other indolent B-cell tumors, lymphoplasmacytic lymphoma is highly responsive to BTK inhibitors, indicating a dependence of the tumor cells on the B-cell receptor signaling pathway. Transformation to large-cell lymphoma occurs but is uncommon. With recent advances in therapy, the median survival is now estimated to be greater than 10 years.

KEY CONCEPTS

Plasma Cell Neoplasms

Multiple Myeloma

- Plasma cell tumor that manifests with multiple lytic bone lesions associated with pathologic fractures and hypercalcemia.
- Neoplastic plasma cells suppress normal humoral immunity and secrete partial immunoglobulins that are nephrotoxic.
- Associated with diverse translocations involving the *IGH* locus; frequent dysregulation and overexpression of D cyclins.
- May be associated with AL amyloidosis (as may other neoplasms later).

Other Plasma Cell Neoplasms and Related Entities

- Monoclonal gammopathy of undetermined significance (MGUS): common in older adults, progresses to myeloma at a rate of 1% per year.
- Smoldering myeloma: disseminated disease that pursues an indolent course.
- Solitary plasmacytoma: solitary bone or soft tissue tumor. Most bone tumors progress to multiple myeloma over 7–10 years, whereas extraosseous tumors rarely progress to systemic disease.
- Lymphoplasmacytic lymphoma: B-cell lymphoma that exhibits plasmacytic differentiation; clinical symptoms dominated by hyperviscosity related to high levels of tumor-derived IgM; highly associated with mutations in the *MYD88* gene.

TABLE 13.7 Differences Between Hodgkin and Non-Hodgkin Lymphomas

Hodgkin Lymphoma	Non-Hodgkin Lymphoma
More often localized to a single axial group of nodes (cervical, mediastinal, para-aortic)	More frequent involvement of multiple peripheral nodes
Orderly spread by contiguity	Noncontiguous spread
Mesenteric nodes and Waldeyer ring rarely involved	Waldeyer ring and mesenteric nodes commonly involved
Extranodal presentation rare	Extranodal presentation common

Hodgkin Lymphoma

Hodgkin lymphoma encompasses a group of lymphoid neoplasms that differ from NHL in several respects (Table 13.7). While NHLs frequently occur at extranodal sites and spread in an unpredictable fashion, **Hodgkin lymphoma arises in a single node or chain of nodes and spreads first to anatomically contiguous lymphoid tissues. Morphologically, the distinctive feature of Hodgkin lymphoma is the presence of neoplastic giant cells called Reed-Sternberg cells.** These cells release factors that induce the accumulation of reactive lymphocytes, macrophages, and granulocytes, which typically make up greater than 90% of the tumor cellularity. Molecular studies have shown that the neoplastic Reed-Sternberg cells are derived from germinal center or post-germinal center B cells.

Hodgkin lymphoma accounts for 0.7% of all new cancers in the United States; there are about 9000 cases each year. The average age at diagnosis is 39 years, but the disease shows a bimodal age distribution, with one peak in young adults and the second in adults older than 55 years. It was the first human cancer to be successfully treated with radiation therapy and chemotherapy and is curable in most cases.

Classification

The WHO classification of Hodgkin lymphoma recognizes five subtypes:

1. Nodular sclerosis
2. Mixed cellularity
3. Lymphocyte-rich
4. Lymphocyte-depleted
5. Nodular lymphocyte predominant

In the first four subtypes—nodular sclerosis, mixed cellularity, lymphocyte-rich, and lymphocyte depleted—the Reed-Sternberg cells have a similar distinctive immunophenotype. These subtypes are lumped together as *classic* forms of Hodgkin lymphoma. In the remaining subtype, lymphocyte predominant, the Reed-Sternberg cells have a B-cell immunophenotype that differs from that of the classic types. Indeed, in the International Consensus Classification (ICC) of lymphoid malignancies this entity has been renamed nodular lymphocyte-predominant B-cell lymphoma, dropping the Hodgkin moniker altogether.

Pathogenesis

The origin of the neoplastic Reed-Sternberg cells of classic Hodgkin lymphoma was solved through elegant molecular studies of single isolated Reed-Sternberg cells. These revealed clonal *IGH* gene rearrangements and the telltale signs of somatic hypermutation, establishing an origin for Reed-Sternberg cells from a germinal center or post-germinal center B cell. Despite their B-cell origin, the Reed-Sternberg cells of classic Hodgkin lymphoma fail to express most B-cell-specific genes, including immunoglobulin genes. The cause of this wholesale reprogramming of gene expression is not understood, but it presumably

results from widespread epigenetic changes. In addition, in a subset of classic Hodgkin lymphoma, the tumor cells are infected with EBV.

Activation of the transcription factor NF- κ B is a common event in classic Hodgkin lymphoma and turns on genes that promote the growth and survival of Reed-Sternberg cells. This can occur by several mechanisms:

- EBV+ tumor cells express latent membrane protein 1 (LMP-1), a protein encoded by the EBV genome that transmits signals that upregulate NF- κ B.
- Activation of NF- κ B may occur in EBV- tumors through acquired loss-of-function mutations in I κ B or TNF- α -induced protein 3, both of which are negative regulators of NF- κ B.

It is hypothesized that activation of NF- κ B rescues “crippled” germinal center B cells that cannot express immunoglobulin from apoptosis, setting the stage for the acquisition of other mutations that collaborate to produce Reed-Sternberg cells. Little is known about the basis for the morphology of Reed-Sternberg cells and variants, but it is intriguing that EBV-infected B cells resembling Reed-Sternberg cells may be found in the lymph nodes of individuals with infectious mononucleosis, strongly suggesting that EBV-encoded proteins play a part in the remarkable metamorphosis of B cells into Reed-Sternberg cells.

Reed-Sternberg cells are aneuploid and possess diverse clonal chromosomal aberrations. Copy number gains in the *REL* proto-oncogene on chromosome 2p are particularly common and may also contribute to increases in NF- κ B activity. Also frequent are copy number gains in genes encoding PD-L1 and PD-L2, located together on chromosome 9p, which you will recall are immune checkpoint proteins that inhibit antitumor T-cell responses (Chapter 7).

The florid accumulation of reactive cells in tissues involved by classic Hodgkin lymphoma occurs in response to a wide variety of cytokines (e.g., IL-5, IL-10, and M-CSF), chemokines (e.g., eotaxin), and other factors that are secreted by Reed-Sternberg cells. Once attracted, the reactive cells produce factors that support the growth and survival of the tumor cells and further modify the reactive cell response. For example, eosinophils and T cells express ligands that activate CD30 and CD40 receptors on Reed-Sternberg cells, producing signals that upregulate NF- κ B. Although Reed-Sternberg cells induce a host immune response, it is ineffective because of factors produced by the Reed-Sternberg cells such as PD-L1 and PD-L2, immune checkpoint molecules that antagonize cytotoxic T-cell responses.

MORPHOLOGY

Identification of Reed-Sternberg cells and their variants is essential for the diagnosis. **Diagnostic Reed-Sternberg cells are large cells (45 μ m in diameter) with multiple nuclei or a single nucleus with multiple nuclear lobes, each with a large inclusion-like nucleolus about the size of a small lymphocyte (5 to 7 μ m in diameter) (Fig. 13.25A).** The cytoplasm is abundant. Several Reed-Sternberg cell variants are also recognized. **Mononuclear variants** contain a single nucleus with a large inclusion-like nucleolus (Fig. 13.25B). **Lacunar cells** (seen in the nodular sclerosis subtype) have more delicate, folded, or multilobate nuclei and abundant pale cytoplasm that is often disrupted during the cutting of sections, leaving the nucleus sitting in an empty space (a lacuna) (Fig. 13.25C). In classic forms of Hodgkin lymphoma, Reed-Sternberg cells undergo a peculiar form of cell death in which the cells shrink and become pyknotic, a process described as “mummification.” **Lymphohistiocytic variants** (L&H cells) with polypoid nuclei, inconspicuous nucleoli, and moderately abundant cytoplasm are characteristic of the lymphocyte predominance subtype (Fig. 13.25D).

Hodgkin lymphoma must be distinguished from other conditions in which cells resembling Reed-Sternberg cells may be seen, such as infectious mononucleosis,

solid tissue cancers, and certain NHLs. The diagnosis of Hodgkin lymphoma depends on the identification of Reed-Sternberg cells in a background of nonneoplastic inflammatory cells. Reed-Sternberg cells also have a characteristic immunohistochemical profile, a feature that is used to confirm the morphologic impression.

With this as background, we turn to the subclasses of Hodgkin lymphoma, pointing out some of the salient morphologic and immunophenotypic features of each (Table 13.8). The clinical manifestations common to all are presented later.

Nodular Sclerosis

This is the most common form of Hodgkin lymphoma, constituting 65% to 70% of cases. It is characterized by the presence of **lacunar variant** Reed-Sternberg cells and the **deposition of collagen in bands that divide involved lymph nodes into circumscribed nodules** (Fig. 13.26). The fibrosis may be scant or abundant. The Reed-Sternberg cells are found in a polymorphous background of T cells, eosinophils, plasma cells, and macrophages. Diagnostic Reed-Sternberg cells are usually uncommon. The Reed-Sternberg cells in this and other classic Hodgkin lymphoma subtypes have a characteristic immunophenotype; they are positive for PAX5 (a B-cell transcription factor), CD15, and CD30 and negative for other B-cell markers, T-cell markers, and CD45 (leukocyte common antigen). As in other forms of Hodgkin lymphoma, involvement of the spleen, liver, bone marrow, and other organs and tissues may appear in due course in the form of irregular tumor nodules resembling those seen in lymph nodes. This subtype is uncommonly associated with EBV.

The nodular sclerosis type occurs with equal frequency in males and females. It has a propensity to involve the lower cervical, supraclavicular, and mediastinal lymph nodes of adolescents or young adults. The prognosis is excellent.

Mixed-Cellularity

This form of Hodgkin lymphoma constitutes about 20% to 25% of cases. Involved lymph nodes are diffusely effaced by a heterogeneous cellular infiltrate, which includes T cells, eosinophils, plasma cells, and benign macrophages admixed with Reed-Sternberg cells (Fig. 13.27). **Diagnostic Reed-Sternberg cells and mononuclear variants are usually plentiful. The Reed-Sternberg cells are infected with EBV in about 70% of cases.** The immunophenotype is identical to that observed in the nodular sclerosis type.

Mixed-cellularity Hodgkin lymphoma is more common in males. Compared with the lymphocyte predominance and nodular sclerosis subtypes, it is more likely to be associated with older age, systemic symptoms such as night sweats and weight loss, and advanced tumor stage. Nonetheless, the overall prognosis is very good.

Lymphocyte-Rich

This is an uncommon form of classic Hodgkin lymphoma in which **reactive lymphocytes make up most of the cellular infiltrate**. In most cases, involved lymph nodes are diffusely effaced, but vague nodularity due to the presence of residual B-cell follicles is sometimes seen. This entity is distinguished from the lymphocyte predominance type by the presence of frequent mononuclear variants and diagnostic Reed-Sternberg cells with a “classic” immunophenotypic profile. It is associated with EBV in about 40% of cases and has a very good to excellent prognosis.

Lymphocyte-Depleted

This is the least common form of classic Hodgkin lymphoma, amounting to less than 5% of cases. It is characterized by a paucity of lymphocytes and a relative abundance of Reed-Sternberg cells or their pleomorphic variants. The immunophenotype of the Reed-Sternberg cells is identical to that seen in other classic types of Hodgkin lymphoma. Immunophenotyping is essential since most tumors suspected of being lymphocyte depletion Hodgkin lymphoma actually prove to be NHL. **The Reed-Sternberg cells are infected with EBV in over 90% of cases.**

Lymphocyte depletion Hodgkin lymphoma occurs predominantly in older adults, in HIV-positive individuals of any age, and in individuals living in developing countries. Advanced stage and systemic symptoms are frequent, and the overall outcome is less favorable than in the other subtypes.

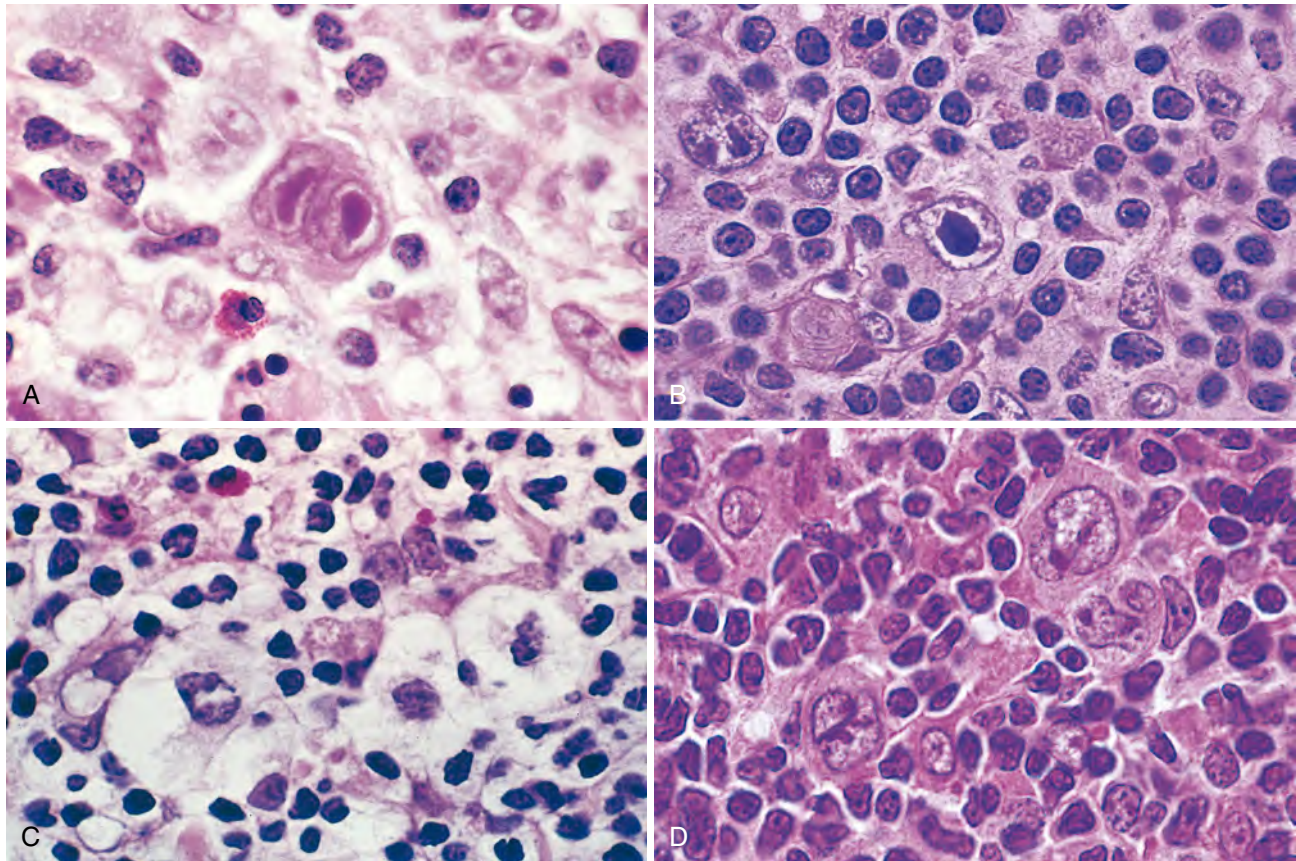


Fig. 13.25 Reed-Sternberg cells and variants. (A) Diagnostic Reed-Sternberg cell, with two nuclear lobes, large inclusion-like nucleoli, and abundant cytoplasm, surrounded by lymphocytes, macrophages, and an eosinophil. (B) Reed-Sternberg cell, mononuclear variant. (C) Reed-Sternberg cell, lacunar variant. This variant has a folded or multilobated nucleus and lies within an open space, which is an artifact created by disruption of the cytoplasm during tissue sectioning. (D) Reed-Sternberg cell, lymphohistiocytic variant. Several such variants with multiple in-folded nuclear membranes, small nucleoli, fine chromatin, and abundant pale cytoplasm are present. (A, Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

TABLE 13.8 Subtypes of Hodgkin Lymphoma

Subtype	Morphology and Immunophenotype	Typical Clinical Features
Nodular sclerosis	Frequent lacunar cells and occasional diagnostic RS cells; background infiltrate composed of T lymphocytes, eosinophils, macrophages, and plasma cells; fibrous bands dividing cellular areas into nodules. RS cells CD15+, CD30+; usually EBV–	Most common subtype; usually stage I or II disease; frequent mediastinal involvement; equal occurrence in males and females, most patients young adults
Mixed cellularity	Frequent mononuclear and diagnostic RS cells; background infiltrate rich in T lymphocytes, eosinophils, macrophages, plasma cells; RS cells CD15+, CD30+; 70% EBV+	More than 50% present as stage III or IV disease; occurrence greater in males than females; biphasic incidence, peaking in young adults and again in adults older than 55 years
Lymphocyte-rich	Frequent mononuclear and diagnostic RS cells; background infiltrate rich in T lymphocytes; RS cells CD15+, CD30+; 40% EBV+	Uncommon; occurrence greater in males than females; tends to be seen in older adults
Lymphocyte-depleted	Reticular variant: Frequent diagnostic RS cells and variants and a paucity of background reactive cells; RS cells CD15+, CD30+; most EBV+	Uncommon; more common in older males, HIV-infected individuals, and people in low-resource countries; often presents with advanced disease
Nodular lymphocyte predominant	Frequent L&H (popcorn cell) variants in a background of follicular dendritic cells and reactive B cells; RS cells CD20+, CD15–, CD30–; EBV–	Uncommon; young males with cervical or axillary lymphadenopathy; mediastinal

EBV, Epstein-Barr virus; HIV, human immunodeficiency virus; L&H, lymphohistiocytic; RS, Reed-Sternberg.

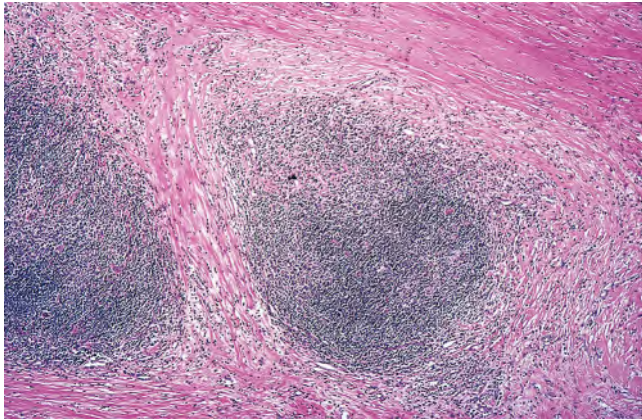


Fig. 13.26 Hodgkin lymphoma, nodular sclerosis subtype. Low-power view shows well-defined bands of pink, acellular collagen that subdivide the tumor into nodules. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

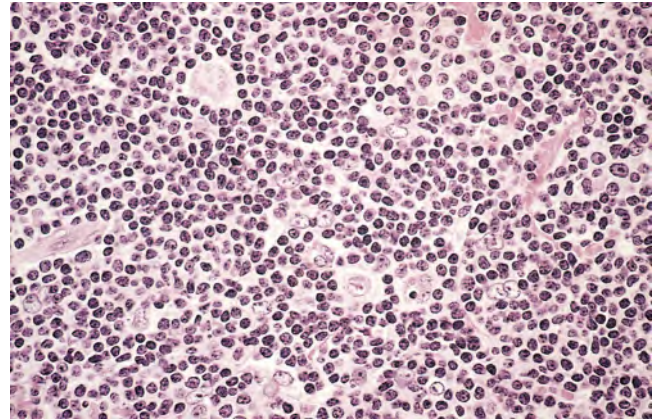


Fig. 13.28 Hodgkin lymphoma, lymphocyte-predominant subtype. Numerous mature-looking lymphocytes surround scattered, large, pale-staining lymphohistiocytic variants (popcorn cells). (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

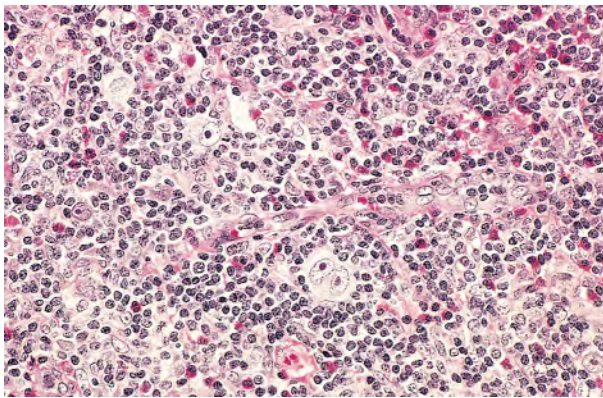


Fig. 13.27 Hodgkin lymphoma, mixed-cellularity subtype. A diagnostic, binucleate Reed-Sternberg cell is surrounded by reactive cells including eosinophils (bright red cytoplasm), lymphocytes, and macrophages. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Nodular Lymphocyte Predominant

This uncommon “nonclassic” variant of Hodgkin lymphoma accounts for about 5% of cases. Involved nodes are effaced by nodules of small lymphocytes admixed with variable numbers of macrophages (Fig. 13.28). “Classic” Reed-Sternberg cells are usually absent. Instead, this tumor contains so-called L&H variants with multilobed nuclei resembling popcorn kernels (**popcorn cells**). Eosinophils and plasma cells are usually scant or absent.

In contrast to the Reed-Sternberg cells found in classic forms of Hodgkin lymphoma, **L&H variants express B-cell markers typical of germinal center B cells**, such as CD20 and BCL6, and are usually negative for CD15 and CD30. The nodular pattern of growth is due to the presence of expanded B-cell follicles populated by L&H variants, numerous reactive B cells, and follicular dendritic cells. The *IGH* genes of the L&H variants show evidence of ongoing somatic hypermutation, further marking these cells as transformed germinal center B cells. In 3% to 5% of cases, this type transforms into a tumor resembling diffuse large B-cell lymphoma. EBV is rarely associated with this subtype.

The majority of patients are males, usually younger than 35 years, presenting with cervical or axillary lymphadenopathy. Mediastinal and bone marrow involvement is rare. In some series, this form of Hodgkin lymphoma is more likely to recur than the classic subtypes, but the prognosis is excellent.

Clinical Features

Like other lymphomas, Hodgkin lymphoma most commonly presents as painless lymphadenopathy. Patients with the nodular sclerosis or lymphocyte predominant subtypes tend to have stage I or II disease and are usually free of systemic manifestations. Patients with disseminated disease (stages III and IV) or the mixed-cellularity or lymphocyte depleted subtypes are more likely to have constitutional symptoms such as fever, night sweats, and weight loss. Cutaneous immune unresponsiveness (*anergy*) resulting from depressed cell-mediated immunity is seen in most cases of classic Hodgkin lymphoma and is attributable to the expression of factors such as IL-10 by Reed-Sternberg cells that suppress Th1 immune responses. This is but one of many examples of crosstalk between Reed-Sternberg cells, various stromal cells, and immune cells (Fig. 13.29).

The spread of Hodgkin lymphoma is remarkably stereotypic: nodal disease first, then splenic disease, hepatic disease, and finally involvement of the marrow and other tissues. Staging involves physical examination; radiologic imaging of the abdomen, pelvis, and chest; and biopsy of the bone marrow (Table 13.9). With current treatment protocols, tumor stage rather than histologic type is the most important prognostic variable. The cure rate of patients with stages I and IIA is close to 90%. Even with advanced disease (stages IVA and IVB), disease-free survival at 5 years is 60% to 70%.

Low-stage, localized Hodgkin lymphoma can be cured with involved field radiotherapy, and indeed cure of such patients was one of the early success stories in oncology. However, it was subsequently recognized that long-term survivors treated with radiotherapy had a much higher incidence of certain other malignancies, including lung cancer, melanoma, and breast cancer. Patients treated with early chemotherapy regimens containing alkylating agents also had a high incidence of secondary tumors, particularly acute myeloid leukemia. These sobering results spurred the development of current treatment regimens, which minimize the use of radiotherapy and employ less genotoxic chemotherapeutic agents; as a result, the incidence of secondary tumors appears to have been reduced markedly, without any loss of therapeutic efficacy.

For patients with classic Hodgkin lymphoma who do not respond to conventional therapy, immune checkpoint inhibitors that block PD-1, the receptor for PD-L1 and PD-L2, have proven to be highly effective. These agents prevent the CD8⁺ cytotoxic T-cell “exhaustion” that is caused by PD-L1 and PD-L2 expressed on Reed-Sternberg cells and lead to sustained responses in almost 90% of cases. Indeed, Hodgkin lymphoma is so sensitive to checkpoint inhibitors that current trials are evaluating its role in treatment of newly diagnosed patients.

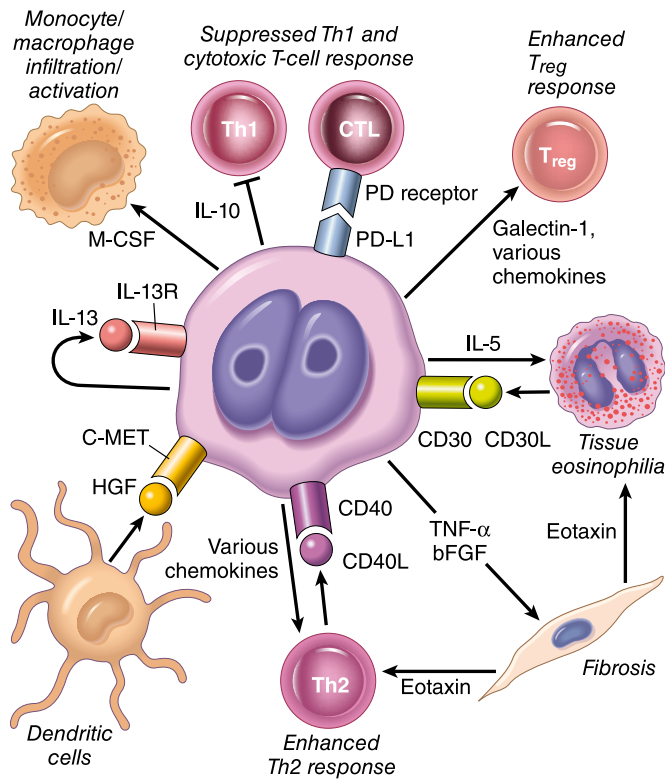


Fig. 13.29 Proposed signals mediating crosstalk between Reed-Sternberg cells and surrounding normal cells in classic forms of Hodgkin lymphoma. *bFGF*, Basic fibroblast growth factor; *CD30L*, CD30 ligand; *CD40L*, CD40 ligand; *CTL*, CD8+ cytotoxic T cell; *HGF*, hepatocyte growth factor (binds to the c-MET receptor); *IL*, interleukin; *M-CSF*, monocyte colony-stimulating factor; *Th1*, *Th2*, CD4+ T-helper cell subsets; *TNFα*, tumor necrosis factor-α; *Treg*, regulatory T cell.

TABLE 13.9 Clinical Staging of Hodgkin and Non-Hodgkin Lymphomas (Ann Arbor Classification)

Stage	Distribution of Disease
I	Involvement of a single lymph node region (I) or a single extralymphatic organ or site (IE)
II	Involvement of two or more lymph node regions on the same side of the diaphragm alone (II) or localized involvement of an extralymphatic organ or site (IIIE)
III	Involvement of lymph node regions on both sides of the diaphragm without (III) or with (IIIE) localized involvement of an extralymphatic organ or site
IV	Diffuse involvement of one or more extralymphatic organs or sites with or without lymphatic involvement

All stages are further divided based on the absence (A) or presence (B) of these symptoms: unexplained fever, drenching night sweats, and/or unexplained weight loss of greater than 10% of normal body weight. Data from Carbone PT, et al. Symposium (Ann Arbor): Staging in Hodgkin's disease. *Cancer Res.* 1971;31:1707.

- Two broad types, classic (which has several subtypes) and nodular lymphocyte predominant, which are distinguished based on morphologic and immunophenotypic grounds.
- Reed-Sternberg cells of classic types express multiple cytokines, chemokines, and ligands that influence the host response, and the responding host cells in turn make factors that support the growth of the tumor cells.
- Classic forms are frequently associated with acquired mutations that activate the transcription factor NF-κB and with EBV infection.
- Lymphocyte-predominant subtype expresses B-cell markers and is not associated with EBV.
- Classic Hodgkin lymphoma is highly responsive to immune checkpoint inhibitors, which antagonize the activity of PD-L1 and PD-L2 expressed on the surface of Reed-Sternberg cells.

Pathogenic Lessons From Lymphoid Neoplasms

Before ending our review of lymphoid neoplasms, it is worthwhile pausing to summarize the manner by which common driver mutations in specific entities produce changes in cellular behavior that exemplify particular hallmarks of cancer. Some of the best characterized pathogenic mechanisms are summarized in Fig. 13.30, including dysregulation of MYC in Burkitt lymphoma (leading to Warburg metabolisms and rapid cell growth); dysregulation of BCL2 in follicular lymphoma (leading to resistance to apoptosis); PD-1 ligand gene amplification in Hodgkin lymphoma (leading to evasion of host immunity); events leading to loss of cell cycle control (cyclin D1 rearrangements in mantle cell lymphoma and loss of the *CDKN2A* gene in acute lymphoblastic leukemia [ALL]); mutations in various transcription factors, particularly in ALL, that block differentiation and enhance “leukemia stem cell” self-renewal; and chronic immune stimulation, in marginal zone lymphoma. Such alterations not only highlight important pathogenic principles, but are increasingly the targets of effective therapies, such as antibodies that block PD-1 (Hodgkin lymphoma) and drugs that antagonize BCL2 (chronic lymphocytic leukemia and other B-cell tumors). By contrast, because lymphoid cells normally circulate throughout the body, there is relatively little selective pressure in lymphoid malignancies for aberrations that increase angiogenesis or activate invasion and metastasis, points of distinction from other forms of cancer.

MYELOID NEOPLASMS

The common feature of this heterogeneous group of neoplasms is their origin from hematopoietic progenitor cells. These diseases primarily involve the marrow and to a lesser degree the secondary hematopoietic organs (spleen, liver, and lymph nodes) and usually present with symptoms related to altered hematopoiesis. Three broad, major categories of myeloid neoplasms exist:

- *Acute myeloid leukemias*, in which an accumulation of immature myeloid forms (blasts) in the bone marrow suppresses normal hematopoiesis
- *Myelodysplastic neoplasms*, in which defective maturation of myeloid progenitors gives rise to ineffective hematopoiesis, leading to cytopenias
- *Myeloproliferative neoplasms*, in which there is usually increased production of one or more types of blood cells

The pathogenesis of myeloid neoplasms is best understood in the context of normal hematopoiesis, which involves a hierarchy of HSCs, committed progenitors, and more differentiated elements (see Fig. 13.1). Normal hematopoiesis is finely tuned by homeostatic feedback mechanisms involving cytokines and growth factors that

KEY CONCEPTS

Hodgkin Lymphoma

- Unusual tumor consisting mostly of reactive lymphocytes, macrophages, eosinophils, plasma cells, and stromal cells mixed with rare tumor giant cells called Reed-Sternberg cells and variants.

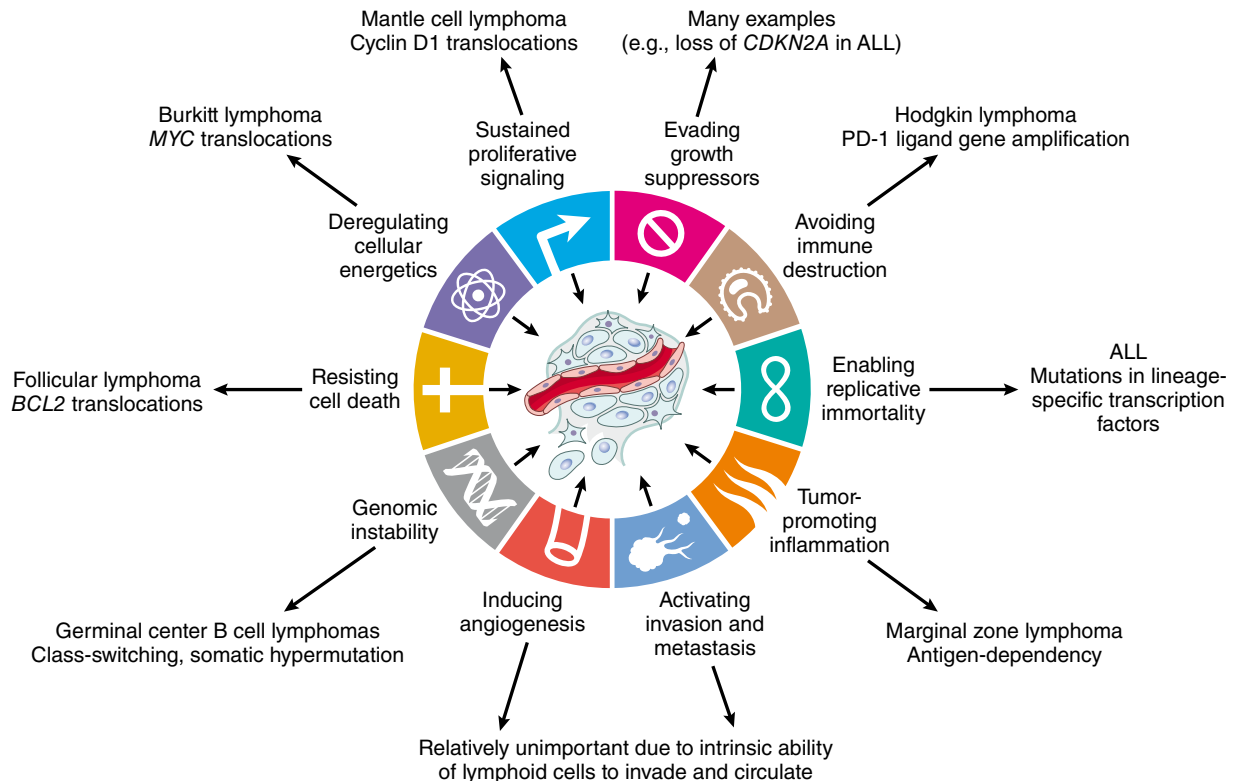


Fig. 13.30 Cancer hallmarks exemplified by particular lymphoid neoplasms. See text for details.

modulate the production of red cells, white cells, and platelets by the marrow. These mechanisms are deranged by myeloid neoplasms, which “escape” from normal homeostatic controls and suppress the function of residual normal HSCs and progenitors. The specific manifestations of the different myeloid neoplasms are influenced by:

- The position of the transformed cell within the hierarchy of progenitors (i.e., a pluripotent HSC versus a more committed progenitor).
- The effect of the transforming events on differentiation, which may be inhibited, skewed, or deranged by specific oncogenic mutations.

Given that all myeloid neoplasms originate from transformed hematopoietic progenitors, it is not surprising that divisions between these neoplasms are sometimes blurred; for example, some tumors have molecular and clinical features that lie between typically myelodysplastic and myeloproliferative neoplasms. Moreover, myeloid neoplasms, like other malignancies, tend to evolve over time to more aggressive forms of disease, and it is common for myelodysplastic and myeloproliferative neoplasms to “transform” to acute myeloid leukemia. In one of the most important myeloproliferative neoplasms, chronic myeloid leukemia, transformation to ALL is also seen, indicating that it originates from a transformed pluripotent HSC.

As a final twist, many older adults with normal blood counts have clonal hematopoiesis, a precursor of myeloid neoplasms and other hematologic malignancies as well. Thus, we will first discuss clonal hematopoiesis before proceeding to the various myeloid neoplasms.

Clonal Hematopoiesis

Clonal hematopoiesis is defined by the presence of a clone of hematopoietic cells containing at least one acquired pathogenic mutation in a cancer gene in an individual with normal blood counts. Using the currently accepted (somewhat arbitrary) standard, a mutated clone must represent at least 2% of the total sequencing “reads” in DNA

extracted from marrow or blood cells for a diagnosis of clonal hematopoiesis to be made. It is rare before the age of 50 years but then steadily increases in incidence. By age 70 years, approximately 10% of individuals are affected, and by age 90 years, the incidence approaches 30%.

Pathogenesis

The mutations associated with clonal hematopoiesis are usually the result of spontaneous deamination of cytosine or methyl-cytosine, mutations that occur in all cells, including hematopoietic stem cells, throughout life at a constant rate. Most are harmless, but by chance some may alter the function of an oncogene or tumor suppressor gene, giving the affected stem cell a growth or survival advantage. The most commonly mutated genes in clonal hematopoiesis are *TET2* and *DNMT3A*, tumor suppressor genes that regulate DNA methylation, an epigenetic modification that influences gene expression and differentiation (Chapter 1).

Overall, individuals with clonal hematopoiesis have approximately a 1% chance per year of developing a hematopoietic neoplasm, but those with large clones (20% or more of marrow cells) or mutations in particular cancer genes, especially those encoding splicing factors, are at substantially higher risk. Myeloid neoplasms are most frequent, but T-cell and B-cell tumors also occur at increased frequencies relative to individuals without clonal hematopoiesis. In general, the genomes of these neoplasms bear the same mutation(s) seen in the prior clonal hematopoiesis, confirming a precursor-product relationship.

In addition to an association with hematologic malignancies, clonal hematopoiesis produces a proinflammatory state that increases the severity of several common chronic inflammatory diseases, including atherosclerosis, nonalcoholic liver disease, and chronic obstructive pulmonary disease. The association with inflammatory disease is particularly strong in clonal hematopoiesis associated with *TET2* mutations, which appear to lead to epigenetic reprogramming of monocytes and macrophages that in turn makes these cells excessively sensitive to

proinflammatory stimuli. In line with this idea, drugs and antibodies that inhibit proinflammatory IL-1 signaling and the activity of the inflammatory are protective against tissue damage in animal models of clonal hematopoiesis.

Clinical Features

By definition, patients have normal blood counts and lack symptoms related to hematologic abnormalities at diagnosis. Despite this, patients with clonal hematopoiesis are at increased risk of noncancer related mortality due to cardiovascular disease and other chronic inflammatory disorders. One laboratory finding associated with clonal hematopoiesis is an increased red cell distribution width (RDW); the basis for this association is uncertain, but an elevated RDW is also an independent predictor of increased mortality. Now that DNA sequencing of cancers has become routine in many centers, clonal hematopoiesis is frequently diagnosed when mutations in *TET2* or *DNMT3A* are identified due to “contamination” of solid tumors such as lung cancer with blood cells.

Much work is now focused on developing models that predict which individuals with clonal hematopoiesis are at greatest risk for clinical complications, who may then become candidates for preventive treatment with antiinflammatory agents.

Acute Myeloid Leukemia

Acute myeloid leukemia (AML) is a tumor of hematopoietic progenitors caused by acquired oncogenic mutations that impede differentiation, leading to the accumulation of immature myeloid blasts in the marrow. The replacement of the marrow with blasts produces marrow failure and complications related to anemia, thrombocytopenia, and

neutropenia. AML occurs at all ages, but the incidence rises throughout life, peaking after 60 years of age. There are about 13,000 new cases each year in the United States.

Classification

AML is genetically heterogeneous, reflecting the complexities of myeloid cell differentiation. The current WHO classification subdivides AML into two categories (Table 13.10), AML associated with specific genetic aberrations, and AML defined by differentiation (tumors in which specific genetic aberrations are absent). This represents a further shift toward using genetics to classify AML, which, given the increasing importance of cytogenetic and molecular features in directing therapy, is both inevitable and desirable.

Pathogenesis

Driver mutations in AML tend to fall into four functional categories:

- *Transcription factor mutations that interfere with normal myeloid differentiation.* For example, the two most common chromosomal rearrangements, t(8;21) and inv(16), disrupt the *RUNX1* and *CBFB* genes, respectively. These two genes encode polypeptides that bind one another to form a *RUNX1/CBFB* transcription factor that is required for normal hematopoiesis. The t(8;21) and the inv(16) create chimeric genes encoding fusion proteins that interfere with the function of *RUNX1/CBFB* and block the maturation of myeloid cells. Another important example is found in acute promyelocytic leukemia, a distinctive subtype of AML associated with the t(15;17). The t(15;17) creates a fusion gene encoding a chimeric protein consisting of the retinoic acid receptor- α (*RAR* α) and a portion of a protein called *PML*. As discussed in Chapter 7, this fusion protein

TABLE 13.10 Major Subtypes of AML in the World Health Organization Classification

Class	Prognosis	Morphology/Comments
I. AML With Specific Genetic Aberrations (Major Subtypes)		
AML with t(8;21)(q22;q22); <i>RUNX1::RUNX1T1</i> fusion gene	Favorable	Full range of myelocytic maturation; Auer rods easily found; abnormal cytoplasmic granules
AML with inv(16)(p13;q22); <i>CBFB::MYH11</i> fusion gene	Favorable	Myelocytic and monocytic differentiation; abnormal eosinophilic precursors with abnormal basophilic granules
AML with t(15;17)(q22;11-12); <i>PML::RARA</i> fusion gene	Very favorable	Numerous Auer rods, often in bundles within individual progranulocytes; primary granules usually very prominent, but inconspicuous in microgranular variant; high incidence of DIC
AML with t(11q23;q); <i>KMT2A</i> rearrangement	Poor	Usually some degree of monocytic differentiation
AML with mutated <i>NPM1</i>	Favorable	Detected by DNA sequencing
AML with myelodysplasia-related genetic abnormalities	Poor	Associated with deletion of chromosome 5q or 7q, or mutations in the genes <i>SRSF2</i> , <i>SF3B1</i> , <i>U2AF1</i> , <i>ZRSR2</i> , <i>ASXL1</i> , <i>EZH2</i> , <i>BCOR</i> , or <i>STAG2</i>
II. AML Defined by Differentiation		
AML, minimally differentiated	Intermediate	Negative for myeloperoxidase; myeloid antigens detected on blasts by flow cytometry
AML without maturation	Intermediate	>3% of blasts positive for myeloperoxidase
AML with maturation	Intermediate	Full range of myelocytic maturation
AML with myelomonocytic maturation	Intermediate	Myelocytic and monocytic differentiation
AML with monocytic maturation	Intermediate	Nonspecific esterase-positive monoblasts and promonocytes predominate in marrow; may see monoblasts or mature monocytes in the blood
AML with erythroid maturation	Intermediate	Erythroid/myeloid subtype defined by >50% dysplastic maturing erythroid precursors and >20% myeloblasts; pure erythroid subtype defined by >80% erythroid precursors without myeloblasts
AML with megakaryocytic maturation	Intermediate	Blasts of megakaryocytic lineage predominate; detected with antibodies against megakaryocyte-specific markers (e.g., GPIIb/IIIa); often associated with marrow fibrosis; most common AML in Down syndrome

AML, Acute myeloid leukemia; DIC, disseminated intravascular coagulation; GPIIb/IIIa, glycoprotein IIb/IIIa.

interferes with the terminal differentiation of granulocytes, an effect that can be overcome by targeting of the fusion protein with all-*trans*-retinoic acid and arsenic trioxide.

- **Mutation of signaling proteins that result in constitutive activation of progrowth/survival pathways.** For example, AMLs with the t(15;17) sometimes also have activating mutations in FLT3, a receptor tyrosine kinase that transmits signals that mimic normal growth factor signaling, thereby increasing cellular proliferation and survival. The combination of PML-RAR α and activated FLT3 is a potent inducer of AML in mice, whereas expression of PML-RAR α alone is only weakly leukemogenic. Mutations in several other genes involved in progrowth signaling, such as RAS, also occur in subsets of AML.
- **Mutation of genes that regulate or maintain the “epigenome.”** Some of these mutations lead to abnormal DNA methylation patterns or involve members of the cohesin family, proteins that regulate the three-dimensional organization of chromatin in the nucleus. Another group of mutations involving the enzymes IDH1 or IDH2 result in the acquisition of a new enzymatic activity that produces the oncometabolite 2-hydroxyglutarate (described in Chapter 7). IDH inhibitors are effective in treating IDH-mutated forms of AML. The precise mechanism by which epigenetic disturbances contribute to the development of AML remains uncertain, but presumably they lead to alterations in gene expression that contribute to the acquisition of one or more cancer hallmarks.
- **Mutation of TP53 or genes that regulate p53.** With increasingly routine sequencing of AML genomes, it has become apparent that AMLs with mutations that impair p53 function have distinctive clinicopathologic features, including associations with complex karyotype, erythroid differentiation, marked dysplasia, and particularly poor prognosis due to resistance to standard therapies.

MORPHOLOGY

All forms of AML are associated with the presence of increased myeloid blasts in the bone marrow. Generally, the number of blasts is 20% or greater, but the WHO classification permits the diagnosis of AML with fewer than 20% blasts if any of the defining genetic aberrations listed in Table 13.10 are present. Several types of myeloid blasts are recognized, and individual tumors may have more than one type of blast or blasts with hybrid features. **Myeloblasts** have delicate nuclear chromatin, two to four nucleoli, and more voluminous cytoplasm than lymphoblasts (Fig. 13.31A). The cytoplasm often contains fine, peroxidase-positive azurophilic granules. **Auer rods**, distinctive needle-like azurophilic granules, are present in many cases; they are particularly numerous in AML with the t(15;17) (acute promyelocytic leukemia) (Fig. 13.32A). **Monoblasts** (Fig. 13.32B) have folded or lobulated nuclei, lack Auer rods, and are nonspecific esterase-positive. In some AMLs, blasts show megakaryocytic differentiation, which is often accompanied by marrow fibrosis caused by the release of fibrogenic cytokines. Rarely, the blasts of AML show erythroid differentiation, a feature that is commonly associated with TP53 mutations.

The number of leukemic cells in the blood is highly variable. Blasts may be more than 100,000/ μ L but are under 10,000/ μ L in about 50% of patients.

Occasionally, blasts are entirely absent from the blood (aleukemic leukemia). For this reason, a bone marrow examination is essential to exclude acute leukemia in pancytopenic patients.

Immunophenotype

Because it can be difficult to distinguish myeloblasts and lymphoblasts morphologically, the diagnosis of AML is confirmed by performing stains for myeloid-specific antigens (Fig. 13.31B–C).

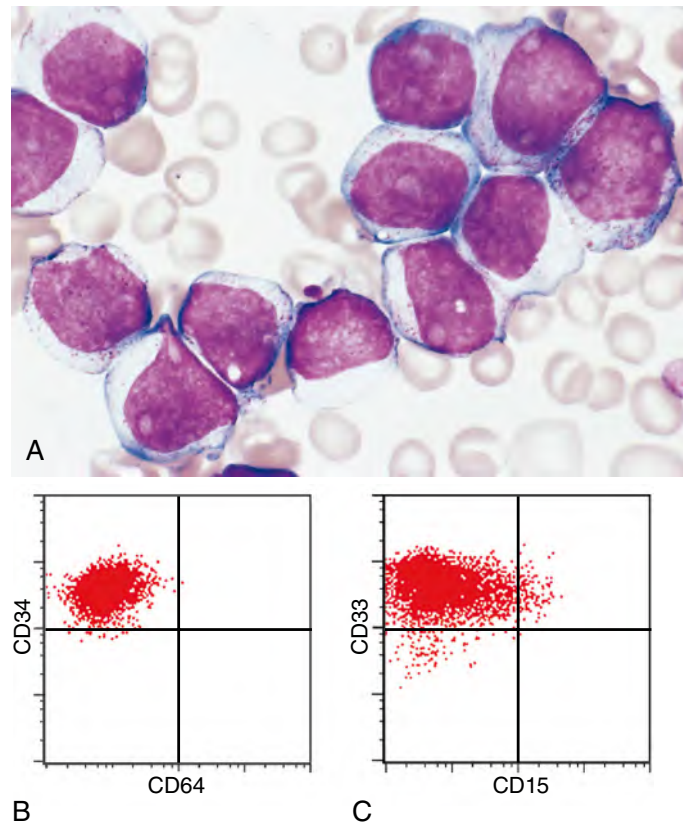


Fig. 13.31 (A) Acute myeloid leukemia. Myeloblasts have delicate nuclear chromatin, prominent nucleoli, and fine azurophilic granules in the cytoplasm. (B) In the flow cytometric analysis shown, myeloid blasts, represented by the *red dots*, express CD34, a marker of multipotent stem cells, but do not express CD64, a marker of mature myeloid cells. (C) The same myeloid blasts express CD33, a marker of immature myeloid cells, and a subset express CD15, a marker of more mature myeloid cells. Thus, these blasts are myeloid cells showing limited maturation. (A, Courtesy Dr. Robert W. McKenna Department of Pathology, University of Texas Southwestern Medical School, Dallas. B–C, courtesy Dr. Louis Picker, Oregon Health Science Center, Portland.)

Cytogenetics and Genetics

Cytogenetic analysis has a central role in the classification of AML. Chromosomal aberrations are detected in 50% to 70% of cases with standard techniques and in approximately 90% of cases using special high-resolution banding. The clinical setting correlates with certain chromosomal and genetic abnormalities. AML arising *de novo* in younger adults is commonly associated with balanced chromosomal translocations, particularly t(8;21), inv(16), and t(15;17). In contrast, AML following exposure to most DNA-damaging agents (such as chemotherapy or radiation therapy), regardless of age, often has deletions or monosomies involving chromosomes 5 and 7 and has a high incidence of mutations affecting the function of p53. The exception to this rule is AML occurring after treatment with topoisomerase II inhibitors, which is strongly associated with translocations involving the *KTM2A* gene on chromosome 11q23. AML in older adults tends to be associated with mutations in genes that are also frequently mutated in clonal hematopoiesis and myelodysplastic neoplasms.

Clinical Features

Most patients present within weeks or a few months of the onset of symptoms with complaints related to anemia, neutropenia, and

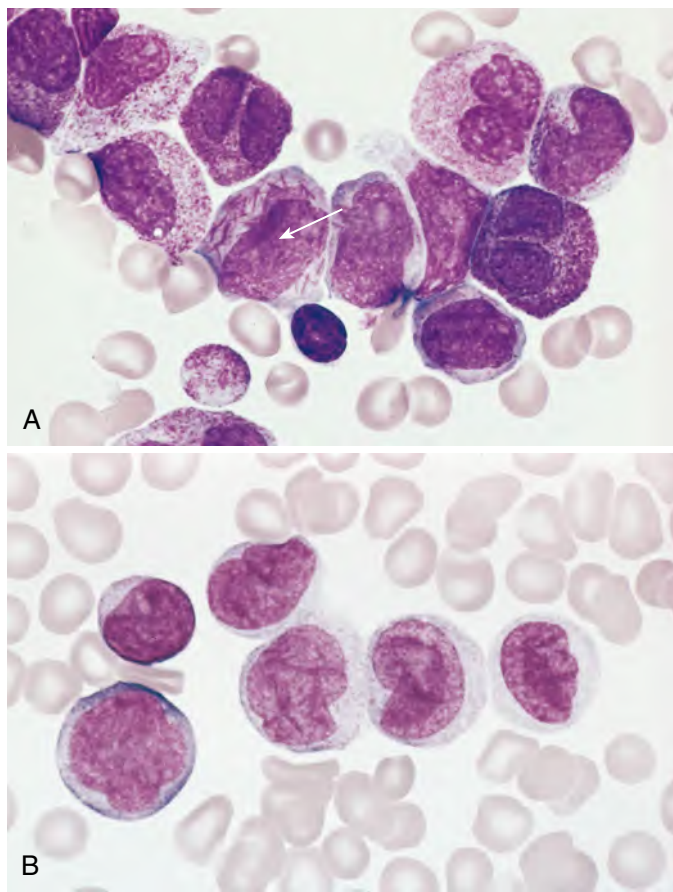


Fig. 13.32 Acute myeloid leukemia subtypes. (A) Acute promyelocytic leukemia with the t(15;17). Bone marrow aspirate shows neoplastic promyelocytes with abnormally coarse and numerous azurophilic granules. Other characteristic findings include the presence of several cells with bilobed nuclei and a cell that contains multiple needle-like Auer rods (arrow). (B) Acute myeloid leukemia with monocytic differentiation. Peripheral smear shows one monoblast and five promonocytes with folded nuclear membranes. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

thrombocytopenia, most notably fatigue, fever, and spontaneous mucosal and cutaneous bleeding. You will remember that these findings are very similar to those produced by ALL because both suppress production of mature red cells, white cells, and platelets. Thrombocytopenia results in abnormal bleeding, which often is prominent. Cutaneous petechiae and ecchymoses, serosal hemorrhages into the linings of the body cavities and viscera, and mucosal hemorrhages into the gingivae and urinary tract are common. Procoagulants and fibrinolytic factors released by leukemic cells, especially in AML with the t(15;17), exacerbate the bleeding tendency. Infections are frequent, particularly in the oral cavity, skin, lungs, kidneys, urinary bladder, and colon, and are often caused by opportunists such as fungi, *Pseudomonas*, and commensal organisms.

Signs and symptoms related to involvement of tissues other than the marrow are usually less striking in AML than in ALL, but tumors with monocytic differentiation often infiltrate the skin (leukemia cutis) and the gingiva; this probably reflects the normal tendency of monocytes to extravasate into tissues. Central nervous system spread is less common than in ALL. AML occasionally presents as a localized soft tissue mass known as a myeloid sarcoma. Without systemic treatment, such tumors inevitably progress to full-blown AML over time.

The overall prognosis is guarded, as AML is a difficult disease to treat; about 60% of patients achieve complete remission with chemotherapy, but only 15% to 30% remain free of disease for 5 years. However, the outcome varies markedly among different molecular subtypes. With targeted therapy using all-*trans*-retinoic acid and arsenic trioxide (described in Chapter 7), AMLs with the t(15;17) have the best prognosis of any type, being curable in more than 90% of patients. AMLs with t(8;21) or inv(16) also have a relatively good prognosis with conventional chemotherapy. By contrast, the prognosis is much worse for patients with AML who are over 60 years of age, or who develop disease following myelodysplasia or genotoxic therapy, in large part because “good” genetic subtypes are rare in these settings and because those that follow genotoxic therapy are often associated with *TP53* mutations. These “high-risk” forms of AML (as well as relapsed AML of all types) are treated with HSC transplantation when possible. Encouragingly, newer therapies have emerged from insights gained from DNA sequencing; one success story is the deployment of inhibitors of mutated forms of *IDH*, which often produce excellent responses in patients with *IDH*-mutated AML, and a second involves the use of inhibitors of the *FLT3* tyrosine kinase in AMLs with activating mutations in *FLT3*. Histone demethylating agents and *BCL2* inhibitors are relatively “gentle” forms of treatment that often produce responses, albeit usually transient, in older adults.

Myelodysplastic Neoplasms

The term *myelodysplastic neoplasm* refers to a group of clonal stem cell disorders characterized by maturation defects that are associated with ineffective hematopoiesis and a high risk of transformation to AML. Because the neoplastic nature of these disorders was not clear when they were initially described, they were formerly called *myelodysplastic syndromes*, a name that is still widely used. In myelodysplastic neoplasms (MDS), the bone marrow is partly or wholly replaced by the clonal progeny of a neoplastic multipotent stem cell that retains the capacity to differentiate but does so in an ineffective and disordered fashion. Because the “birthing process” of blood cells is defective in MDS, patients have peripheral blood cytopenias of varying severity.

MDS may be either primary (idiopathic) or secondary to previous genotoxic drug or radiation therapy (t-MDS). t-MDS usually appears from 2 to 8 years after the genotoxic exposure. All forms of MDS can transform to AML, but transformation occurs with highest frequency and most rapidly in t-MDS. Although characteristic morphologic changes are typically seen in the marrow and the peripheral blood, the diagnosis frequently requires correlation with other laboratory tests. Cytogenetic analysis is often helpful, since certain chromosomal aberrations (discussed later) are often observed, and DNA sequencing is now routinely used to help establish the diagnosis.

Pathogenesis

MDS is associated with driver mutations that partially overlap with those seen in AML, which is unsurprising given that MDN often evolves to AML. The affected proteins can be lumped into three major functional categories:

- **Epigenetic factors.** Frequent mutations are seen involving many of the same epigenetic factors that are mutated in AML, including factors that regulate DNA methylation, histone modifications, and chromatin looping; thus like AML, dysregulation of the epigenome appears to be important in the genesis of MDS.
- **RNA splicing factors.** A subset of tumors has mutations involving components of the 3' end of the RNA splicing machinery. Precisely how these mutations contribute to the development of MDS has yet to be determined. Notably, patients with clonal hematopoiesis associated with mutations in splicing factor genes are at particularly high risk for progression to MDS.

- **Transcription factors.** These mutations affect transcription factors that are required for normal myelopoiesis and may contribute to the deranged differentiation that characterizes MDS. Common chromosomal rearrangements seen in de novo AML, such as the t(8;21), inv(16), and t(15;17), are not seen in MDS; instead, most of the mutations are loss-of-function mutations in genes such as *RUNX1*.

In addition, roughly 10% of MDS cases have loss-of-function mutations in *TP53*, which as in AML, correlate with the presence of a complex karyotype and particularly poor clinical outcomes. Both primary MDS and t-MDS are associated with similar recurrent chromosomal abnormalities, including monosomies 5 and 7; deletions of 5q, 7q, and 20q; and trisomy 8. As with aneuploidy in other cancers, how these aberrations contribute to MDS is incompletely understood. One idea is that the gain or loss of single copies of key genes is sufficient to give cells a growth advantage and that aneuploidy is one way to achieve this result. For example, subtle increases in the notorious oncoprotein MYC is sufficient to stimulate cell growth. The *MYC* gene is located on chromosome 8, and trisomy 8 (and therefore *MYC* copy number gain) is one of the most common forms of aneuploidy in a wide range of myeloid tumors. Similarly, the region that is commonly lost on chromosome 5q contains a gene encoding the ribosomal protein RPS14. In experimental systems, loss of one copy of *RPS14* produces ineffective erythropoiesis, one of the hallmarks of MDS.

MORPHOLOGY

Although the marrow is usually hypercellular at diagnosis, it is sometimes normocellular or, less commonly, hypocellular. The most characteristic finding is disordered (dysplastic) differentiation affecting the erythroid, granulocytic, monocytic, and megakaryocytic lineages to varying degrees (Fig. 13.33). Within the erythroid series, common abnormalities include **ring sideroblasts**, erythroblasts with iron-laden mitochondria visible as perinuclear granules in Prussian blue–stained aspirates or biopsies; **megaloblastoid maturation**, resembling that seen in vitamin B₁₂ and folate deficiency (Chapter 14); and **nuclear budding abnormalities**, recognized as nuclei with misshapen, often polyploid, outlines. Neutrophils frequently contain decreased numbers of secondary granules, toxic granulations, and/or Döhle bodies. **Pseudo–Pelger-Hüet cells**, neutrophils with only two nuclear lobes, are commonly observed, and neutrophils are seen occasionally that completely lack nuclear segmentation. Megakaryocytes with single nuclear lobes or multiple separate nuclei (**pawn ball megakaryocytes**) are also characteristic. **Myeloid blasts** may be increased but make up less than 20% of the overall marrow cellularity. The blood often contains pseudo–Pelger-Hüet cells, giant platelets, macrocytes, and poikilocytes, accompanied by a relative or absolute monocytosis. Myeloid blasts usually make up less than 10% of the leukocytes in the blood.

Clinical Features

Primary MDS is predominantly a disease of older adults; the mean age of onset is 70 years. In up to half of the cases, it is discovered incidentally

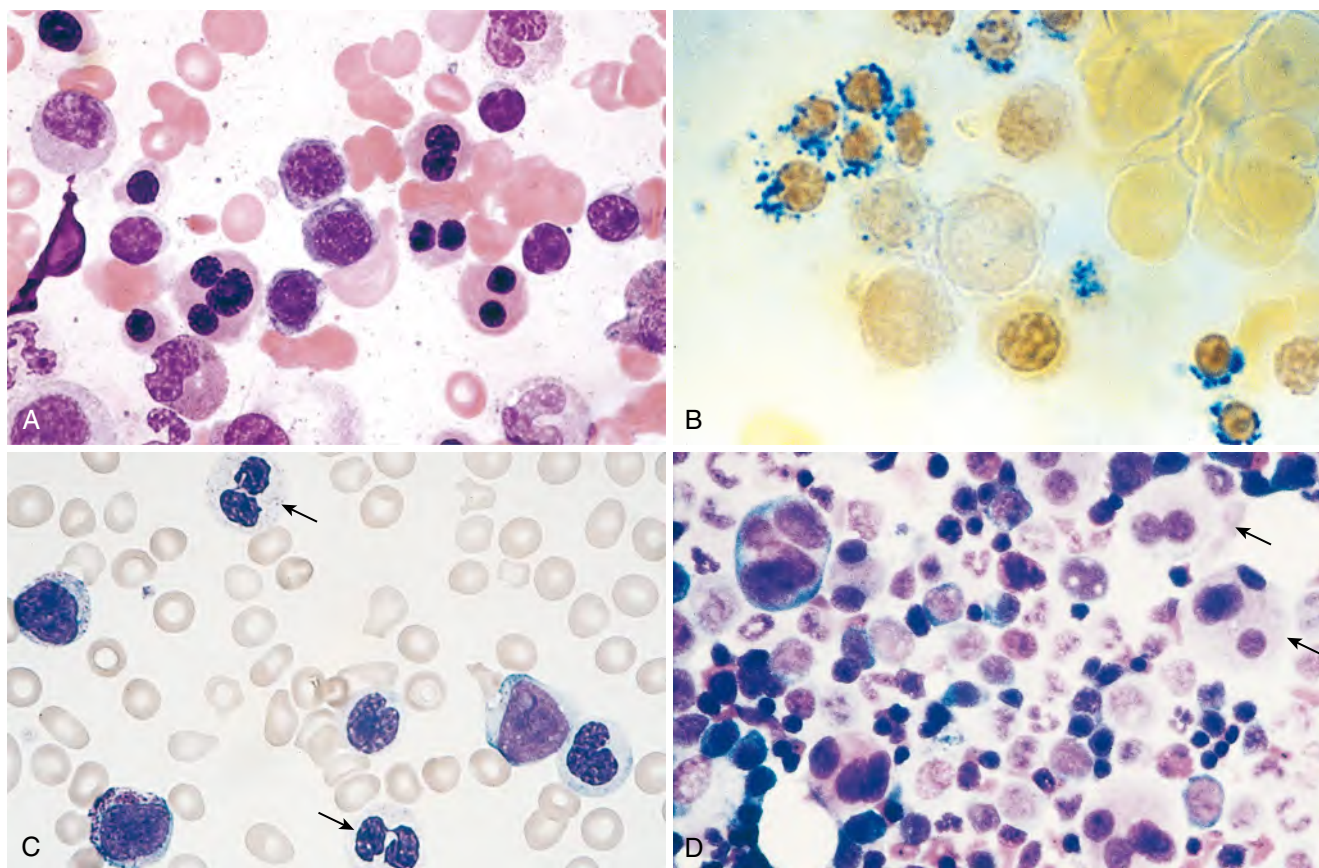


Fig. 13.33 Myelodysplastic neoplasms. Characteristic forms of dysplasia are shown. (A) Nucleated red cell progenitors with multilobated or multiple nuclei. (B) Ringed sideroblasts, erythroid progenitors with iron-laden mitochondria seen as blue perinuclear granules (Prussian blue stain). (C) Pseudo–Pelger-Hüet cells (arrows), neutrophils with only two nuclear lobes instead of the normal three to five (arrows). (D) Megakaryocytes with multiple nuclei instead of the normal single multilobated nucleus (arrows). A, B, and D, Marrow aspirates; C, peripheral blood smear.

on routine blood testing. When symptomatic, it presents with weakness, infections, and hemorrhages, all due to pancytopenia.

Like AML, in the WHO classification primary MDS is divided into two major classes, one defined by specific genetic abnormalities and the second by the presence of certain morphologic features. Each class contains several subtypes, details of which are beyond our scope. Several prognostic scoring systems have been developed. In each, worse outcomes are (understandably) predicted by higher blast count, more severe cytopenias, and complex cytogenetics, defined by the presence of multiple chromosomal abnormalities.

Depending on the prognostic group, the median survival in primary MDS varies from less than 6 months to greater than 5 years. Progression to AML is more likely and occurs more rapidly in poor prognosis groups and is often accompanied by the appearance of additional cytogenetic abnormalities and driver mutations. Patients frequently succumb to the complications of thrombocytopenia (bleeding) and neutropenia (infection). The outlook is even grimmer in t-MDS, which has a median survival of only 4 to 8 months and often progresses to AML within 2 to 3 months of diagnosis.

Treatment options vary. In relatively young, healthy patients, allogeneic HSC transplantation can reconstitute normal hematopoiesis and may be curative. Older patients with MDS and multiple infirmities may be treated supportively with antibiotics and blood product transfusions. Drugs that lead to DNA hypomethylation (hypomethylating agents) improve the effectiveness of hematopoiesis and peripheral blood counts in a subset of patients. The presence of isolated 5q deletion is correlated with response to thalidomide-like drugs.

Myeloproliferative Neoplasms

The common pathogenic feature of myeloproliferative neoplasms is the presence of mutated, constitutively activated tyrosine kinases or other acquired aberrations in signaling pathways that lead to growth factor independence. Hematopoietic growth factors act on normal progenitors by binding to surface receptors and activating tyrosine kinases, which turn on pathways that promote growth and survival (Chapter 7). The mutated tyrosine kinases found in myeloproliferative neoplasms circumvent normal controls and lead to growth factor-independent proliferation and survival of marrow progenitors. Because the tyrosine kinase mutations underlying the various myeloproliferative neoplasms do not impair differentiation, the most common consequence

is an increase in the production of one or more mature blood elements. Most myeloproliferative neoplasms originate in multipotent myeloid progenitors, whereas others arise in pluripotent stem cells that give rise to both lymphoid and myeloid cells.

There is a considerable degree of clinical and morphologic overlap among myeloproliferative neoplasms. The common features include:

- Increased proliferative drive in the bone marrow
- Variable homing of the neoplastic stem cells to secondary hematopoietic organs, producing *extramedullary hematopoiesis*
- Variable transformation to a spent phase characterized by *marrow fibrosis* and peripheral blood *cytopenias*
- Variable transformation to *acute leukemia*

Certain myeloproliferative neoplasms are strongly associated with activating mutations in specific tyrosine kinases. This insight and the availability of kinase inhibitors have increased the importance of molecular tests for tyrosine kinase mutations, both for purposes of diagnosis and the selection of therapy. This discussion is confined to the more common myeloproliferative neoplasms, which are classified based on clinical, laboratory, and molecular criteria. Systemic mastocytosis, a distinctive neoplasm that is associated with mutations in the KIT tyrosine kinase, is discussed under disorders of the skin (Chapter 25). The association of various myeloproliferative neoplasms with specific tyrosine kinase mutations (including rare disorders not discussed here) is summarized in Table 13.11.

Chronic Myeloid Leukemia

Chronic myeloid leukemia (CML) is distinguished from other myeloproliferative neoplasms by the presence of a chimeric *BCR::ABL* gene derived from portions of the *BCR* gene on chromosome 22 and the *ABL* gene on chromosome 9. *BCR::ABL* directs the synthesis of a constitutively active BCR-ABL tyrosine kinase, which in CML is usually 210 kDa in size. In more than 90% of cases, *BCR::ABL* is created by a reciprocal (9;22)(q34;q11) translocation (the so-called *Philadelphia chromosome* [Ph]). In the remaining cases the *BCR::ABL* fusion gene is formed by cytogenetically complex or cryptic rearrangements and must be detected by other methods, such as fluorescence in situ hybridization or polymerase chain reaction (PCR)-based tests. The cell of origin is a pluripotent HSC.

TABLE 13.11 Mutations That Activate Tyrosine Kinase Signaling in Myeloproliferative Neoplasms

Disorder	Mutation	Frequency ^a	Consequences
Chronic myeloid leukemia	<i>BCR::ABL</i> fusion gene	100%	Constitutive ABL kinase activation
Polycythemia vera	<i>JAK2</i> mutations	>95%	Constitutive <i>JAK2</i> kinase activation
Essential thrombocythemia	<i>JAK2</i> mutations <i>CALR</i> mutations <i>MPL</i> mutations	50%–60% 25%–35% 5%–10%	Constitutive <i>JAK2</i> kinase activation Alternative MPL ligand Constitutive MPL kinase activation
Primary myelofibrosis	<i>JAK2</i> mutations <i>CALR</i> mutations <i>MPL</i> mutations	50%–60% 25%–35% 5%–10%	Constitutive <i>JAK2</i> kinase activation Alternative MPL ligand Constitutive MPL kinase activation
Systemic mastocytosis	<i>KIT</i> mutations	>90%	Constitutive KIT kinase activation
Chronic eosinophilic leukemia	<i>FIP1L1::PDGFRα</i> fusion gene <i>PDE4DIP::PDGFRβ</i> fusion gene	Common Rare	Constitutive PDGFRα kinase activation Constitutive PDGFRβ kinase activation
Myeloid/lymphoid neoplasms with eosinophilia and tyrosine kinase gene mutations ^b	Various fusion genes involving <i>FGFR1</i> , <i>PDGFRα</i> , <i>PDGFRβ</i> , or <i>JAK2</i>	100%	Constitutive tyrosine kinase activation

^aRefers to frequency within a diagnostic category.

^bRare disorder originating in pluripotent hematopoietic stem cells that presents with concomitant myeloproliferative neoplasm and lymphoblastic leukemia/lymphoma.

Pathogenesis

Tyrosine kinases are normally regulated by ligand-mediated dimerization and autophosphorylation, which creates an activated kinase capable of phosphorylating other protein substrates (Chapters 1 and 7). The BCR moiety of BCR-ABL contains a dimerization domain that self-associates, leading to the activation of the ABL tyrosine kinase moiety (Fig. 13.34). The ABL kinase in turn phosphorylates proteins that induce signaling through the same progrowth and prosurvival pathways that are turned on by hematopoietic growth factors, including the RAS and JAK/STAT pathways. For unknown reasons, BCR-ABL preferentially drives the proliferation of granulocytic and megakaryocytic progenitors and also causes the abnormal release of immature granulocytic forms from the marrow into the blood.

MORPHOLOGY

The marrow is markedly **hypercellular** because of massively increased numbers of maturing granulocytic precursors, which usually include an elevated proportion of eosinophils and basophils. Megakaryocytes are also increased and usually include small, dysplastic forms. Erythroid progenitors are present in normal or mildly decreased numbers. A characteristic finding is the presence of scattered macrophages with abundant wrinkled, green-blue cytoplasm—so-called sea-blue histiocytes. Increased deposition of reticulin is typical, but overt marrow fibrosis is rare. The blood reveals a **leukocytosis, often exceeding 100,000 cells/ μ L** (Fig. 13.35), which consists predominantly of neutrophils, band forms, metamyelocytes, myelocytes, eosinophils, and basophils. Blasts usually make up less than 10% of the circulating cells. Platelets are also usually increased, sometimes markedly. The spleen is often greatly enlarged owing to extensive extramedullary hematopoiesis (Fig. 13.36) and often contains infarcts of varying age. Extramedullary hematopoiesis may also produce mild hepatomegaly and lymphadenopathy.

Clinical Features

CML is primarily a disease of adults but also occurs in children and adolescents. The peak incidence is in the fifth to sixth decades of life. There are about 4800 new cases per year in the United States.

The onset is insidious. Mild-to-moderate anemia and hypermetabolism due to increased cell turnover lead to fatigability, weakness, weight loss, and anorexia. Sometimes the first symptom is a dragging sensation in the abdomen caused by splenomegaly or the acute onset of left upper quadrant pain due to splenic infarction. CML is best differentiated from other myeloproliferative neoplasms by detection of the *BCR::ABL* fusion gene through either chromosomal analysis or PCR-based tests.

The natural history of untreated disease is one of slow progression; without any treatment, the median survival is about 3 years. After a variable period averaging 3 years, about 50% of patients enter an “accelerated phase” marked by increasing anemia and thrombocytopenia, sometimes accompanied by a rise in the number of basophils in the blood. Additional clonal cytogenetic abnormalities, such as trisomy 8, isochromosome 17q, or duplication of the Ph chromosome, often appear. Within 6 to 12 months, the accelerated phase terminates in a picture resembling acute leukemia (*blast crisis*). In the other 50% of patients, blast crises occur abruptly without an accelerated phase. In 70% of crises, the blasts are of myeloid origin (myeloid blast crisis), whereas in most of the remainder the blasts are of pre-B cell origin (lymphoid blast crisis). This is taken as evidence that CML originates from a pluripotent stem cell with both myeloid and lymphoid potential.

Given that acute leukemias often stem from complementary mutations involving a transcription factor and a tyrosine kinase, it might be predicted that blast crisis would be caused by an acquired mutation in a key transcriptional regulator. This prediction has been realized in

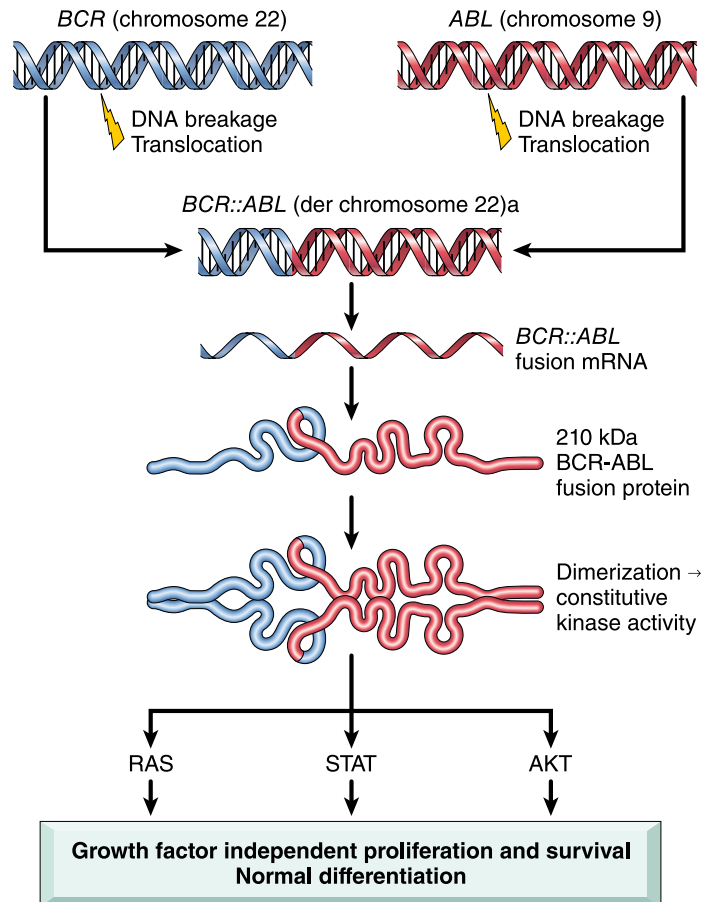


Fig. 13.34 Pathogenesis of chronic myeloid leukemia. Breakage and joining of *BCR* and *ABL* creates a chimeric *BCR::ABL* fusion gene that encodes a constitutively active BCR-ABL tyrosine kinase. BCR-ABL activates multiple downstream pathways, which drive growth factor-independent proliferation and survival of bone marrow progenitors. Because BCR-ABL does not interfere with differentiation, the net result is an increase in mature elements in the peripheral blood, particularly granulocytes and platelets. *der*, Derivative chromosome.

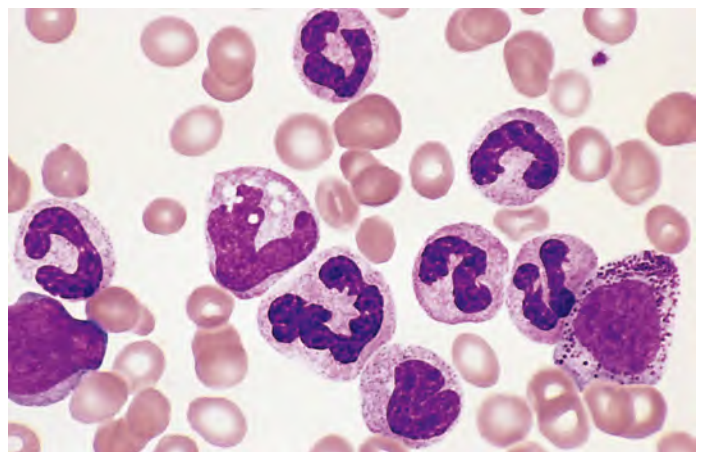


Fig. 13.35 Chronic myeloid leukemia. Peripheral blood smear shows many mature neutrophils, some metamyelocytes, and a myelocyte. (Courtesy Dr. Robert W. McKenna, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)



Fig. 13.36 Chronic myeloid leukemia (spleen). Enlarged spleen (2630 g; normal: 150 to 200 g) with greatly expanded red pulp stemming from neoplastic hematopoiesis. (Courtesy Dr. Daniel Jones, Department of Pathology, Ohio State University, Columbus.)

lymphoid blast crisis, in which 85% of cases are associated with mutations that interfere with the activity of Ikaros, a transcription factor encoded by the gene *IKZF1* that regulates the differentiation of hematopoietic progenitors. The same types of Ikaros mutations are also seen in BCR-ABL–positive B-ALL, suggesting that these two varieties of aggressive leukemia have a similar molecular origin.

Understanding of the pathogenesis of CML has led to the use of drugs that target BCR-ABL. Treatment with BCR-ABL inhibitors results in sustained hematologic remissions in greater than 90% of patients, with generally tolerable side effects. These inhibitors markedly decrease the number of BCR-ABL–positive cells in the marrow and elsewhere, but usually do not extinguish the CML “stem cell,” which persists at low levels. However, this form of targeted therapy controls blood counts and substantially decreases the risk of transformation to the accelerated phase and blast crisis, which is the greatest threat to the patient. It may be that by lowering the proliferative drive of the BCR-ABL–positive progenitors, BCR-ABL inhibitors decrease the rate at which these cells acquire mutations that lead to disease progression.

The other major threat to the patient is the emergence of resistance to first-generation BCR-ABL inhibitors, which in about 50% of cases stems from mutations in BCR-ABL and in the remaining cases mutations in other kinases. This problem has been overcome in part by development of second- and third-generation kinase inhibitors that are active against mutated forms of BCR-ABL. Because it is easy to monitor for evidence of recurrence with very sensitive molecular tests, treatment can be discontinued in a subset of patients with very deep remissions, which may persist off-therapy for at least several years. HSC transplantation is restricted to otherwise healthy, relatively young patients with suboptimal response to tyrosine kinase inhibitors.

Polycythemia Vera

Polycythemia vera (PCV) is characterized by increased marrow production of red cells, granulocytes, and platelets (*panmyelosis*), but it is the increase in red cells (*polycythemia*) that is responsible for most of the clinical symptoms. PCV must be differentiated from relative polycythemia resulting from hemoconcentration and other causes of absolute polycythemia (Chapter 14).

Pathogenesis

In PCV, the transformed progenitor cells have markedly decreased requirements for erythropoietin and other hematopoietic growth factors due to activating mutations in the tyrosine kinase JAK2. JAK2 participates in the JAK/STAT pathway, which lies downstream of multiple hematopoietic growth factor receptors, including the erythropoietin receptor. Because the pathway is constitutively active and red

cell numbers are abnormally high, serum erythropoietin levels in PCV are low. This is in contrast to secondary forms of polycythemia, in which erythropoietin levels are high. The elevated hematocrit leads to increased blood viscosity and sludging. These hemodynamic factors, together with thrombocytosis and abnormal platelet function, make patients with PCV prone to both thrombosis and bleeding.

More than 97% of cases are associated with a mutation in *JAK2* that results in a valine-to-phenylalanine substitution at residue 617; other *JAK2* mutations are found in most (and perhaps all) of the remaining cases. The mutated forms of *JAK2* found in PCV render hematopoietic cell lines growth factor–independent and when expressed in murine bone marrow progenitors cause a PCV-like disease. In 25% to 30% of cases the tumor cells contain two mutated copies of *JAK2*, a genotype that is associated with higher white cell counts, more significant splenomegaly, symptomatic pruritus, and a greater rate of progression to a spent phase marked by marrow fibrosis (described later).

The proliferative drive in PCV (and other myeloproliferative neoplasms associated with *JAK2* mutations) is less than in CML, which is associated with more pronounced marrow hypercellularity, leukocytosis, and splenomegaly. Presumably, *JAK2* signals are quantitatively weaker or qualitatively different from those produced by BCR-ABL (see Fig. 13.34).

MORPHOLOGY

The marrow is hypercellular, but some residual fat is usually present. **The increase in red cell progenitors is subtle and usually accompanied by an increase in granulocytic precursors and megakaryocytes.** At diagnosis, a moderate to marked increase in reticulin fibers is seen in about 10% of marrows. Mild organomegaly is common, being caused early in the course largely by congestion; at this stage extramedullary hematopoiesis is minimal. The peripheral blood often contains increased numbers of basophils and abnormally large platelets.

Late in the course, PCV often progresses to a **spent phase** characterized by extensive marrow fibrosis that displaces hematopoietic cells. This is accompanied by increased extramedullary hematopoiesis in the spleen and liver, often leading to prominent organomegaly (Fig. 13.37). Transformation to AML, with its typical features, occurs in about 1% of patients.

Clinical Features

PCV is uncommon, having an incidence of 1 to 3 per 100,000 people per year in the United States. It appears insidiously, usually in adults



Fig. 13.37 Polycythemia vera, spent phase. Massive splenomegaly (3020 g; normal: 150 to 200 g) largely due to extramedullary hematopoiesis occurring in the setting of advanced marrow myelofibrosis. (Courtesy Dr. Mark Fleming, Department of Pathology, Children's Hospital, Boston.)

of late middle age. Most symptoms are related to the increased red cell mass and hematocrit. Usually, there is also an increased total blood volume. Together, these factors cause abnormal blood flow, particularly on the low-pressure venous side of the circulation, which becomes greatly distended. Patients are plethoric and cyanotic due to stagnation and deoxygenation of blood in peripheral vessels. Headache, dizziness, hypertension, and gastrointestinal symptoms are common. Intense pruritus and peptic ulceration may occur, both possibly resulting from the release of histamine from basophils. High cell turnover gives rise to hyperuricemia; symptomatic secondary gout is seen in 5% to 10% of cases.

More ominously, the abnormal blood flow and platelet function that accompany PCV lead to an increased risk of both major bleeding and thrombotic episodes. About 25% of patients first come to attention due to deep venous thrombosis, myocardial infarction, or stroke. Thromboses sometimes also occur in the hepatic veins (producing *Budd-Chiari syndrome*, Chapter 4) and the portal and mesenteric veins (leading to bowel infarction). It must be remembered that thrombotic complications may precede the appearance of the typical hematologic findings. Minor hemorrhages (epistaxis, bleeding gums) are common, and life-threatening hemorrhages occur in 5% to 10% of cases.

The hemoglobin concentration is typically greater than 16 g/dL, and the hematocrit is often 55% or more. Sometimes chronic bleeding leads to iron deficiency, which can suppress erythropoiesis sufficiently to lower the hematocrit into the normal range, an example of two defects counteracting one another to “correct” a laboratory abnormality. The white cell count ranges from 12,000 to 50,000 cells/mm³, and the platelet count is often greater than 500,000 platelets/mm³. The platelets usually exhibit morphologic abnormalities such as giant forms and are often defective in functional aggregation studies.

Without treatment, death from bleeding or thrombosis occurs within months of diagnosis. However, simply maintaining the red cell mass at nearly normal levels by phlebotomy and treatment with low doses of aspirin extends the median survival to about 10 years. Although JAK2 inhibitors are an obvious targeted therapy for PCV, because simple phlebotomy and aspirin treatment is so effective, use of JAK2 inhibitors is restricted to patients with evidence of progression to the “spent phase” of myelofibrosis. The disease undergoes this transition in about 15% to 20% of patients after an average period of 10 years. It is marked by the appearance of oblitative fibrosis in the bone marrow (myelofibrosis) and extensive extramedullary hematopoiesis, principally in the spleen, which enlarges greatly. The mechanisms underlying the progression to the spent phase are not known.

In about 2% of patients, PCV transforms to AML. Surprisingly, the AML clone often lacks JAK2 mutations, suggesting that the causative JAK2 mutations occur in an abnormal stem cell (possibly a product of clonal hematopoiesis) that already harbors other oncogenic mutations and therefore is “at risk” for giving rise to several different myeloid tumors. Unlike CML, transformation to ALL is rarely observed, consistent with the cell of origin being a progenitor committed to myeloid differentiation.

Essential Thrombocythemia

Essential thrombocythemia (ET) is associated with diverse mutations that increase JAK-STAT signaling and mimic constitutive growth factor receptor signaling. Over 90% of cases have either activating mutations in JAK2 (approximately 50% to 60% of cases); MPL (5% to 10% of cases), a receptor tyrosine kinase that is normally activated by thrombopoietin and that signals through JAK2; or CALR (calreticulin, approximately 30% of cases). Remarkably, the mutated forms of CALR seen in ET are secreted and bind and activate the thrombopoietin receptor, an example of an oncogenic autocrine feedback loop.

ET manifests clinically with elevated platelet counts sometimes accompanied by abnormal thrombosis or bleeding and is separated from PCV and primary myelofibrosis based on the absence of polycythemia and marrow fibrosis, respectively. In cases without tyrosine kinase mutations, causes of reactive thrombocytosis, such as inflammatory disorders and iron deficiency, must be excluded before the diagnosis can be established.

Pathogenesis

Constitutive JAK2 or MPL signaling renders the progenitors thrombopoietin-independent and leads to hyperproliferation. The JAK2 mutation is the same as that found in almost all cases of PCV. Why some patients with JAK2 mutations present with PCV and others with ET is not understood. Some cases thought to be ET may in fact be PCV disguised by iron deficiency (which is more common in individuals diagnosed with ET), but this is probably true of only a small fraction of patients. As mentioned, most cases without JAK2 or MPL mutations have calreticulin mutations instead.

MORPHOLOGY

Bone marrow cellularity is usually only mildly increased, but megakaryocytes are often markedly increased in number and include abnormally large forms. Delicate reticulin fibrils are often seen, but the overt fibrosis of primary myelofibrosis (see later) is absent. Peripheral smears usually reveal abnormally large platelets (Fig. 13.38), often accompanied by mild leukocytosis. Modest degrees of extramedullary hematopoiesis may occur, producing mild organomegaly in about 50% of patients. Uncommonly, a spent phase of marrow fibrosis or transformation to AML supervenes.

Clinical Features

The incidence of ET is 1 to 3 per 100,000 people per year in the United States. It usually occurs past the age of 60 years but may also be seen in young adults. Dysfunctions of platelets derived from the neoplastic clone can lead to thrombosis and hemorrhage, the major clinical manifestations. Platelets are not only increased in number but also frequently demonstrate qualitative abnormalities in functional tests. The types of thrombotic events resemble those observed in PCV; they include deep venous thrombosis, portal and hepatic vein thrombosis, myocardial infarction, and abnormal bleeding. Of interest, up to one-third of patients with ET have *acquired von Willebrand disease* of

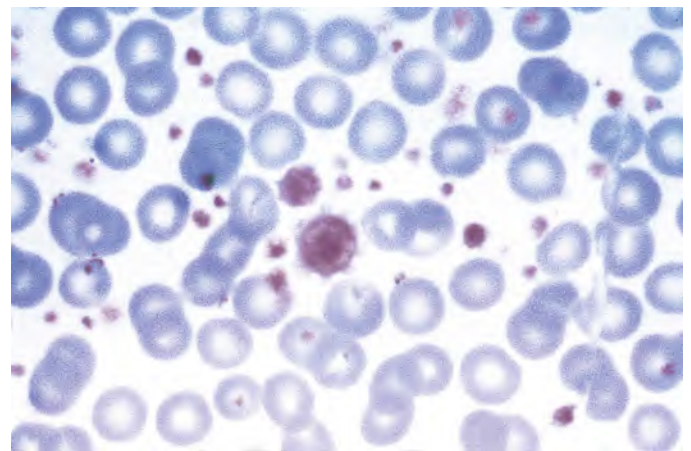


Fig. 13.38 Essential thrombocythemia. Peripheral blood smear shows marked thrombocytosis including giant platelets approximating the size of surrounding red cells.

uncertain etiology that may at least partially explain the risk of bleeding. One characteristic symptom is *erythromelalgia*, a throbbing and burning of hands and feet caused by occlusion of small arterioles by platelet aggregates, which also may be seen in PCV.

ET is an indolent disorder with long asymptomatic periods punctuated by occasional thrombotic or hemorrhagic crises. Median survival times are 12 to 15 years. Thrombotic complications are most likely in patients with very high platelet counts and homozygous *JAK2* mutations. Therapy consists of “gentle” chemotherapeutic agents that suppress thrombopoiesis and (in those without evidence of bleeding) low-dose aspirin.

Primary Myelofibrosis

The hallmark of primary myelofibrosis is the development of obliterative marrow fibrosis. The replacement of the marrow by fibrous tissue reduces bone marrow hematopoiesis, leading to cytopenias and extensive extramedullary hematopoiesis. Histologically, the appearance is identical to the spent phase that occurs late in the course of other myeloproliferative neoplasms. The genetics of primary myelofibrosis is very similar to ET; approximately 90% of cases have activating mutations of *JAK2*, *CALR*, or *MPL*.

Pathogenesis

The chief pathologic feature is the extensive deposition of collagen in the marrow by nonneoplastic fibroblasts. The fibrosis inexorably displaces hematopoietic elements, including stem cells, from the marrow and eventually leads to marrow failure. It is probably caused by the inappropriate release of fibrogenic factors from neoplastic megakaryocytes. Two factors synthesized by megakaryocytes have been implicated: *platelet-derived growth factor* and *TGF-β*. As you recall, platelet-derived growth factor and *TGF-β* are fibroblast mitogens. In addition, *TGF-β* promotes collagen deposition and causes angiogenesis, both of which are observed in myelofibrosis (Chapter 3).

As marrow fibrosis progresses, circulating neoplastic HSCs take up residence in niches in secondary hematopoietic organs, such as the spleen, the liver, and the lymph nodes, leading to the appearance of extramedullary hematopoiesis. For incompletely understood reasons, red cell production at extramedullary sites is disordered. This factor and the concomitant suppression of marrow function result in moderate to severe anemia. It is not clear whether primary myelofibrosis is truly distinct from PCV and ET or merely reflects unusually rapid progression to the spent phase.

MORPHOLOGY

Early in the course, the marrow is often hypercellular due to increases in maturing cells of all lineages, a feature reminiscent of PCV. Morphologically, the erythroid and granulocytic precursors appear normal, but megakaryocytes are large, have unusual bulbous, hyperchromatic, or “cloud-like” nuclei, and are abnormally clustered. At this stage, fibrosis is minimal, and the blood may show leukocytosis and thrombocytosis. **With progression, the marrow becomes more hypocellular and diffusely fibrotic.** Clusters of atypical hyperchromatic megakaryocytes with unusual nuclear shapes are seen and hematopoietic elements are often found within dilated sinusoids, a manifestation of severe architectural distortion caused by the fibrosis. Very late in the course, the fibrotic marrow space may be converted into bone, a change called “osteosclerosis.” These features are identical to those seen in the spent phase of other myeloproliferative neoplasms.

Fibrotic obliteration of the marrow space leads to extensive extramedullary hematopoiesis, principally in the spleen, which is usually markedly enlarged, sometimes up to 4000 g. Grossly, such spleens are firm and diffusely red to gray. As in CML, subcapsular infarcts are common (see Fig. 13.42). Initially,

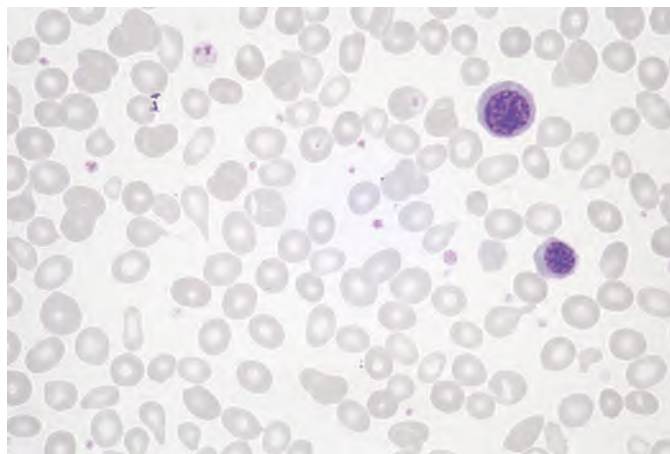


Fig. 13.39 Primary myelofibrosis (peripheral blood smear). Two nucleated erythroid precursors and several teardrop-shaped red cells (dacryocytes) are evident. Immature myeloid cells were present in other fields. An identical picture can be seen in other diseases producing marrow distortion and fibrosis.

extramedullary hematopoiesis is confined to the sinusoids, but later it expands into the cords. The liver may be enlarged moderately by sinusoidal foci of extramedullary hematopoiesis. Hematopoiesis may also appear within lymph nodes, but significant lymphadenopathy is uncommon.

The marrow fibrosis is reflected in several characteristic blood findings (Fig. 13.39). Marrow distortion leads to the premature release of nucleated erythroid and early granulocyte progenitors (**leukoerythroblastosis**), and immature cells also enter the circulation from sites of extramedullary hematopoiesis. **Teardrop-shaped red cells** (dacryocytes), cells that were probably damaged during the birthing process in the fibrotic marrow, are often seen. Although characteristic of primary myelofibrosis, leukoerythroblastosis and teardrop red cells are seen in other infiltrative disorders of the marrow, including granulomatous diseases and metastatic tumors. Other common, albeit nonspecific, blood findings include abnormally large platelets and basophilia.

Clinical Features

Primary myelofibrosis is less common than PCV and ET and usually occurs in individuals older than 60 years of age. Except when preceded by another myeloproliferative neoplasm, it comes to attention because of progressive anemia and splenomegaly, which produces a sensation of fullness in the left upper quadrant. Nonspecific symptoms such as fatigue, weight loss, and night sweats result from an increase in metabolism associated with the expanding mass of hematopoietic cells. Hyperuricemia and secondary gout due to a high rate of cell turnover can complicate the picture.

Laboratory studies typically show a moderate to severe normochromic normocytic anemia accompanied by leukoerythroblastosis. The white cell count is usually normal or mildly reduced but can be markedly elevated (80,000 to 100,000 cells/ μ L) early in the course. The platelet count is usually normal or elevated at the time of diagnosis, but thrombocytopenia may supervene as the disease progresses. These blood findings are not specific; bone marrow biopsy is essential for diagnosis.

Primary myelofibrosis is a much more difficult disease to treat than PCV or ET. The course is variable, but the median survival is in the range of 3 to 5 years. Threats to life include intercurrent infections, thrombotic episodes, bleeding related to platelet abnormalities, and transformation to AML, which occurs in 5% to 20% of cases. When myelofibrosis is extensive, AML sometimes arises at

extramedullary sites, including lymph nodes and soft tissues. JAK2 inhibitors are approved to treat this disease and are effective at decreasing the splenomegaly and constitutional symptoms. HSC transplantation is a treatment option for patients young and fit enough to withstand the procedure.

KEY CONCEPTS

Myeloid Neoplasms

Myeloid tumors occur mainly in adults and fall into three major groups.

Acute Myeloid Leukemia

- Aggressive tumor composed of immature myeloid lineage blasts, which replace the marrow and suppress normal hematopoiesis.
- Associated with diverse acquired mutations that lead to expression of abnormal transcription factors, which interfere with myeloid differentiation.
- Often also associated with mutations in genes encoding growth factor receptor signaling pathway components or regulators of the epigenome.

Myelodysplastic Neoplasms

- Myeloid tumors characterized by disordered and ineffective hematopoiesis and dysmaturation.
- May occur de novo or following genotoxic exposures.
- Frequently harbor mutations in splicing factors, epigenetic regulators, and transcription factors.
- Manifest with one or more cytopenias and often progress to AML.

Myeloproliferative Neoplasms

- Myeloid tumors in which production of formed myeloid elements is initially increased, leading to high blood counts and extramedullary hematopoiesis.
- Commonly associated with acquired mutations that lead to constitutive activation of tyrosine kinases, which mimic signals from normal growth factors. The most common pathogenic kinases are BCR-ABL (associated with CML) and mutated *JAK2* (associated with PCV and primary myelofibrosis).
- All can transform to acute leukemia and to a spent phase of marrow fibrosis associated with anemia, thrombocytopenia, and splenomegaly.

Langerhans Cell Histiocytosis

The term *histiocytosis* is an “umbrella” designation for a variety of proliferative disorders of dendritic cells or macrophages. Some, such as rare “histiocytic” sarcomas and blastic plasmacytoid dendritic cell neoplasms, are highly malignant, whereas others, such as reactive proliferations of macrophages in lymph nodes, are completely benign. Lying between these two extremes are the Langerhans cell histiocytoses, a spectrum of clonal proliferations of a special type of immature dendritic cell called the Langerhans cell. Because Langerhans cells are part of the innate immune system, these tumors (as well as other clonal histiocytoses) can be considered unusual myeloid neoplasms.

Pathogenesis

Most cases of Langerhans cell histiocytosis have driver mutations in cancer-associated genes and are best considered neoplasms, albeit with an unusual proclivity for spontaneous remission. Over 85% of cases have driver mutations involving some component of the RAS signaling pathway. The most common mutation is an activating valine-to-glutamate substitution at residue 600 in *BRAF*, already discussed for its role in hairy cell leukemia, which is present in 55% to 60% of cases, while the next most commonly mutated gene, *MAPK1*, encodes a kinase that lies downstream of *BRAF*.

One factor that contributes to the homing of neoplastic Langerhans cells is the aberrant expression of chemokine receptors. For example, while normal epidermal Langerhans cells express CCR6, their neoplastic

counterparts express both CCR6 and CCR7. This allows the neoplastic cells to migrate into tissues that express the relevant chemokines—CCL20 (a ligand for CCR6) in skin and bone and CCL19 and CCL21 (ligands for CCR7) in lymphoid organs.

MORPHOLOGY

Regardless of the clinical picture, the proliferating Langerhans cells have abundant, often vacuolated cytoplasm and vesicular nuclei containing linear grooves or folds (Fig. 13.40A). The presence of **Birbeck granules** in the cytoplasm is characteristic. Birbeck granules are pentalaminar tubules, often with a dilated terminal end producing a tennis racket–like appearance (Fig. 13.40B), which contain the protein langerin. In addition, the tumor cells also typically express HLA-DR, S-100, and CD1a.

Clinical Features

Langerhans cell histiocytosis presents as several clinicopathologic entities:

- *Multifocal multisystem Langerhans cell histiocytosis (Letterer-Siwe disease)* occurs most frequently before 2 years of age but occasionally

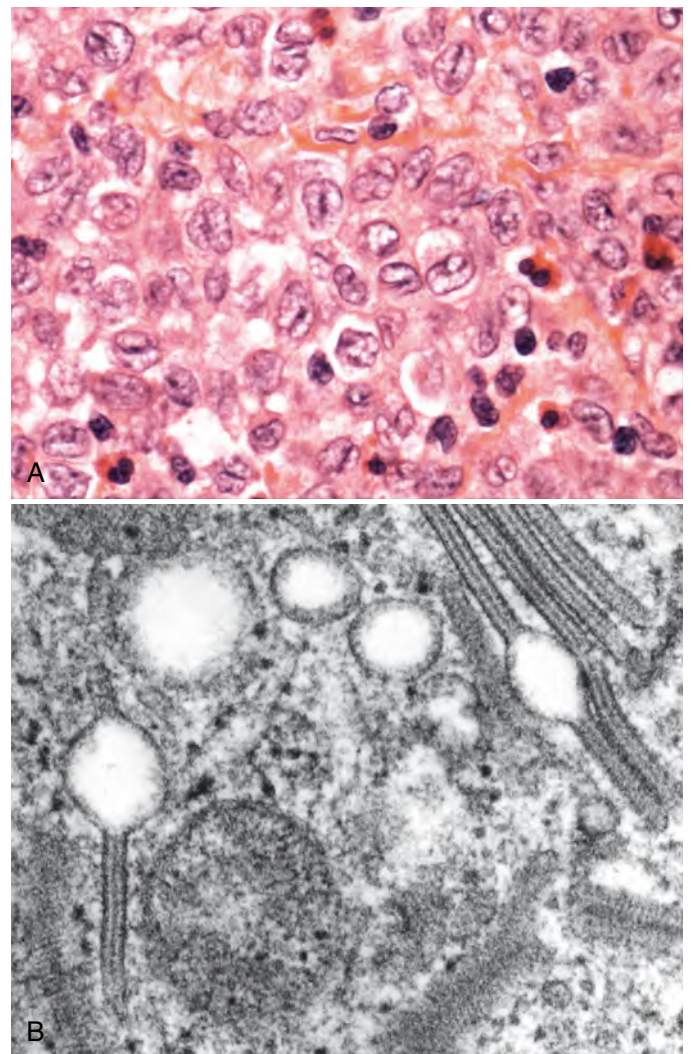


Fig. 13.40 Langerhans cell histiocytosis. (A) Langerhans cells with folded or grooved nuclei and moderately abundant pale cytoplasm are mixed with a few eosinophils. (B) An electron micrograph shows rodlike Birbeck granules with characteristic periodicity and dilated terminal end. (B, Courtesy Dr. George Murphy, Department of Pathology, Brigham and Women's Hospital, Boston.)

affects adults. A dominant clinical feature is the development of cutaneous lesions resembling a seborrheic eruption, which is caused by infiltrates of Langerhans cells over the front and back of the trunk and on the scalp. Most of those affected have concurrent hepatosplenomegaly, lymphadenopathy, pulmonary lesions, and (eventually) destructive osteolytic bone lesions. Extensive infiltration of the marrow often leads to anemia, thrombocytopenia, and a predisposition to recurrent infections, such as otitis media and mastoiditis. In some instances, the proliferating cells are quite anaplastic; such tumors are sometimes referred to as Langerhans cell sarcoma. The course of untreated disease is rapidly fatal. With chemotherapy, approximately 50% of patients survive 5 years.

- *Unifocal and multifocal unisystem Langerhans cell histiocytosis* is characterized by proliferations of Langerhans cells admixed with variable numbers of eosinophils, lymphocytes, plasma cells, and neutrophils. Eosinophils are usually, but not always, a prominent component of the infiltrate. It typically arises within the medullary cavities of bones, most commonly the calvaria, ribs, and femur. Less commonly, unisystem lesions of identical histology arise in the skin, lungs, or stomach. *Unifocal lesions* (previously called *eosinophilic granuloma*) most commonly affect the skeletal system in older children or adults. Bone lesions can be asymptomatic or cause pain, tenderness, and, in some instances, pathologic fractures. Unifocal disease is indolent and may heal spontaneously or may be cured by local excision or irradiation. *Multifocal unisystem disease* usually affects young children (peak onset between 2 and 10 years of age), who present with multiple erosive bony masses that sometimes expand into adjacent soft tissue. Involvement of the posterior pituitary stalk of the hypothalamus leads to diabetes insipidus in about 50% of patients. The combination of calvarial bone defects, diabetes insipidus, and exophthalmos is referred to as the *Hand-Schüller-Christian* triad. Many patients experience spontaneous regression; others can be treated successfully with chemotherapy. BRAF inhibitors are active against *BRAF*-mutated disease.
- *Pulmonary Langerhans cell histiocytosis* represents a special category of disease, most often seen in adult smokers, which may regress spontaneously upon cessation of smoking. These lesions have been described as reactive proliferations of Langerhans cells, but 40% are associated with *BRAF* mutations, suggesting that they too are neoplastic in origin.

SPLEEN

The spleen is an ingeniously designed filter for the blood and a site of immune responses to blood-borne antigens. Normally in the adult it weighs about 150 g and is enclosed within a thin, glistening, slate-gray connective tissue capsule. Its cut surface reveals extensive red pulp dotted with gray specks, which are the white pulp follicles. These consist of an artery with an eccentric collar of T lymphocytes, the so-called periarteriolar lymphatic sheath. At intervals this sheath expands to form lymphoid follicles containing mainly B lymphocytes, which upon antigenic stimulation may proliferate and form germinal centers identical to those seen in lymph nodes (Fig. 13.41).

The red pulp of the spleen is traversed by numerous thin-walled vascular sinusoids, separated by the splenic cords or “cords of Billroth.” The endothelial lining of the sinusoid is discontinuous, providing a passage for blood cells between the sinusoids and cords. The cords contain a labyrinth of macrophages loosely connected through long dendritic processes. As it traverses the red pulp, the blood takes two routes to reach the splenic veins. Some flows through capillaries into the cords, from which blood cells squeeze through gaps in the discontinuous basement of the endothelial lining to reach the sinusoids; this is the so-called

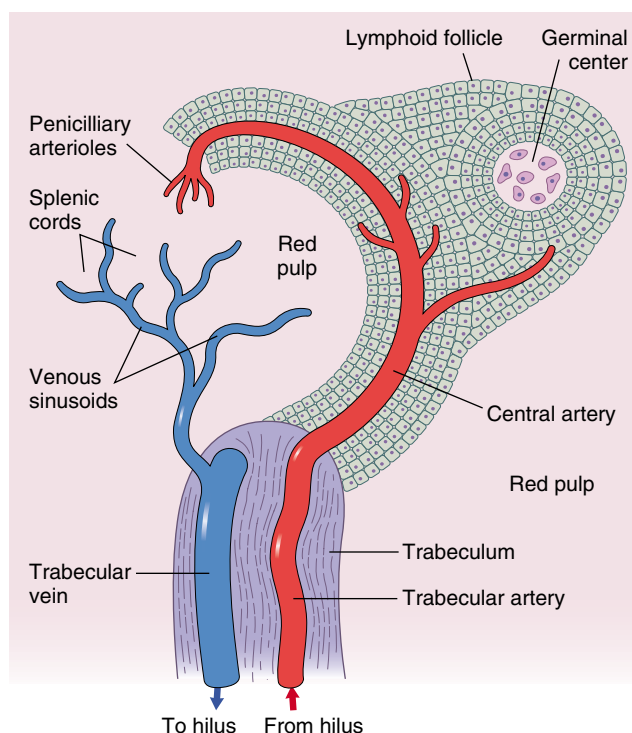


Fig. 13.41 Normal splenic architecture. (Modified from Faller DV. Diseases of the spleen. In: Wyngaarden JB, Smith LH, eds. *Cecil Textbook of Medicine*. 18th ed. WB Saunders; 1988:1036.)

open circulation or slow compartment. In the other “closed circuit,” blood passes rapidly and directly from the capillaries to the splenic veins. Although only a small fraction of splenic blood pursues the “open” route, during the course of a day the entire blood volume passes through the cords, where it is closely examined by macrophages.

The spleen has four functions that impact disease states:

- *Phagocytosis of blood cells and particulate matter.* As is discussed under the hemolytic anemias (Chapter 14), red cells undergo extreme deformation during passage from the splenic cords into the sinusoids. In conditions in which red cell deformability is decreased, red cells become trapped in the cords and are more readily phagocytosed by macrophages. Splenic macrophages also are responsible for “pitting” of red cells, the process by which inclusions such as *Heinz bodies* and *Howell-Jolly bodies* are excised, and for the removal of particles, such as bacteria, from the blood.
- *Antibody production.* Dendritic cells in the periarterial lymphatic sheath trap antigens and present them to T lymphocytes. T- and B-cell interaction at the edges of white pulp follicles leads to the generation of antibody-secreting plasma cells, which are found mainly within the sinuses of the red pulp. The spleen seems to be an important site of production of antibodies against microbial polysaccharides as well as autoantibodies against a variety of self-antigens.
- *Hematopoiesis.* During fetal development, the spleen may be a minor site of hematopoiesis, but this normally disappears by birth. However, the spleen can become a major site of compensatory extramedullary hematopoiesis in the setting of severe chronic anemia (e.g., in patients with thalassemia, described in Chapter 14) and in patients with myeloproliferative neoplasms, such as CML and primary myelofibrosis.
- *Sequestration of formed blood elements.* The normal spleen contains only about 30 to 40 mL of red cells, but this volume increases greatly with splenomegaly. The spleen also normally harbors approximately

30% to 40% of the total platelet mass in the body. With splenomegaly, up to 80% to 90% of the total platelet mass may be sequestered in the interstices of the red pulp, producing thrombocytopenia. Similarly, the enlarged spleen may trap white cells and thereby induce leukopenia.

The spleen is involved secondarily in all systemic inflammations, generalized hematopoietic disorders, and many metabolic disturbances. In each, the spleen undergoes enlargement (*splenomegaly*), which is the major manifestation of splenic disorders. It is only rarely the primary site of disease.

Splenic insufficiency due to splenectomy or autoinfarction (as in sickle-cell disease) has one major clinical manifestation, an increased susceptibility to sepsis caused by encapsulated bacteria such as pneumococci, meningococci, and *Haemophilus influenzae*. The decreases in phagocytic capacity and antibody production that result from asplenia both contribute to the increased risk of sepsis, which may be fatal. All asplenic individuals should be vaccinated against these agents to reduce the risk of this tragic complication.

SPLENOMEGALY

When sufficiently enlarged, the spleen causes a dragging sensation in the left upper quadrant and, through pressure on the stomach, discomfort after eating. In addition, enlargement can cause a syndrome known as *hypersplenism*, which is characterized by anemia, leukopenia, and/or thrombocytopenia. The probable cause of the cytopenias is increased sequestration of formed elements and the consequent enhanced phagocytosis by the splenic macrophages.

Table 13.12 lists major disorders associated with splenomegaly. Splenomegaly in virtually all the conditions mentioned has been discussed elsewhere. A few disorders are left to consider.

Nonspecific Acute Splenitis

Enlargement of the spleen occurs in any blood-borne infection. The nonspecific splenic reaction in these infections is caused both by the microbiologic agents themselves and by cytokines that are released as part of the immune response.

MORPHOLOGY

The spleen is enlarged (200 to 400 g) and soft. Microscopically, the major feature is acute congestion of the red pulp, which may encroach on and virtually efface the lymphoid follicles. Neutrophils, plasma cells, and occasionally eosinophils are usually present throughout the white and red pulp. At times, the white pulp follicles may undergo necrosis, particularly when the causative agent is a hemolytic streptococcus. Rarely, abscess formation occurs.

Congestive Splenomegaly

Chronic venous outflow obstruction causes a form of splenic enlargement referred to as *congestive splenomegaly*. Venous obstruction may be caused by intrahepatic disorders that retard portal venous drainage or extrahepatic disorders that directly impinge upon the portal or splenic veins. All these disorders ultimately lead to portal or splenic vein hypertension. Systemic, or central, venous congestion is encountered in cardiac decompensation involving the right side of the heart, as can occur in tricuspid or pulmonic valvular disease, chronic cor pulmonale, or following left-sided heart failure. Systemic congestion is associated with only moderately enlarged spleens that rarely exceed 500 g in weight.

Cirrhosis of the liver is the main cause of massive congestive splenomegaly. The “pipestem” hepatic fibrosis of schistosomiasis causes

TABLE 13.12 Disorders Associated With Splenomegaly

Infections
Nonspecific splenitis of various blood-borne infections (particularly infectious endocarditis)
Infectious mononucleosis
Tuberculosis
Typhoid fever
Brucellosis
Cytomegalovirus
Syphilis
Malaria
Histoplasmosis
Toxoplasmosis
Kala-azar
Trypanosomiasis
Schistosomiasis
Leishmaniasis
Echinococcosis
Congestive States Related to Portal Hypertension
Cirrhosis of the liver
Portal or splenic vein thrombosis
Cardiac failure
Lymphohematogenous Disorders
Hodgkin lymphoma
Non-Hodgkin lymphomas and lymphocytic leukemias
Multiple myeloma
Myeloproliferative neoplasms
Hemolytic anemias
Immunologic-Inflammatory Conditions
Rheumatoid arthritis
Systemic lupus erythematosus
Storage Diseases
Gaucher disease
Niemann-Pick disease
Mucopolysaccharidoses
Miscellaneous Disorders
Amyloidosis
Primary neoplasms and cysts
Secondary neoplasms

particularly severe congestive splenomegaly, but other causes such as alcohol-related cirrhosis may also evoke profound enlargements.

Congestive splenomegaly may also be caused by obstruction of the extrahepatic portal vein or splenic vein. This can stem from *portal vein thrombosis*, which is usually associated with some intrahepatic obstructive disease or inflammation of the portal vein (*pyelophlebitis*), such as follows intraperitoneal infections. Thrombosis of the splenic vein can be caused by infiltrating tumors arising in neighboring organs, such as carcinomas of the stomach or pancreas.

MORPHOLOGY

Long-standing splenic congestion produces marked enlargement (1000 to 5000 g). The organ is firm, and the capsule is usually thickened and fibrous. Microscopically, the red pulp is congested early in the course but becomes increasingly fibrotic and cellular with time. The elevated portal venous pressure stimulates the deposition of collagen in the basement membrane of the sinusoids, which appear dilated because of the rigidity of their walls. The resultant slowing of blood flow from the cords to the sinusoids prolongs the exposure of the blood cells to macrophages, resulting in excessive destruction (**hypersplenism**).

Splenic Infarcts

Because the spleen lacks any significant collateral blood supply, it is predisposed to infarction following obstruction of the splenic artery or its branches. The spleen, along with kidneys and brain, ranks as one of the most frequent sites where emboli lodge. In normal-sized spleens, infarcts are most often caused by emboli that arise from the heart. The infarcts may be small or large, single or multiple, or even involve the entire organ. They are usually bland except in individuals with infectious endocarditis of the mitral or aortic valves, in whom septic infarcts are common. Infarcts also are common in markedly enlarged spleens, regardless of cause, presumably because the blood supply in such spleens is tenuous and easily compromised. Finally, you may recall that autoinfarction of the spleen, typically during childhood, is an inevitable consequence of sickle cell disease (Chapter 14).

MORPHOLOGY

Bland infarcts are characteristically pale, wedge-shaped, and subcapsular in location. The overlying capsule is often covered with fibrin (Fig. 13.42). In septic infarcts this appearance is modified by the development of suppurative necrosis. During healing, large, depressed scars often develop.

NEOPLASMS

Neoplastic involvement of the spleen is rare except for myeloid and lymphoid tumors, which often cause splenomegaly (discussed earlier). The most common primary splenic tumors are hemangioma, often of the cavernous type, and lymphangioma. Many other types of mesenchymal tumors, both benign and malignant, may arise in the spleen, but all are very rare.

CONGENITAL ANOMALIES

Complete absence of the spleen is rare and is usually associated with other congenital abnormalities, such as situs inversus and cardiac malformations. Hypoplasia is a more common finding.

Accessory spleens (splenunculi) are common, being present singly or multiply in 20% to 35% of postmortem examinations. They are small, spherical structures that are histologically and functionally

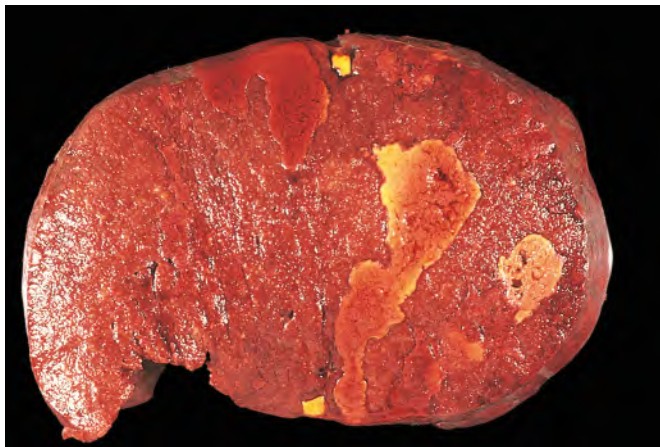


Fig. 13.42 Splenic infarcts. Multiple well-circumscribed infarcts are present in this spleen, which is massively enlarged (2820 g; normal: 150 to 200 g) by extramedullary hematopoiesis secondary to a myeloproliferative neoplasm (myelofibrosis). Recent infarcts are hemorrhagic, whereas older, more fibrotic infarcts are a pale yellow-gray color.

identical to the normal spleen. They can be found at any place within the abdominal cavity. Accessory spleens are of clinical importance in some hematologic disorders, such as hereditary spherocytosis (Chapter 14), where splenectomy may be used as a treatment. If an accessory spleen is overlooked, the therapeutic benefit of removal of the definitive spleen may be reduced or lost entirely.

RUPTURE

Splenic rupture is usually precipitated by blunt trauma. Much less often, it occurs in the apparent absence of a physical blow. Such “spontaneous ruptures” never involve truly normal spleens but rather stem from some minor physical insult to a spleen made fragile by an underlying condition. The most common predisposing conditions are infectious mononucleosis, malaria, typhoid fever, and lymphoid neoplasms, all of which may cause the spleen to enlarge rapidly, producing a thin, tense capsule that is susceptible to rupture. This dramatic event often precipitates intraperitoneal hemorrhage, which must be treated by prompt splenectomy to prevent death from blood loss. Chronically enlarged spleens are unlikely to rupture because of the toughening effect of extensive reactive fibrosis.

THYMUS

Once an organ buried in obscurity, the thymus has a starring role in cell-mediated immunity (Chapter 6). Here, our interest centers on the disorders of the gland itself.

The thymus is embryologically derived from the third and, inconsistently, the fourth pair of pharyngeal pouches. At birth it weighs 10 to 35 g. It grows until puberty, achieving a maximum weight of 20 to 50 g, and thereafter undergoes progressive involution to little more than 5 to 15 g in older adults. The thymus may also involute in children and young adults in response to severe illness and HIV infection.

The fully developed thymus is composed of two fused, well-encapsulated lobes. Fibrous extensions of the capsule divide each lobe into numerous lobules, each with an outer cortical layer enclosing the central medulla. Diverse types of cells populate the thymus, but thymic epithelial cells and immature T lymphocytes, also called thymocytes, predominate. Epithelial cells in the cortex are polygonal in shape and have an abundant cytoplasm with dendritic extensions that contact adjacent cells. In contrast, the epithelial cells in the medulla are densely packed, often spindle-shaped, and have scant cytoplasm devoid of interconnecting processes. Whorls of degenerated medullary epithelial cells create *Hassall corpuscles*, with their characteristic keratinized cores.

As you know from consideration of the thymus in relation to immunity (Chapter 6), progenitor cells migrate from the marrow to the thymus and mature into T cells, which are exported to the periphery, but only after they have been educated in the “thymic university” to distinguish between self-antigens and non-self-antigens. During adulthood the thymic production of T cells slowly declines as the organ atrophies.

Macrophages, dendritic cells, a minor population of B lymphocytes, rare neutrophils and eosinophils, and scattered myoid (muscle-like) cells are also found within the thymus. The myoid cells are of particular interest because they likely play some role in the development of myasthenia gravis, a musculoskeletal disorder of immune origin.

Pathologic changes within the thymus are limited and will be described here. The changes associated with myasthenia gravis are considered in Chapter 27.

DEVELOPMENTAL DISORDERS

Thymic hypoplasia or *aplasia* is seen in DiGeorge syndrome, which is marked by severe defects in cell-mediated immunity. As discussed in Chapter 6, DiGeorge syndrome is often associated with other developmental defects, particularly parathyroid hypoplasia and malformations of the heart and palate, as part of the chromosome 22q11 deletion syndrome.

Isolated *thymic cysts* are uncommon lesions that are usually discovered incidentally postmortem or during surgery. They rarely exceed 4 cm in diameter, can be spherical or arborizing, and are lined by stratified to columnar epithelium. The fluid contents can be serous or mucinous and are often modified by hemorrhage.

While isolated cysts are not clinically significant, neoplastic thymic masses (whatever their origin) compress and distort adjacent normal thymus and sometimes cause cysts to form. Therefore, the presence of a cystic thymic lesion in a symptomatic patient should provoke a thorough search for a neoplasm, particularly a lymphoma or a thymoma.

THYMIC HYPERPLASIA

The term thymic hyperplasia is misleading, since it usually applies to the appearance of B-cell germinal centers within the thymus, a finding that is referred to as *thymic follicular hyperplasia*. Such B-cell follicles are present in only small numbers in the normal thymus. Although follicular hyperplasia can occur in several chronic inflammatory and immunologic states, it is most frequently encountered in myasthenia gravis, in which it is found in 65% to 75% of cases (Chapter 27). Similar thymic changes are sometimes encountered in Graves disease, systemic lupus erythematosus, scleroderma, rheumatoid arthritis, and other autoimmune disorders.

In other instances, a morphologically normal thymus is simply large for the age of the patient. As mentioned, the size of the thymus varies widely, and whether this constitutes a true hyperplasia or is merely a variant of normal is unclear. The main significance of this form of thymic “hyperplasia” is that it may be mistaken radiologically for a thymoma, leading to unnecessary surgical procedures.

THYMOMA

A diversity of neoplasms may arise in the thymus—germ cell tumors, lymphomas, carcinoids, and others—but the designation “**thymoma**” is restricted to tumors of thymic epithelial cells. Thymoma accounts for about 80% to 85% of all thymic neoplasms. Such tumors typically also contain benign immature T cells (thymocytes).

The WHO has created a classification system based on histology for thymomas, but its clinical utility remains uncertain. We instead use a classification that relies on the most important prognostic features, the surgical stage and the presence or absence of overt cytologic features of malignancy. In this simple system there are only three histologic subtypes:

- Tumors that are cytologically benign and noninvasive
- Tumors that are cytologically benign but invasive or metastatic
- Tumors that are cytologically malignant (typically referred to as *thymic carcinoma*)

In all categories the tumors usually occur in adults older than 40 years; thymomas are rare in children. Males and females are affected equally. Most tumors arise in the anterior superior mediastinum, but sometimes they occur in the neck, thyroid, pulmonary hilus, or elsewhere. They are uncommon in the posterior mediastinum. Thymomas account for 20% to 30% of tumors in the anterosuperior mediastinum, which is also a common location for certain lymphomas.

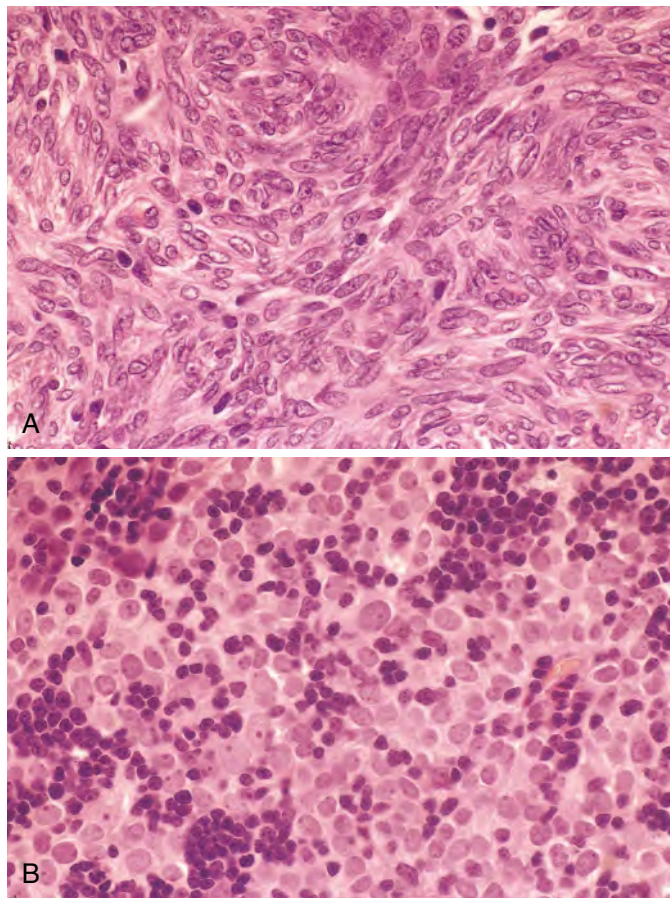


Fig. 13.43 Thymoma. (A) Benign thymoma (medullary type). The neoplastic epithelial cells are arranged in a swirling pattern and have bland, oval to elongated nuclei with inconspicuous nucleoli. Only a few small, non-neoplastic lymphoid cells are interspersed. (B) Malignant thymoma. The neoplastic epithelial cells are polygonal and have round to oval, bland nuclei with inconspicuous nucleoli. Numerous small lymphoid cells representing immature T cells (thymocytes) are interspersed. The morphologic appearance of this tumor is identical to that of benign thymomas of the cortical type. In this case, however, the tumor was locally aggressive, invading adjacent lung and pericardium.

MORPHOLOGY

Macroscopically, thymomas are lobulated, firm, gray-white masses of up to 15 to 20 cm in size. They sometimes have areas of cystic necrosis and calcification. Most are encapsulated, but 20% to 25% of tumors penetrate the capsule and infiltrate perithymic tissues and structures.

Noninvasive thymomas are most often composed of medullary-type epithelial cells or a mixture of medullary- and cortical-type epithelial cells. The medullary-type epithelial cells are elongated or spindle-shaped (Fig. 13.43A). There is usually a sparse infiltrate of thymocytes, which often recapitulate the phenotype of medullary thymocytes. In mixed thymomas there is an admixture of polygonal cortical-type epithelial cells and a denser infiltrate of thymocytes. The medullary and mixed patterns together account for about 50% of all thymomas. Tumors that have a substantial proportion of medullary-type epithelial cells are usually noninvasive.

Invasive thymoma refers to a tumor that is cytologically benign but locally invasive. These tumors are much more likely to metastasize. The epithelial cells are most commonly of the cortical variety, with abundant cytoplasm and rounded vesicular nuclei (Fig. 13.43B) and are usually mixed with numerous

thymocytes. In some cases, the neoplastic cells show cytologic atypia, a feature that correlates with a propensity for more aggressive behavior. These tumors account for about 20% to 25% of all thymomas. **By definition, invasive thymomas penetrate through the capsule into surrounding structures.** The extent of invasion has been subdivided into various stages, which are beyond our scope. With minimal invasion, complete excision yields a 5-year survival rate of greater than 90%, whereas extensive invasion is associated with a 5-year survival rate of less than 50%.

Thymic carcinoma represents about 10% of all thymic neoplasms. Macroscopically, they are usually fleshy, obviously invasive masses, sometimes accompanied by metastases to sites such as the lungs. Microscopically, most are **squamous cell carcinomas**. Another distinctive variant is **lymphoepithelioma-like carcinoma**, a tumor composed of sheets of cells that bears a close histologic resemblance to nasopharyngeal carcinoma. About 50% of lymphoepithelioma-like carcinomas contain monoclonal EBV genomes, consistent with a role for EBV in their pathogenesis. A variety of other, less common histologic patterns of thymic carcinoma have been described; all exhibit cytologic atypia seen in other carcinomas.

Clinical Features

About 40% of thymomas present with symptoms stemming from impingement on mediastinal structures. Another 30% to 45% are detected in the course of evaluating patients with myasthenia gravis. The rest are discovered incidentally during imaging studies or cardiothoracic surgery. In addition to myasthenia gravis, other associated autoimmune disorders include hypogammaglobulinemia, pure red cell aplasia, Graves disease, pernicious anemia, dermatomyositis-polymyositis, and Cushing syndrome. The basis for these associations is uncertain, but the thymocytes that arise within thymomas give rise to long-lived CD4+ and CD8+ T cells, and cortical thymomas rich in thymocytes are more likely to be associated with autoimmune disease. Hence it seems likely that abnormalities in the selection or “education” of T cells maturing within the environment of the neoplasm contribute to the development of diverse autoimmune disorders.

SELECTED READINGS

- Alaggio R, Amador C, Anagnostopoulos I, et al. The 5th edition of the World Health Organization Classification of Haematolymphoid Tumours: Lymphoid Neoplasms. *Leukemia*. 2022;36:1720-1748. [Overview of the most recent version of the classification system used for lymphoid neoplasms.]
- Bielamowicz K, Dimitrion P, Abba O, et al. Langerhans cell histiocytosis: NACHO update on progress, chaos, and opportunity on the path to rational cures. *Cancer*. 2024;130:2416-2439. [Review of the pathogenesis and clinical features of Langerhans cell histiocytosis.]
- Bosch F, Dalla-Favera R. Chronic lymphocytic leukaemia: from genetics to treatment. *Nat Rev Clin Oncol*. 2019;16:684-701. [A review of the impact of driver mutations on the pathogenesis and clinical management of CLL.]
- Gao X, Xu C, Asada N, Frenette PS. The hematopoietic stem cell niche from embryo to adult. *Development*. 2018;145:dev139691. [Discussion of origins of hematopoietic stem cells.]
- Gordon MJ, Smith MR, Nastoupil LJ. Follicular lymphoma: the long and winding road leading to your cure? *Blood Rev*. 2023;57:100992. [Summary of follicular lymphoma pathobiology and new approaches to therapy.]
- Jain P, Wang ML. Mantle cell lymphoma in 2022—A comprehensive update on molecular pathogenesis, risk stratification, clinical approach, and current and novel treatments. *Am J Hematol*. 2022;97:638-656. [A comprehensive update on molecular pathogenesis, risk stratification, clinical approach, and current and novel treatments.]
- Jaiswal S, Ebert BL. Clonal hematopoiesis in human aging and disease. *Science*. 2019;366:eaan4673. [Excellent review from leaders in this field.]
- Khoury JD, Solary E, Abba O, et al. The 5th edition of the World Health Organization Classification of Haematolymphoid Tumours: Myeloid and Histiocytic/Dendritic Neoplasms. *Leukemia*. 2022;36:1703-1719. [Overview of the most recent version of the classification system used for myeloid and histiocytic/dendritic cell neoplasms.]
- Küppers R, Stevenson FK. Critical influences on the pathogenesis of follicular lymphoma. *Blood*. 2018;131:2297-2306. [Discussion of new insights into the pathogenesis of this disease, including its genetics and its microenvironmental influences.]
- Ogawa S. Genetics of MDS. *Blood*. 2019;133:1049-1059. [Comprehensive review of genetic drivers in myelodysplastic syndromes.]
- Pinho S, Frenette PS. Haematopoietic stem cell activity and interactions with the niche. *Nat Rev Mol Cell Biol*. 2019;20:303-320. [Discussion of the nature and biology of the marrow stem cell niche.]
- Rajkumar SV. Multiple myeloma: 2024 update on diagnosis, risk stratification, and management. *Am J Hematol*. 2024;99:1802-1824. [A review focused on myeloma diagnosis, classification, prognostication, and therapy.]
- Roschewski M, Staudt LM, Wilson WH. Burkitt's lymphoma. *New Engl J Med*. 2022;387:1111-1122. [Review of the pathophysiology, diagnosis, and treatment of this aggressive non-Hodgkin lymphoma.]
- Sallman DA, List A. The central role of inflammatory signaling in the pathogenesis of myelodysplastic syndromes. *Blood*. 2019;133:1039-1048. [Review of emerging link between myelodysplastic syndromes and inflammation.]
- Terwilliger T, Abdul-Hay M. Acute lymphoblastic leukemia: a comprehensive review. *Blood Canc J*. 2017;7:e577. [Overview of the genetics, clinical features, and treatment of childhood acute lymphoblastic leukemia.]
- Vainchenker W, Kralovics R. Genetic basis and molecular pathophysiology or classical myeloproliferative neoplasms. *Blood*. 2017;129:667-679. [Review of the pathogenesis of the major subtypes of myeloproliferative neoplasms.]
- Sehn LH, Salles G. Diffuse large B-cell lymphoma. *New Engl J Med*. 2021;384:842-858. [Review of the pathogenesis, diagnosis, and treatment of diffuse large B-cell lymphoma, the most common lymphoma.]